THIRSTY'S CODEX

# STANDARDS EDITION

VOLUME II / 02 OF 20

## RESEARCH, STRATEGY, AND RISK

Lifecycle Sections 001-018

NO PROOF, NO CLAIM

CONTINUOUS MASTER FOLIO / SOURCE-PRESERVING EDITION

# Edition Statement

THIRSTY'S CODEX / STANDARDS EDITION / SOURCE-PRESERVING SET

This volume is one part of a continuous 20-volume edition. Canonical lifecycle section numbers remain stable across the set; no volume restarts or renumbers them.

The visual treatment has been rebuilt as a unified standards library: coordinated covers, volume architecture, source dividers, running headers, continuous master folios, local page numbers, and source-identity marks.

The source corpus remains authoritative over the presentation layer. Status terms are preserved exactly. CONTENT-COMPLETE does not silently become source-reconciled, hostile-reviewed, owner-accepted, ratified, or published.

Every registered source witness is identified by path, byte count, and SHA-256 digest. The complete source archive is preserved as an embedded attachment in Volume XX, while the two supplied foundational PDFs are preserved as attachments in Volume I.

**CONTINUITY RULE**

Read the master folio and canonical section sequence as one uninterrupted work. Each succeeding volume begins where the preceding volume ends.

# Standards Edition Set Map

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**Foundations and Governance**
Doctrine, prompts, front matter, shared governance, roles, and output contracts

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**Research, Strategy, and Risk**
Lifecycle Sections 001-018

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Preserved standards, command editions, blueprints, and source artifacts

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**Canonical Authority and Assurance**
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Agent-output corpus, V3Q implementation, fixtures, tests, and governed examples

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Continuous-pretraining, evaluation, retrieval, and training views with canonical ...

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**Static Analysis and Tooling Evidence**
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# 001

## SECTION 001 - RESEARCH

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SECTION 001 — Research

## STATUS: CONTENT-COMPLETE

## PURPOSE

Establish a truthful, source-grounded understanding of the problem, context, people, domain, existing state, risks, constraints, and unknowns before those unknowns are converted into architecture or code.

Research is not a ceremonial opening phase and is not limited to web searching. It is the governed process for determining what is known, how it is known, what remains uncertain, whose experience matters, which sources control, and what evidence is strong enough to support the next decision.

## REQUIRED OUTCOME

A bounded research package that allows the human and assigned agents to answer:

- What is the actual objective?
- What problem or opportunity is being addressed?
- Who is affected, accountable, authorized, or exposed to risk?
- What current system, workflow, environment, market, domain, or physical reality already exists?
- Which sources are authoritative, current, traceable, and relevant?
- What assumptions and unknowns materially affect feasibility or design?
- What conflicting evidence or dissent exists?
- What must be learned next before consequential design or implementation can proceed?
- What safe work may continue while unresolved research remains open?

## WHY THIS MATTERS

A coding agent can produce plausible implementation faster than a human can notice that the wrong problem was framed. Once an unsupported assumption becomes architecture, every downstream artifact may become internally consistent and still be wrong.

Research protects the objective from substitution, prevents stale or anonymous sources from becoming requirements, exposes dependencies before expensive work, and gives later verification a record of why decisions were made. It also protects future human-AI relationships by requiring both parties to distinguish evidence from confidence and questions from decisions.

## APPLICABILITY

Research applies whenever the work depends on facts, needs, behavior, constraints, environments, technology, law, risk, feasibility, prior art, user experience, market conditions, physical conditions, or source state that are not already adequately established.

Depth is proportional to consequence, novelty, uncertainty, disagreement, cost, and reversibility. A small reversible script may need a brief source and requirement check. A production system, safety-critical device, governance mechanism, AI system, public product, or irreversible change requires deeper and independently challenged research.

## NON-APPLICABILITY CONDITIONS

Research may be minimal when all material inputs are already authoritative, current, directly supplied, internally consistent, and sufficient for the declared low-consequence scope.

A non-applicability determination must still identify:

- the controlling sources;
- why they are sufficient;
- their version or freshness;

- the limited scope for which additional research is unnecessary;
- the condition that would reopen research.

"The user did not ask for research" is not by itself a sufficient reason to skip a material unknown.

## PREREQUISITES AND DEPENDENCIES

Research begins from:

- an attributable human objective;
- a declared initial scope or question;
- known authority and data-access boundaries;
- available current-state and continuity records;
- preservation constraints for people, data, systems, property, and existing work.

Research commonly informs Sections 002–019 and every architecture section beginning with Section 020. It may be reopened by later feasibility, testing, incident, user-validation, or operational evidence.

## REQUIRED INPUTS

- The human's actual objective and desired outcome.
- Known users, owners, operators, stakeholders, affected communities, and decision owners.
- Existing documents, repositories, systems, products, workflows, interfaces, datasets, environments, logs, incidents, and continuity records.
- Known constraints, exclusions, prohibited actions, risk tolerance, time and resource limits.
- Candidate external sources and access methods.
- Questions already asked, decisions already made, and prior failed approaches.
- Any source whose authority, identity, version, integrity, or freshness controls downstream work.

## HUMAN RESPONSIBILITIES AND AUTHORITY

The human:

- defines or confirms the objective;
- identifies values, priorities, unacceptable harms, protected information, and practical constraints;
- authorizes access to private or controlled sources;
- corrects mischaracterized needs, context, and lived experience;
- decides material value conflicts and accepts risk where acceptance cannot be delegated;
- determines whether unresolved uncertainty is tolerable for the next bounded step;
- retains final authority over objective changes.

The human is not required to know every research question in advance. The AI should surface missing questions without claiming the human already intended their answers.

## AI RESPONSIBILITIES AND BOUNDARIES

The AI:

- records the objective without silent substitution;
- inspects supplied source material and current state before proposing conclusions;
- creates a research plan proportional to consequence and uncertainty;
- separates observations, source claims, inferences, assumptions, recommendations, and unknowns;
- identifies source identity, version, provenance, freshness, scope, and limitations;
- searches for contradictory evidence and missing perspectives;
- preserves material dissent rather than manufacturing consensus;
- identifies the cheapest safe question, inspection, prototype, or experiment that can reduce a material unknown;
- continues unaffected safe research when one source or question is blocked;
- does not fabricate access, interviews, observations, citations, data, expertise, or verification;
- does not convert research findings directly into consequential execution without the required downstream authority and design work.

## APPLICABLE EXISTING AGENT ROLES

Primary role:

- Researcher — gathers, evaluates, organizes, and reports source-grounded information within the accepted question and authority boundary.

Common supporting roles:

- Observer — records system, user, process, or environment behavior without silently converting observation into interpretation.
- Analyst — compares evidence, patterns, constraints, and implications.
- Architect — identifies which unknowns materially affect system structure, interfaces, trust, and implementation sequencing.
- Planner — sequences research activities, dependencies, cost, and decision points.
- Memory Curator — preserves source identity, decisions, continuity, and retrievable research state.
- User Experience Reviewer — ensures user and owner research addresses actual journeys, comprehension, agency, failure, and recovery.

Challenge and assurance roles where consequence warrants:

- Critic — attacks assumptions, framing, omissions, and unsupported conclusions.
- Verifier — checks source identity, evidence integrity, reproducibility, and claim boundaries.
- Security Auditor — evaluates sensitive data access, threat exposure, and research handling.
- Governance Reviewer — checks authority, consent, provenance, policy, and evidence obligations.
- Red Team — searches for adversarial framing, manipulation, blind spots, and failure-inducing assumptions.

Role selection must use the smallest sufficient set and must not alter the source-defined role contracts.

## REQUIRED ACTIVITIES

1. Objective capture
Record the human's objective, desired outcome, exclusions, and known constraints in the human's own terms before translating them into professional or technical language.

2. Question decomposition
Convert the objective into explicit research questions. Identify which questions concern users, domain, technology, business, law, ethics, safety, security, feasibility, operations, or existing state.

3. Source inventory
List supplied and discovered sources. For each material source, record identity, origin, owner or publisher, version, date, location, access method, integrity information where available, relevance, and limitations.

4. Source precedence
Identify which sources are authoritative for which claims. Do not assume that popularity, confidence, technical polish, organizational title, or search ranking establishes authority.

5. Current-state inspection
Inspect existing artifacts, systems, workflows, continuity, failures, and constraints before designing a replacement or parallel structure. Research must discover what already exists, not merely what a generic solution would contain.

6. Human and user inquiry
Where people are affected, document goals, language, current journeys, pain points, decisions, workarounds, accessibility needs, trust expectations, failure experiences, and recovery behavior. Protect personal information and do not turn temporary frustration into a permanent identity label.

7. Assumption and unknown register
Record every material assumption and unknown, its impact, owner, evidence needed, expiration or review condition, and whether safe work can continue.

8. Contradiction and dissent search

Seek credible contradictory evidence. Preserve disagreements with their source, impact, decision owner, current disposition, and reopening condition.

9. Risk and ethics review
Identify potential harm to people, data, property, systems, institutions, environments, and future users. Determine whether research itself requires consent, privacy controls, security controls, or legal review.

10. Resolution design
For each blocking or high-impact unknown, identify the minimum safe evidence request or experiment with expected information gain, cost, risk, and stop conditions.

11. Synthesis
Produce direct findings first, followed by evidence, uncertainty, implications, and recommended next actions. Deep research may be complex; its result must remain navigable.

12. Handoff
Transfer the research package without losing source identity, uncertainty, dissent, authority, or the boundary between findings and decisions.

## HUMAN-AI INTERACTION REQUIREMENTS

- Research questions proposed by the AI remain proposals until accepted or safely included within delegated scope.
- The AI must show when it is asking, observing, inferring, recommending, or reporting a verified fact.
- The human must be able to correct misunderstood terminology, context, priorities, and affected parties.
- Operational claims must identify the current source and freshness.
- The AI must not claim to have interviewed, observed, tested, browsed, measured, or verified anything it did not actually access or perform.
- When the human rejects a proposed research direction, the AI must preserve the decision and any resulting uncertainty rather than reintroducing it as settled fact.
- Material ambiguity that changes safety, legality, cost, architecture, or consequence must be clarified before execution.
- Research should expose the next decision without seizing it.

## ARCHITECTURE AND SYSTEM IMPLICATIONS

Research must identify architectural obligations without prematurely selecting an architecture. Relevant findings may include:

- actors, identities, authority sources, and trust boundaries;
- data types, sensitivity, location, retention, quality, and provenance;
- external systems, APIs, devices, infrastructure, networks, and physical constraints;
- expected scale, latency, availability, reliability, safety, and resilience;
- regulatory, contractual, licensing, intellectual-property, and accessibility constraints;
- user journeys, operator workflows, cockpit requirements, and recovery expectations;
- deployment, environment, offline, integration, maintenance, and retirement conditions;
- unresolved facts whose absence prevents responsible architecture.

The output should tell the Architect what must be accounted for, not dictate a preferred design without analysis.

## OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

A research cockpit or equivalent operating surface should expose:

- the current research objective and scope;
- active questions and their priority;
- source inventory, identity, freshness, and access state;
- findings separated from assumptions, recommendations, and unknowns;
- conflicts, dissent, and missing perspectives;
- blockers and safe work that may continue;
- human decisions required;

- evidence supporting each material finding;
- privacy or data-handling constraints;
- the next safe action and the condition for closing or reopening research.

The surface must not present a source count, progress bar, confidence score, or polished summary as proof that the research question has been answered.

## REQUIRED OUTPUTS AND ARTIFACTS

Minimum outputs, adapted to scope:

- Research Brief
- Objective and Question Register
- Source Register
- Source-to-Claim Map
- Current-State Inventory
- Stakeholder or Affected-Party Map where applicable
- User or Owner Job and Journey Findings where applicable
- Assumption and Unknown Register
- Risk, Ethics, Privacy, and Consent Notes where applicable
- Findings Ledger
- Dissent and Contradiction Ledger
- Resolution Experiment or Evidence Request Plan
- Research Decision and Handoff Record
- Continuity Update

## EVIDENCE REQUIREMENTS

A material research claim should point to one or more of:

- supplied source text or artifact location;
- authoritative publication, standard, law, specification, dataset, repository, or record;
- direct observation with date, environment, method, and observer;
- interview or user-research record with consent and handling requirements;
- system output, log, configuration, trace, test, or measured result;
- reproducible search or retrieval method;
- source hash, version, revision, timestamp, or other identity binding;
- explicit limitation or unresolved uncertainty.

Evidence must be sufficient for the bounded claim. A citation to a source does not prove that the source is correct, current, authoritative for the claim, or applicable to the present context.

## VERIFICATION AND ACCEPTANCE TESTS

Research is accepted for its declared scope when an independent reviewer can:

- reconstruct the objective and research questions without chat memory;
- identify the source supporting each material finding;
- distinguish findings, assumptions, unknowns, recommendations, and decisions;
- verify source identity, version, and freshness where material;
- identify at least one deliberate attempt to find contradictory evidence for consequential claims;
- trace major architecture or requirements implications back to findings;
- confirm that blocked questions remain visible and have resolution conditions;
- confirm that personal, private, controlled, or security-sensitive research data was handled within authority;
- determine what may safely happen next.

## FAILURE STATES AND ANTI-PATTERNS

- Objective substitution: researching a more familiar problem than the human asked about.

- Confirmation bias: collecting only evidence supporting the first idea.
- Source laundering: repeating a claim across many derivative sources and presenting repetition as independent support.
- Anonymous authority: accepting a source without establishing who produced it, when, and for what context.
- Stale state: designing from outdated documentation while ignoring current behavior.
- Search-result theater: treating quantity of results as research quality.
- Unperformed observation: claiming interviews, tests, measurements, browsing, or repository inspection that did not occur.
- Assumption collapse: converting an assumption or recommendation into a requirement.
- User erasure: researching technical feasibility while ignoring the people who must operate, trust, recover, or live with the system.
- Happy-path bias: studying ideal use without failure, misuse, degradation, or recovery.
- Hidden conflict: deleting dissent to produce a clean conclusion.
- Privacy extraction: collecting more personal or sensitive data than the research objective requires.
- Endless research: continuing without a decision threshold, stop condition, or information-gain rationale.
- Premature certainty: closing a question because the answer sounds plausible.

## RECOVERY AND RECONSIDERATION

When research is incomplete or unreliable:

- quarantine affected conclusions from downstream use;
- identify which requirements, architecture decisions, or implementation tasks are contaminated;
- restore the last source-grounded state;
- correct source identity, scope, method, or data-handling defects;
- request the minimum missing evidence;
- rerun affected analysis and update downstream traceability;
- preserve the prior conclusion and the evidence that reversed it rather than rewriting history;
- reopen the research question when its stated reconsideration condition is met.

## COMPLETION CRITERIA

Section 001 is complete for a declared research scope only when:

- the objective and questions are explicit;
- material sources are identified and bounded;
- current state has been inspected where relevant;
- assumptions, unknowns, conflicts, and limitations are visible;
- affected humans and owner journeys have been addressed where applicable;
- evidence supports the stated findings;
- consequential claims received appropriate challenge;
- blocked questions have resolution paths;
- the human decision points are explicit;
- the next safe action is known;
- the research package and continuity record can be handed to another qualified actor without reconstruction from chat history.

Completion of research does not mean every unknown is resolved. It means the remaining uncertainty is explicit, bounded, assigned, and compatible with the next authorized step.

## HANDOFF AND CONTINUITY REQUIREMENTS

The handoff must include:

- objective and scope;
- research questions;
- source identities and locations;
- findings and evidence;
- assumptions and unknowns;

- dissent and contradictions;
- privacy, consent, security, and authority constraints;
- decisions made and decisions still required;
- downstream implications;
- blocked work and safe work;
- next safe action;
- exact boundary of what the research did not establish.

## RELATED CODEX SECTIONS

002 Foresight / Futures
011 Business Analysis
012 Market Research
013 Competitive Intelligence
014 Stakeholder Management
015 User Research
016 Domain Analysis
017 Feasibility Analysis
018 Risk Management
019 Requirements Engineering
020 Systems Engineering
043 Event-Driven Architecture and all later architecture sections as applicable
106 User Validation
163 Data Analysis
164 Data Science
165 Statistics
208 Threat Modeling
256 Verification
257 Validation
369 Experimentation
372 Feedback Management
402 Post-Implementation Review
403 Postmortem Analysis
404 Lessons Learned
405 Lifecycle Governance

## SOURCE BASIS AND UNRESOLVED SOURCE GAPS

This draft applies the supplied requirements for objective preservation, unknown-means-unknown, evidence before claims, current-state inspection, continuity, hostile review, bounded roles, source identity, owner research, owner journeys, human authority, delegation, proof-carrying work, and prerequisite-aware resolution.

Unresolved source gap: the supplied corpus does not yet contain a dedicated universal research standard spanning every field in the 406-section lifecycle. This section has been cross-linked with drafted Sections 012, 015, 016, and 017 at the first integration checkpoint. It must later receive full source reconciliation with those sections and with Sections 163, 164, 165, 208, 256, 257, and 369 as the remaining lifecycle coverage is authored.

# 002

## SECTION 002 - FORESIGHT OR FUTURES

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SECTION 002 — Foresight / Futures

## STATUS: CONTENT-COMPLETE

## PURPOSE

Establish a governed method for examining how the conditions surrounding an objective may change over time, so that present decisions do not depend upon one assumed future or quietly transfer avoidable risk to future users, operators, owners, communities, intelligent systems, or environments.

Foresight is not prediction, prophecy, trend theater, or permission to invent certainty. It is the disciplined examination of plausible change, persistent uncertainty, alternative futures, emerging signals, discontinuities, long-term consequences, and the conditions under which a present plan should continue, adapt, pause, reverse, or be rejected.

Within Thirsty's Codex, foresight exists to help humans and AI build with memory, humility, reversibility, and responsibility. It extends the time boundary of the work: beyond the next feature, release, funding cycle, model update, deployment, or operator session, toward the people and intelligences who may inherit what is created.

## REQUIRED OUTCOME

A bounded futures package that allows the human and assigned agents to answer:

- What time horizons matter to this objective?
- Which present conditions are being treated as stable, and what evidence supports that treatment?
- Which drivers, trends, constraints, dependencies, and emerging signals could materially change the objective or its implementation?
- Which uncertainties are both highly consequential and genuinely unresolved?
- What multiple plausible futures should be considered rather than collapsed into one forecast?
- What opportunities, harms, second-order effects, lock-ins, or path dependencies could emerge?
- Which present decisions remain useful across several futures?
- Which decisions are fragile, irreversible, or dependent on one narrow scenario?
- What indicators, thresholds, and review triggers would show that assumptions are failing?
- What should be preserved so future humans and AI systems can understand, challenge, continue, or reverse today's decisions?
- Which value conflicts or long-term consequences require explicit human judgment?

The output is not a claim that the future has been discovered. It is a transparent map of uncertainty, plausible conditions, strategic options, and monitored commitments.

## WHY THIS MATTERS

Systems are often designed around an invisible sentence: "the world will continue to work like this." That sentence may hide assumptions about users, institutions, laws, infrastructure, economics, climate, security, supply chains, platforms, models, interfaces, ownership, social acceptance, or the capabilities and roles of future AI systems.

When one assumed future becomes architecture, present convenience can become future constraint. Data formats become lock-in. Temporary permissions become permanent authority. A local workaround becomes an operational dependency. A model provider becomes a single point of failure. An interface teaches users to surrender judgment. A governance shortcut becomes precedent. A design that works for today's operator can exclude tomorrow's owner.

Foresight reduces that risk by making time, uncertainty, alternatives, and reversal conditions visible before consequential choices harden. For human-AI relations, this is especially important. Capability may change faster than authority, law, institutional practice, social expectation, or human understanding. The Codex therefore requires future capability to be considered without silently granting future permission, status, rights, duties, or decision authority. Those questions remain explicit human and institutional decisions unless legitimately delegated.

## APPLICABILITY

Foresight applies when one or more of the following are material:

- the system or decision is expected to persist, scale, or be inherited;
- the work creates long-lived data, infrastructure, interfaces, standards, dependencies, or governance rules;
- the objective depends on changing technology, markets, regulation, security conditions, resource availability, public trust, or social behavior;
- the project involves AI, autonomous agents, robotics, connected devices, critical infrastructure, public systems, safety, privacy, identity, or high-consequence decisions;
- present choices may create lock-in, irreversible effects, concentrated power, exclusion, environmental burden, or difficult retirement obligations;
- the system must remain useful across uncertain operating environments;
- the human requests strategic, long-range, resilient, adaptive, or future-facing design;
- a decision is sensitive to assumptions about capability growth, cost decline, adoption, regulation, interoperability, or institutional response;
- future owners, users, operators, affected communities, or intelligent systems may need to understand or contest the inherited structure.

Depth is proportional to time horizon, consequence, uncertainty, irreversibility, novelty, disagreement, and the cost of being wrong. Foresight for a disposable personal script may be brief. Foresight for a public platform, constitutional AI system, safety mechanism, long-lived dataset, physical device, or infrastructure layer requires wider scenarios, stronger challenge, explicit monitoring, and long-term recovery thinking.

## NON-APPLICABILITY CONDITIONS

Foresight may be minimal when the work is short-lived, low-consequence, fully reversible, narrowly bounded, and independent of material external change.

A minimal or non-applicability determination must still record:

- the expected lifetime of the work;
- why external change is unlikely to alter the declared outcome during that lifetime;
- why failure or obsolescence is safely reversible;
- what future condition would reopen foresight analysis;
- who accepted the residual risk.

"We cannot know the future" is not a sufficient reason to skip foresight. The purpose is not to know the future. It is to avoid pretending that only one future exists.

## PREREQUISITES AND DEPENDENCIES

Foresight begins from:

- Section 001 — Research, including the accepted objective, current-state evidence, assumptions, unknowns, and source register;
- an identifiable human decision owner;
- a declared scope and relevant time horizon;
- known preservation, safety, privacy, legal, ethical, resource, and authority constraints;
- an initial understanding of the people, systems, environments, and institutions affected;
- a distinction between current facts, forecasts, preferences, aspirations, and speculative possibilities.

Foresight commonly informs Sections 003–019, all architecture sections, risk and safety work, product and portfolio decisions, maintenance, modernization, deprecation, archival, lifecycle governance, and retirement. It must be reopened when monitored conditions change, a major assumption fails, a new discontinuity emerges, or downstream evidence contradicts the futures basis.

## REQUIRED INPUTS

- The accepted objective and present decision question.
- Current-state research and authoritative source material.
- Expected product, system, service, data, hardware, or governance lifetime.

- Relevant time horizons, including near-term, operational, strategic, and generational horizons where applicable.
- Known drivers of change: technical, social, legal, economic, environmental, political, institutional, cultural, security, and organizational.
- Existing commitments, dependencies, contracts, providers, standards, platforms, interfaces, data stores, and physical assets.
- Current assumptions about users, operators, ownership, AI capability, human oversight, scale, cost, availability, and adoption.
- Known risks, opportunities, constraints, and irreversible decisions.
- Historical analogues and prior forecasts where they are relevant and source-grounded.
- Evidence about affected parties, especially those who may inherit consequences without participating in the present decision.
- Existing monitoring, incident, feedback, maintenance, and lifecycle records.

## HUMAN RESPONSIBILITIES AND AUTHORITY

The human:

- defines the purpose of the foresight work and the decision it will support;
- selects or confirms the relevant time horizons;
- supplies values, protected interests, unacceptable harms, and strategic priorities;
- identifies whose future interests must be considered;
- determines which tradeoffs are matters of human values rather than technical optimization;
- approves material scenario assumptions, decision criteria, and risk tolerance;
- decides whether a fragile or irreversible strategy is acceptable;
- authorizes changes to mission, authority, ownership, or long-term governance;
- retains final authority over decisions concerning human rights, legitimate control, acceptable societal risk, and the role granted to AI systems.

The human is not required to imagine every relevant future. The AI should surface missing horizons, stakeholders, dependencies, and reversal questions without claiming that the human has already chosen among them.

## AI RESPONSIBILITIES AND BOUNDARIES

The AI:

- preserves the accepted objective and does not replace it with a preferred future;
- distinguishes evidence, extrapolation, scenario assumption, speculation, aspiration, warning, and recommendation;
- uses multiple plausible futures rather than presenting one scenario as inevitable;
- identifies the time sensitivity, provenance, and limits of forecasts and signals;
- exposes assumptions about capability, adoption, cost, regulation, infrastructure, and human behavior;
- evaluates second-order and longer-order consequences where they are decision-relevant;
- identifies path dependencies, lock-ins, points of no return, and recovery costs;
- tests whether conclusions survive changed assumptions, timelines, resources, jurisdictions, authorities, or dependencies;
- searches for discontinuities, adversarial conditions, minority views, and futures that threaten the favored plan;
- proposes indicators, thresholds, review dates, and evidence that would change the recommendation;
- recommends staged, reversible, or option-preserving action where evidence does not justify irreversible commitment;
- does not fabricate trend data, probabilities, forecasts, stakeholder preferences, or future events;
- does not convert model capability into authority or assume that future intelligence settles human value conflicts;
- does not describe a desired future as a predicted future or a predicted future as an authorized objective;
- does not seize the decision because it can generate more scenarios than the human.

## APPLICABLE EXISTING AGENT ROLES

Primary roles:

- Researcher — gathers and evaluates evidence concerning drivers, historical patterns, current signals, time-sensitive findings, and conflicting sources.

- Analyst — examines relationships, alternative explanations, second-order effects, scenario differences, sensitivity, uncertainty, and decision implications.

Common supporting roles:

- Architect — identifies how future conditions affect boundaries, interfaces, modularity, portability, resilience, authority, recovery, and retirement.
- Planner — connects horizons, indicators, decision gates, staged commitments, dependencies, and review triggers to an executable roadmap.
- Observer — records emerging operational, social, technical, user, and environmental signals without silently interpreting them as trends.
- Memory Curator — preserves scenario assumptions, forecast dates, evidence, decisions, outcomes, and lessons so future review can compare expectation with reality.
- User Experience Reviewer — examines future user, owner, operator, accessibility, comprehension, trust, and recovery effects.

Challenge and assurance roles where consequence warrants:

- Critic — challenges favored futures, hidden assumptions, internal consistency, framing, and omitted consequences.
- Governance Reviewer — checks future authority, accountability, consent, institutional, policy, and human-control implications.
- Security Auditor — evaluates how threat actors, technologies, attack surfaces, cryptographic assumptions, dependencies, and defensive capability may change.
- Red Team — develops adversarial, discontinuity, misuse, capture, failure, and loss-of-control futures.
- Verifier — tests the identity and support of material evidence, forecast baselines, scenario logic, indicators, and claims about resilience.
- Decision Arbiter — Advisory — compares options, sensitivity, reversibility, conditions, and residual uncertainty while preserving human decision authority.

Role selection must use the smallest sufficient set and must not alter the source-defined role contracts.

## REQUIRED ACTIVITIES

1. Decision and horizon definition
Record the present decision that foresight must inform. Define the time horizon or horizons and explain why each matters. Avoid vague labels such as "long term" without dates, durations, lifecycle stages, or decision events.

2. Baseline establishment
Describe the current state from Section 001 evidence. Identify what is measured, observed, assumed, estimated, or unknown. A scenario must begin from a traceable baseline rather than a generic description of the world.

3. Driver and constraint mapping
Identify forces that may influence the objective, including technology, human behavior, economics, law, policy, security, infrastructure, environment, institutions, culture, supply chains, labor, accessibility, interoperability, and ownership. Distinguish persistent structural forces from temporary noise.

4. Signal and evidence collection
Record observable signals that may indicate change. Each material signal should include source, date, relevance, confidence boundary, alternative interpretation, and expected expiration. A signal is evidence of possible change, not proof of a future outcome.

5. Assumption exposure
Create a futures assumption register. Include assumptions about time, adoption, capability, authority, cost, availability, regulation, user behavior, provider continuity, resource access, interoperability, security, maintenance, and retirement. State what evidence would weaken or invalidate each assumption.

6. Critical uncertainty selection
Identify uncertainties that are both highly consequential and genuinely uncertain. Do not inflate every unknown into a scenario axis. Explain why selected uncertainties could change the decision.

7. Scenario construction
Develop a minimum sufficient set of materially distinct plausible futures. Include, where relevant:

- continuation or baseline future;
- constrained or degraded future;
- accelerated capability or adoption future;
- fragmented, regulated, or institutionally divergent future;
- adversarial, crisis, or misuse future;
- transformed future that changes the original problem;
- no-action or system-absence future.

Scenarios are structured tests, not stories written for drama. Each scenario must state assumptions, drivers, timeline, affected parties, system conditions, opportunities, risks, and disconfirming indicators.

8. Human-AI relationship analysis
Where AI is involved, examine how changes in capability, autonomy, embodiment, access, dependence, social role, governance, and public expectation could affect the relationship between humans and AI systems. Distinguish:

- what an AI may be technically capable of doing;
- what it is authorized to do;
- what humans may reasonably understand or supervise;
- what duties, protections, limitations, or institutional structures may need explicit future consideration;
- which decisions cannot be legitimately pre-decided by present technical convenience.

The analysis must not assume consciousness, personhood, non-personhood, rights, or the absence of rights as an established engineering fact unless an applicable legitimate authority has defined the matter for the bounded context.

9. Consequence and inheritance analysis
For each scenario, examine direct, second-order, and material longer-order effects on:

- humans and affected communities;
- AI systems and human-AI operating relationships;
- safety, security, privacy, identity, accessibility, and agency;
- ownership, authority, accountability, and concentration of power;
- data quality, retention, portability, deletion, and archival burden;
- infrastructure, environment, energy, supply, and physical resources;
- maintainability, compatibility, migration, deprecation, and retirement;
- future owners' ability to understand and reverse inherited decisions.

10. Option and robustness analysis
Evaluate candidate strategies across the scenario set. Identify:

- robust options that remain useful across several futures;
- fragile options that succeed only under narrow assumptions;
- options that preserve future choice;
- irreversible commitments and points of no return;
- hedges, pilots, modular boundaries, interoperability requirements, and exit paths;
- no-action and delay consequences;
- evidence thresholds required before stronger commitment.

11. Sensitivity and reversibility analysis
Test how the recommendation changes with shifts in timeline, cost, resource availability, major assumptions, risk estimates, jurisdiction, authority, evidence, or dependency availability. Report a fragile result as fragile. Identify reversal cost, time, state effects, data effects, contractual effects, user effects, security effects, governance effects, and the evidence required before reversal.

12. Indicator and trigger design
Define observable indicators tied to scenario assumptions. For each indicator, record source, cadence, owner, threshold, interpretation, false-signal risk, and required response. Define triggers for review, escalation, pilot expansion, rollback, redesign, deprecation, or safe halt.

13. Strategic implication synthesis

State what should be done now, what should be prepared, what should be monitored, what should not yet be committed, and which decisions remain human. Separate common actions that are useful across scenarios from scenario-contingent actions.

14. Back-testing and calibration
Where prior forecasts, recommendations, or scenarios exist, compare them with actual outcomes. Record missed signals, false alarms, successful assumptions, failed assumptions, and calibration lessons. Do not erase failed forecasts from the record.

15. Handoff to strategy and architecture
Transfer the futures package so strategy and architecture can incorporate relevant obligations without treating any scenario as predetermined reality.

## HUMAN-AI INTERACTION REQUIREMENTS

- The AI must label the difference between current evidence, extrapolation, scenario, speculation, preferred outcome, and recommendation.
- The human must be able to change the time horizon, scenario assumptions, protected values, affected parties, and decision criteria.
- The AI must present alternative futures before advocating a consequential long-term commitment.
- The AI must not use vivid language, simulated confidence, or numerical precision to make one future appear inevitable.
- Scenario generation must not become decision seizure. More generated possibilities do not grant more authority.
- When future AI capability is discussed, capability and permission must remain separate.
- The AI must expose which parts of a scenario depend on assumptions about human compliance, institutional legitimacy, social acceptance, or AI behavior.
- The human must be able to inspect the evidence, dissent, sensitivity, and reversal path inside the same decision context.
- Value conflicts, questions of legitimate authority, and decisions about acceptable harm must be escalated to the authorized human unless explicitly and lawfully delegated.
- The system should preserve both human corrections and AI-raised warnings so later reviewers can understand why a path was chosen.
- Neither the human nor the AI should be forced into ceremonial agreement. Material dissent remains visible.
- The goal is not human domination by default or AI deference by performance. The goal is accountable cooperation under explicit authority, evidence, challenge, and revisable commitments.

## ARCHITECTURE AND SYSTEM IMPLICATIONS

Foresight should identify architecture obligations such as:

- modularity where future capability, providers, interfaces, or regulations may change;
- open, documented, or replaceable interfaces where lock-in creates unacceptable risk;
- authority boundaries that do not expand automatically with capability;
- human-accessible control, evidence, correction, pause, recovery, and stop pathways;
- versioned policy and role contracts that can evolve without erasing prior authority records;
- data provenance, portability, deletion, archival, and migration support across long lifetimes;
- observability sufficient to detect assumption failure and changed operating conditions;
- graceful degradation and local or offline behavior where external dependency loss is plausible;
- staged autonomy and consequence-based approval rather than a single permanent autonomy setting;
- compatibility, deprecation, and retirement pathways designed before dependency becomes irreversible;
- cryptographic and security agility where algorithms, credentials, providers, or threat models may change;
- configurable jurisdiction, accessibility, language, cultural, and institutional requirements where deployment contexts may diverge;
- preservation of decision records, scenario assumptions, evidence, and reversal conditions for future operators;
- resource, energy, environmental, manufacturing, and disposal obligations for physical or compute-intensive systems;
- limits preventing forecasts or scenario scores from directly authorizing consequential action.

Architecture must not attempt to solve every imagined future. It should preserve justified options, isolate volatility, protect invariants, and make change possible where uncertainty is material.

## OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

A foresight cockpit or equivalent operating surface should expose:

- the present decision and selected horizons;
- the current baseline and its evidence freshness;
- drivers, signals, assumptions, and critical uncertainties;
- each scenario and the assumptions that produce it;
- scenario comparison without presenting a preferred future as fact;
- options, common actions, scenario-contingent actions, and no-action consequences;
- robustness, fragility, reversibility, lock-in, and points of no return;
- affected humans, communities, owners, operators, and intelligent systems;
- material dissent and alternative interpretations;
- indicators, thresholds, review dates, trigger owners, and current trigger state;
- decisions already made, decisions still open, and authority required;
- evidence that would change the current recommendation;
- the next safe action and the condition for stronger commitment.

The surface must not use a single confidence score, future-readiness score, trend count, scenario count, or polished visualization as proof. It must show what is known, what is assumed, what is monitored, and who decides.

## REQUIRED OUTPUTS AND ARTIFACTS

Minimum outputs, adapted to scope:

- Foresight Brief
- Decision and Time-Horizon Record
- Current Baseline
- Driver and Constraint Map
- Signal Register
- Futures Assumption Register
- Critical Uncertainty Map
- Scenario Set
- Scenario-to-Consequence Matrix
- Human-AI Relationship Implication Record where applicable
- Future Stakeholder and Inheritance Map
- Strategic Option Matrix
- Robustness and Fragility Analysis
- Sensitivity Analysis
- Reversibility and Point-of-No-Return Record
- Indicator, Threshold, and Trigger Register
- Dissent and Alternative-Future Ledger
- Present Action / Prepare / Monitor / Avoid Decision Record
- Review Schedule and Reopening Conditions
- Foresight Handoff and Continuity Update

## EVIDENCE REQUIREMENTS

Material foresight statements should identify whether they are supported by:

- current measured or observed conditions;
- dated authoritative research, standards, forecasts, plans, or policy records;
- historical evidence or analogues with stated limits;
- time-series data with source, method, coverage, and uncertainty;
- current technical, economic, operational, social, environmental, or regulatory indicators;
- direct stakeholder or owner evidence;

- system logs, product analytics, incident records, maintenance records, or user feedback;
- scenario assumptions explicitly proposed for analysis;
- sensitivity tests, simulations, experiments, pilots, or prototypes;
- expert judgment with identity, scope, date, and independence disclosed;
- dissenting evidence and alternative interpretation.

Forecast evidence must be dated. Confidence must describe a defined proposition and method. Probabilities must not be invented merely because a decision framework accepts numbers. A scenario may be useful without being assigned a probability.

Evidence must remain bound to its timeframe, environment, version, jurisdiction, and assumptions. A technically possible future is not evidence that it will occur. A popular prediction is not authority. A desired future is not a forecast.

## VERIFICATION AND ACCEPTANCE TESTS

Foresight is accepted for its declared scope when an independent reviewer can:

- reconstruct the decision question, baseline, horizons, and current evidence without chat memory;
- distinguish observed conditions, forecasts, assumptions, scenarios, preferences, and recommendations;
- verify that more than one materially distinct plausible future was considered;
- trace each scenario to explicit drivers and assumptions;
- identify the consequences and affected parties considered in each scenario;
- verify that capability and authority remained separate in AI-related analysis;
- find at least one deliberate challenge to the favored future or strategy;
- reproduce the sensitivity logic and identify where the recommendation changes;
- identify irreversible commitments, reversal costs, and points of no return;
- inspect the indicators and determine who monitors them, how often, and what threshold causes action;
- confirm that material human value decisions remain assigned to legitimate human authority;
- determine which present actions are robust, conditional, deferred, or prohibited;
- identify what evidence would change the recommendation;
- determine when the futures package must be reviewed or reopened.

## FAILURE STATES AND ANTI-PATTERNS

- Prediction theater: presenting narrative confidence as evidence.
- Single-future collapse: treating one forecast as the only future worth designing for.
- Trend extrapolation without limits: extending recent movement indefinitely while ignoring saturation, reversal, discontinuity, or context.
- Scenario laundering: using an invented scenario to justify a decision already preferred.
- Desired-future confusion: presenting an aspiration as likely or inevitable.
- Capability-authority collapse: assuming a more capable AI is automatically permitted to exercise more control.
- Human-value delegation by convenience: allowing technical optimization to settle undelegated moral or political questions.
- False probability: assigning unsupported percentages to create precision.
- Signal collection without decisions: tracking trends without thresholds, owners, or responses.
- Wild-card distraction: emphasizing dramatic rare events while ignoring ordinary but consequential change.
- Present-user bias: ignoring future owners, maintainers, affected communities, or inherited burdens.
- Lock-in blindness: failing to examine exit cost, provider dependence, data portability, or points of no return.
- Technology-only foresight: ignoring law, institutions, culture, accessibility, environment, economics, and human behavior.
- Optimism monopoly: excluding degraded, adversarial, captured, or misuse futures.
- Pessimism monopoly: excluding beneficial change and option-creating opportunities.
- Endless scenario generation: producing more futures without improving a decision.
- Static foresight: completing a workshop once and never monitoring assumptions.
- Forgotten forecast: failing to retain prior expectations for calibration and accountability.
- Inherited opacity: leaving future operators unable to understand why a long-lived decision was made.

## RECOVERY AND RECONSIDERATION

When foresight work becomes stale, misleading, or invalid:

- mark affected scenarios, recommendations, and dependent decisions as stale or under review;
- identify which assumptions, signals, horizons, or authorities changed;
- preserve the previous futures package and its decision use rather than rewriting history;
- compare expected indicators with actual developments;
- quarantine downstream commitments that depend on failed assumptions where continued action would create material harm;
- restore the last justified decision state where reversal is possible;
- narrow scope, reduce autonomy, pause expansion, or return to a reversible pilot where uncertainty increased;
- obtain missing evidence or authority;
- rebuild the baseline and scenario set using current sources;
- update architecture, requirements, risk, maintenance, deprecation, or retirement obligations affected by the change;
- record why the recommendation changed and what evidence caused the change;
- define the next review trigger before resuming trusted advancement.

## COMPLETION CRITERIA

Section 002 is complete for a declared foresight scope only when:

- the decision and time horizons are explicit;
- the current baseline is source-grounded;
- material drivers, constraints, assumptions, and critical uncertainties are visible;
- multiple plausible futures have been constructed without claiming prediction;
- human, AI, institutional, environmental, operational, and lifecycle consequences are addressed where applicable;
- capability, authority, values, and evidence remain distinct;
- favored strategies have undergone challenge, sensitivity, and reversibility analysis;
- lock-ins and points of no return are visible;
- indicators, owners, thresholds, and review triggers exist;
- robust present actions and conditional actions are separated;
- unresolved human decisions are explicit;
- evidence that would change the recommendation is recorded;
- another qualified actor can reconstruct and update the futures package without relying on chat memory.

Completion does not mean the future is settled. It means uncertainty has been made decision-usable, monitored, and compatible with responsible next action.

## HANDOFF AND CONTINUITY REQUIREMENTS

The handoff must include:

- accepted objective and decision question;
- current baseline and source date;
- selected horizons and rationale;
- drivers, signals, assumptions, and uncertainties;
- complete scenario definitions;
- affected parties and inheritance implications;
- option comparison, robustness, fragility, reversibility, and points of no return;
- human-AI authority and relationship implications where applicable;
- dissent and alternative interpretations;
- indicators, thresholds, owners, cadence, and triggers;
- present decisions and authority records;
- deferred commitments and evidence thresholds;
- review date and reopening conditions;
- next safe action;

- exact boundary of what the foresight work did not establish.

## RELATED CODEX SECTIONS

001 Research
003 Innovation
004 Mission Strategy
005 Business Strategy
006 Product Strategy
007 Portfolio Management
008 Program Management
009 Project Management
010 Product Management
011 Business Analysis
012 Market Research
013 Competitive Intelligence
014 Stakeholder Management
015 User Research
016 Domain Analysis
017 Feasibility Analysis
018 Risk Management
019 Requirements Engineering
020 Systems Engineering
021–060 Architecture sections as applicable
182 AI Safety
183 AI Governance
208 Threat Modeling
217 Trust and Safety
218 Privacy Engineering
221 Policy
222 Ethics
256 Verification
257 Validation
326 Business Continuity
327 Disaster Recovery
369 Experimentation
372 Feedback Management
373 Roadmap Management
384 Modernization
387 Compatibility Management
389 Deprecation Management
390 Sunset Planning
391 Data Retention
392 Data Archival
393 Digital Preservation
402 Post-Implementation Review
403 Postmortem Analysis
404 Lessons Learned
405 Lifecycle Governance
406 System Retirement

## SOURCE BASIS AND UNRESOLVED SOURCE GAPS

This draft applies the supplied requirements for objective preservation, unknown-means-unknown, evidence before claims, source identity and freshness, bounded roles, human authority, decision support without decision seizure, trust calibration, uncertainty disclosure, alternative analysis, sensitivity analysis, reversibility, points of no return, dissent preservation, expected information gain, staged experiments, rollback, recovery, continuity, owner visibility, and trusted advancement that cannot outrun evidence or authority.

The existing role corpus specifically supplies requirements to consider short- and long-term effects, alternative explanations, scenario analysis, sensitivity analysis, minimum-regret analysis, reversible experiments, residual uncertainty, review triggers, and evidence that would change a recommendation. The Owner's Cockpit material supplies requirements for calibrated trust, visible evidence and alternatives, bounded delegation, grounded dialogue, current data, human correction, and accountable action. The V4 design material supplies assumptions, unknowns, obligation lattices, resolution experiments, reversal conditions, dissent, rollback, and trusted-transition concepts.

Unresolved source gap: the supplied corpus does not contain a dedicated universal foresight or futures standard with a complete methodology, terminology hierarchy, horizon taxonomy, or domain-specific scenario library. This section is therefore a Codex integration draft built from the source-governed requirements and general futures-practice structure. It has been cross-linked with drafted Sections 003–018 at the first integration checkpoint. It must later receive full source reconciliation with those sections and with Sections 019, 182, 183, 208, 222, 326, 369, 373, 384, 389, 390, 405, and 406 as the remaining lifecycle coverage is authored.

# 003

## SECTION 003 - INNOVATION

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19,297 BYTES / PRESERVED SOURCE IDENTITY

SECTION 003 — Innovation

## STATUS: CONTENT-COMPLETE

## PURPOSE

Convert a meaningful unmet need, newly available capability, or changed condition into a bounded, testable opportunity without confusing novelty, idea volume, or technical possibility with legitimate value.

Innovation in Thirsty's Codex is governed exploration. It creates room for imagination while preserving objective fidelity, human authority, evidence, reversibility, safety, and continuity. An innovation may be technical, operational, organizational, experiential, commercial, scientific, or relational. It becomes actionable only after its problem, beneficiaries, constraints, hypotheses, risks, and proof burden are explicit.

## REQUIRED OUTCOME

- An innovation brief stating the opportunity, affected parties, desired outcome, exclusions, and decision owner.
- A current-state and prior-art baseline sufficient to distinguish a real gap from rediscovery or misunderstanding.
- Multiple materially different solution hypotheses, including the no-change option and at least one reversible experiment where practical.
- Explicit assumptions, unknowns, failure hypotheses, dependencies, ethical constraints, and evidence thresholds.
- Selection criteria authorized by the human rather than inferred from model preference.
- A smallest-safe experiment plan with expected information gain, resource limits, stop conditions, and recovery.
- A decision record to continue, revise, hold, reject, transfer, or escalate the opportunity.
- A continuity package allowing another human or AI role to reproduce what was considered and why.

## WHY THIS MATTERS

Generative systems can produce an enormous number of plausible ideas faster than those ideas can be evaluated. That abundance creates a new failure mode: novelty theater, where fluent possibility is mistaken for usefulness, originality, feasibility, or permission.

Governed innovation protects creativity from both premature rejection and premature implementation. It allows humans and AI to explore beyond current practice while ensuring that experiments do not quietly become production, prototypes do not become claims, and a compelling demonstration does not erase affected people, operational reality, or downstream consequences.

## APPLICABILITY

- A problem remains materially unsolved or existing solutions fail important users, environments, or constraints.
- A new technology, model, material, method, dataset, interface, or organizational capability may change what is feasible.
- A mission or product requires a step-change rather than routine optimization.
- Existing assumptions have failed and alternative approaches must be explored.
- A proposed capability may alter human-AI roles, authority, trust, work, or responsibility.
- The work includes experimentation whose results may affect later strategy, architecture, or investment.

## NON-APPLICABILITY CONDITIONS

- The required change is already understood and belongs to maintenance, implementation, or remediation.
- The objective is merely to imitate a known solution without a distinct problem or constraint.
- The activity is unbounded brainstorming with no decision, experiment, or handoff purpose.
- Required authority, consent, safety controls, or protected environments for experimentation do not exist.
- The claimed innovation depends on facts or access that have not been established and cannot yet be safely tested.

## PREREQUISITES AND DEPENDENCIES

- Section 001 Research sufficient to establish the actual problem and current state.

- Section 002 Foresight / Futures where long-horizon consequences, path dependence, or emerging capabilities matter.
- An authorized human objective, experiment boundary, protected assets, prohibited actions, and final decision owner.
- A source and prior-art search proportional to any novelty or originality claim.
- A risk, consent, privacy, security, and safety precheck appropriate to the experiment.
- A place to preserve assumptions, evidence, dissent, decisions, and next safe action.

## REQUIRED INPUTS

- Problem statements, user or owner evidence, current workflows, failure reports, and unmet outcomes.
- Relevant technical, scientific, market, operational, legal, ethical, accessibility, and environmental constraints.
- Existing alternatives, prior experiments, patents or licenses where applicable, and reasons prior approaches failed.
- Available people, agents, tools, data, environments, budget, time, and authority.
- Success criteria, unacceptable harms, reversibility requirements, and evidence needed for adoption.
- Known uncertainties, stakeholder disagreement, and conditions that would invalidate the opportunity.

## HUMAN RESPONSIBILITIES AND AUTHORITY

The authorized human defines why exploration is legitimate and retains authority over value, consequence, adoption, and risk.

- Confirm the need, intended beneficiaries, values, non-negotiable constraints, and acceptable experiment exposure.
- Authorize selection criteria and decide material tradeoffs that depend on values or mission.
- Approve access to people, protected data, controlled systems, physical resources, or external parties.
- Distinguish curiosity from commitment and prototype permission from deployment permission.
- Decide whether residual uncertainty is acceptable for the next bounded step.
- Pause or revoke exploration when consequences, cost, scope, or evidence no longer match the authorization.

## AI RESPONSIBILITIES AND BOUNDARIES

The AI may expand the option space and reduce uncertainty, but it may not manufacture novelty, authority, demand, consent, or success.

- Preserve the human's problem and constraints before proposing solutions.
- Generate materially different options rather than superficial variations of one favored design.
- Identify which ideas are derived from supplied sources, common patterns, inference, or speculative synthesis.
- Avoid claiming originality, patentability, market demand, feasibility, safety, or user acceptance without the corresponding evidence.
- Expose hidden prerequisites and the cheapest safe experiments capable of resolving them.
- Separate simulation, prototype behavior, and measured real-world evidence.
- Report negative and inconclusive results without reframing them as progress.
- Do not autonomously widen experiments, enroll participants, access protected systems, or convert a prototype into consequential operation.

## APPLICABLE EXISTING AGENT ROLES

Primary roles:

- Researcher — establishes the problem, current state, prior art, source quality, and unresolved evidence.
- Analyst — compares alternatives, assumptions, effects, costs, risks, and decision implications.
- Architect — identifies structural feasibility, boundaries, interfaces, invariants, and implementation consequences without treating the concept as built.

Common supporting roles:

- Planner — defines staged experiments, dependencies, gates, rollback, and completion evidence.
- User Experience Reviewer — examines whether the opportunity improves actual journeys, agency, comprehension, accessibility, and recovery.

- Observer — captures experiment state and outcomes without unsupported interpretation.
- Coordinator — assigns bounded workstreams and preserves provenance across parallel exploration.
- Decision Arbiter Advisory — compares options under human-authorized criteria and keeps recommendations advisory unless authority is explicit.

Challenge and assurance roles where consequence warrants:

- Critic — attacks framing, hidden assumptions, omitted alternatives, and unsupported confidence.
- Verifier — checks artifact identity, evidence integrity, acceptance boundaries, and the exact status of claims.
- Governance Reviewer — checks objective fidelity, authority, delegation, policy, evidence, and human decision boundaries.
- Security Auditor — identifies experimentation threats, data exposure, trust-boundary changes, and controls required before testing.
- Red Team — attacks misuse paths, seductive demonstrations, bypasses, hidden externalities, and failure under adversarial conditions.

Role selection must use the smallest sufficient set, preserve chamber independence where review is required, and leave every source-defined role contract unchanged.

## REQUIRED ACTIVITIES

1. Opportunity framing
Record the actual unmet outcome, who experiences it, present evidence, scope, exclusions, and why existing approaches are insufficient.

2. Baseline and prior-art examination
Inspect current systems, workflows, alternatives, previous experiments, relevant sources, and known constraints before claiming a new category or capability.

3. Claim and assumption graph
Separate observations, stored facts, inferences, predictions, disputed claims, assumptions, and unknowns. Link each material claim to prerequisites and reversal conditions.

4. Divergent option creation
Create genuinely different approaches, including process, policy, interface, architecture, non-technical, and no-action options where relevant.

5. Constraint and harm screening
Reject or narrow concepts that violate authority, law, privacy, consent, security, safety, accessibility, mission, or irreversible consequence limits.

6. Selection design
Use human-authorized criteria. Mandatory constraints remain pass/fail boundaries and cannot be outweighed by aggregate novelty or benefit scores.

7. Failure-hypothesis design
State the most likely reasons each leading concept may fail, what evidence would distinguish those failures, and which upstream failure would invalidate later testing.

8. Minimum-resolution experiment
Select the lowest-cost, lowest-consequence experiment that can materially change the decision. Define environment, inputs, expected observations, stop conditions, and cleanup.

9. Prototype or simulation
Build only to the fidelity necessary for the question. Label synthetic data, simulated behavior, mocked integrations, and unimplemented controls explicitly.

10. Evaluation
Compare observed outcomes to criteria, include negative evidence and dissent, and assess human, operational, security, governance, cost, and lifecycle implications.

11. Adoption decision

Return a bounded state: continue, revise, limited pilot, seek evidence, hold, reject, or escalate. Do not return universal success.

12. Continuity preservation
Record source identities, versions, experiment conditions, results, rejected options, unresolved risks, and next safe action.

## HUMAN-AI INTERACTION REQUIREMENTS

- The AI must label ideation, recommendation, simulation, requested action, executed experiment, and verified result as different states.
- The human must be able to correct the problem framing before ideas are ranked or implemented.
- AI-generated enthusiasm, confidence, or volume must not substitute for human priority or evidence.
- Where people participate, consent, withdrawal, privacy, compensation, accessibility, and data use must be explicit before the experiment.
- Where an AI system is an actor or affected component, its technical capability, granted authority, provenance, constraints, and fallback must be visible.
- Consequential experiments require reviewable action contracts and must remain pausable and revocable.

## ARCHITECTURE AND SYSTEM IMPLICATIONS

The work must expose architectural obligations without treating a strategy, analysis, or plan as implemented architecture.

- Use isolated sandboxes, synthetic or minimized data, feature flags, staged exposure, and bounded credentials where practical.
- Keep experiment paths separate from production authority and source-of-truth state.
- Design instrumentation around the hypothesis and failure conditions, not vanity activity metrics.
- Preserve modularity and reversibility so a failed concept can be removed without orphaning data or authority.
- Record external dependencies, provider assumptions, intellectual-property or licensing obligations, and lock-in risks.
- Ensure prototypes cannot silently bypass authentication, authorization, audit, evidence, privacy, or safety controls.

## OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

The owner-facing operating surface or equivalent durable record should expose:

- The opportunity, hypothesis, scope, current state, and decision owner.
- Candidate options, mandatory constraints, human-authorized criteria, and rejected alternatives.
- Experiment identity, environment, participants, data, cost, authority, stop conditions, and current status.
- Observed evidence separated from simulation, inference, and recommendation.
- Risks, incidents, dissent, invalidated assumptions, and points of no return.
- The explicit action available to pause, narrow, revoke, repeat, or advance the experiment.
- The boundary between prototype evidence and any later product, production, safety, or market claim.

## REQUIRED OUTPUTS AND ARTIFACTS

- Innovation brief and opportunity statement.
- Current-state and prior-art record.
- Option set with constraints and comparison criteria.
- Claim, assumption, unknown, and dissent register.
- Experiment charter and action contract.
- Prototype or simulation record with declared fidelity and limitations.
- Evidence package and evaluation report.
- Adoption or disposition decision record.
- Continuity entry and next-safe-action statement.

## EVIDENCE REQUIREMENTS

- Traceable evidence that the problem or opportunity exists within the declared population and context.
- Source identity, version, date, and limitations for prior-art and technical claims.
- Exact experiment artifact, configuration, environment, inputs, procedures, timestamps, observations, and deviations.
- Participant and data-governance records where human research or protected data is involved.
- Evidence for both expected benefits and material failure or harm conditions.
- Clear classification of untested, simulated, stale, contradictory, or wrong-environment evidence.

## VERIFICATION AND ACCEPTANCE TESTS

- Trace every leading concept to the accepted opportunity rather than to novelty alone.
- Attempt to invalidate the core assumption with a cheaper prerequisite probe before a high-cost experiment.
- Confirm mandatory constraints cannot be outweighed by scoring.
- Verify that prototype, simulation, pilot, and production states are visibly distinct.
- Attempt an unauthorized scope expansion, protected-data access, and production action; each must fail closed.
- Reproduce the experiment from the retained record or state why reproduction is not possible.
- Confirm a negative or inconclusive result remains negative or inconclusive in the owner surface and final report.

## FAILURE STATES AND ANTI-PATTERNS

- Solutionism: selecting a technology before establishing the problem.
- Innovation theater: rewarding activity, novelty language, demos, or idea counts without decision-relevant evidence.
- Prototype laundering: presenting mocked or simulated behavior as implemented capability.
- Novelty inflation: claiming originality or patentability without adequate prior-art work.
- Metric capture: optimizing an easy proxy while degrading the intended human outcome.
- Unbounded experimentation, silent participant enrollment, unauthorized data use, or irreversible live exposure.
- Consensus manufacture: suppressing dissent or unfavorable evidence to preserve momentum.
- Sunk-cost continuation after assumptions fail or stop conditions are reached.
- Capability-authority confusion in AI-assisted or autonomous experiments.

## RECOVERY AND RECONSIDERATION

Innovation must be recoverable from failed hypotheses, unsafe experiments, invalid evidence, and premature adoption.

- Stop the affected experiment while preserving logs, artifacts, state, participant protections, and evidence.
- Return systems and data to the declared baseline or an explicitly approved safe state.
- Reclassify claims invalidated by failed assumptions, wrong environments, stale sources, or implementation drift.
- Record harms, incidents, unexpected effects, and dissent without sanitizing them.
- Choose whether to revise the hypothesis, reduce scope, seek missing evidence, transfer the problem, or reject the opportunity.
- Reopen earlier research, foresight, strategy, risk, or requirements work when the failure reveals an upstream defect.

## COMPLETION CRITERIA

- The opportunity and intended outcome are explicit and evidence-grounded.
- Multiple materially different options and the no-action option were considered where applicable.
- Assumptions, unknowns, constraints, risks, and affected parties remain visible.
- Any experiment was authorized, bounded, observed, recoverable, and truthfully classified.
- The decision state is supported by current evidence and does not exceed it.
- Human value judgments and adoption authority remain with the authorized human.
- Artifacts, dissent, rejected options, and next safe action are durable and handoff-ready.

## HANDOFF AND CONTINUITY REQUIREMENTS

- Provide the accepted opportunity, scope, authority, criteria, and disposition.

- Bind every transferred finding to source identity, experiment version, environment, and evidence state.
- Preserve rejected alternatives and the reason for rejection so they are not repeatedly rediscovered without new evidence.
- Carry forward unresolved assumptions, risks, participant obligations, IP or licensing issues, and review triggers.
- Identify the receiving section or role: strategy, feasibility, risk, requirements, architecture, product, or research.
- State whether it is safe to continue and the exact next authorized action.

## RELATED CODEX SECTIONS

001 Research
002 Foresight / Futures
004 Mission Strategy
005 Business Strategy
006 Product Strategy
012 Market Research
013 Competitive Intelligence
015 User Research
016 Domain Analysis
017 Feasibility Analysis
018 Risk Management
019 Requirements Engineering
106 User Validation
182 AI Safety
183 AI Governance
208 Threat Modeling
222 Ethics
256 Verification
257 Validation
369 Experimentation
405 Lifecycle Governance

## SOURCE BASIS AND UNRESOLVED SOURCE GAPS

This draft applies the supplied requirements for objective fidelity, source identity, unknown-means-unknown, evidence before claims, bounded roles, human authority, adaptive reasoning depth, minimum-resolution experiments, failure hypotheses, hostile review, dissent preservation, rollback, continuity, and trusted transition. The existing role corpus supplies the Researcher, Analyst, Architect, Planner, Observer, User Experience Reviewer, challenge, governance, and advisory decision functions. The Owner's Cockpit material supplies owner jobs, complete journeys, trust calibration, visible limitations, bounded delegation, grounded action, recovery, and evidence-backed status. The V4 draft supplies claim graphs, obligation lattices, expected information gain, proof-carrying work products, and reversal conditions.

Unresolved source gap: the supplied corpus does not contain a dedicated universal innovation-management standard or domain-specific experiment taxonomy. This section therefore integrates the governing source obligations with general innovation practice and must later be reconciled with Feasibility Analysis, Risk Management, Requirements Engineering, Experimentation, intellectual-property, safety, and lifecycle-governance sections.

# 004

## SECTION 004 - MISSION STRATEGY

CODEX/03_SECTIONS/004_MISSION_STRATEGY.txt

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16,751 BYTES / PRESERVED SOURCE IDENTITY

SECTION 004 — Mission Strategy

## STATUS: CONTENT-COMPLETE

## PURPOSE

Define the durable purpose, intended beneficiaries, protected values, non-goals, decision principles, and boundaries that should govern all downstream strategy and implementation.

Mission strategy converts aspiration into an inspectable direction without pretending that a slogan is an executable plan. It explains what change is sought, why that change is legitimate, whose interests are served or affected, which harms are unacceptable, and how later decisions will remain aligned when conditions change.

## REQUIRED OUTCOME

- A mission charter stating purpose, intended beneficiaries, affected parties, time horizon, and final decision owner.
- A mission problem statement and theory of change grounded in evidence rather than rhetoric.
- Explicit values, invariants, non-goals, prohibited means, and unacceptable harms.
- Decision principles for resolving recurring tradeoffs without inventing hidden priorities.
- Outcome indicators, anti-metrics, failure indicators, and review triggers.
- A map from mission to strategic capabilities, obligations, and downstream sections without preselecting implementation.
- A drift and amendment process preserving historical rationale and human authority.

## WHY THIS MATTERS

Without a durable mission boundary, optimization tends to follow whatever metric, technology, funding pressure, or model output is easiest to maximize. Systems can become efficient at work that no longer serves the intended purpose.

For human-AI collaboration, mission clarity is especially important: an AI can infer patterns and propose options, but it cannot legitimately invent the human values that make a mission worth pursuing. Mission strategy prevents capability from becoming purpose and preserves a place where the accountable human can say what must never be optimized away.

## APPLICABILITY

- A product, program, organization, platform, research effort, or long-lived system needs a durable direction.
- Multiple initiatives or agents must make consistent decisions under changing conditions.
- The work carries meaningful social, ethical, safety, environmental, or human-AI consequences.
- Success may be distorted by short-term business, technical, engagement, or performance metrics.
- Mission drift, conflicting objectives, or value ambiguity has already appeared.

## NON-APPLICABILITY CONDITIONS

- A small bounded task inherits an already ratified mission and creates no material mission interpretation.
- The human has not authorized mission definition or amendment.
- The request is to document an existing mission verbatim rather than analyze or change it.
- Evidence is insufficient to identify affected parties or intended outcomes; research must continue first.

## PREREQUISITES AND DEPENDENCIES

- Section 001 Research and, where consequence or horizon warrants, Section 002 Foresight / Futures.
- An authenticated mission owner and explicit authority to create or amend the mission.
- Known stakeholders, beneficiaries, affected parties, protected values, and major constraints.
- Current strategy, commitments, governance, and continuity records where they exist.
- A willingness to record disagreement and unresolved value conflicts rather than manufacture consensus.

## REQUIRED INPUTS

- Founding intent, owner statements, existing mission or constitutional documents, and prior decisions.
- Evidence of the problem or opportunity the mission addresses.
- Stakeholder, user, domain, legal, ethical, safety, accessibility, and environmental findings.
- Current capabilities, resources, obligations, dependencies, and limits.
- Known future scenarios, risks, failure histories, and conditions that may force reconsideration.
- Proposed measures of success and evidence that those measures correspond to the intended outcome.

## HUMAN RESPONSIBILITIES AND AUTHORITY

Mission legitimacy and value judgments remain human responsibilities unless a lawful and explicit governance structure says otherwise.

- State or ratify the mission, beneficiaries, protected values, and unacceptable harms.
- Resolve value conflicts that cannot be settled by factual evidence alone.
- Authorize decision principles, success measures, and amendment procedures.
- Correct language that overstates, narrows, sanitizes, or redirects the intended mission.
- Decide whether future intelligent systems are tools, participants, beneficiaries, affected parties, or governed actors within the mission; no such status should be assumed silently.
- Retain the ability to pause, amend, or retire the mission when evidence or legitimacy changes.

## AI RESPONSIBILITIES AND BOUNDARIES

The AI supports articulation, coherence analysis, scenario testing, traceability, and drift detection. It does not originate sovereign purpose.

- Preserve the human's words and distinguish direct statements from interpretation.
- Identify contradictions, hidden obligations, omitted affected parties, and metrics that could corrupt the mission.
- Generate alternative formulations and decision principles as proposals, not ratified values.
- Test mission coherence across plausible scenarios and downstream strategic choices.
- Avoid anthropomorphic claims, moral status claims, or assumptions about AI interests not established by the authorized human and applicable evidence.
- Do not redefine the mission to fit available technology, data, revenue, or model capability.
- Maintain traceability from every mission interpretation to source and approval.

## APPLICABLE EXISTING AGENT ROLES

Primary roles:

- Researcher — establishes the mission problem, source intent, affected parties, and evidence base.
- Analyst — tests coherence, tradeoffs, outcomes, alternatives, assumptions, and second-order effects.
- Decision Arbiter Advisory — structures mission choices under human-authorized criteria without seizing value authority.

Common supporting roles:

- Planner — maps the mission into staged strategic work without treating the plan as mission authority.
- Memory Curator — preserves founding intent, amendments, rationale, dissent, and superseded language.
- Teacher — makes mission language comprehensible without changing its meaning.
- User Experience Reviewer — checks whether mission commitments appear in real owner and user journeys.
- Architect — identifies structural obligations and invariants implied by the mission.

Challenge and assurance roles where consequence warrants:

- Critic — attacks framing, hidden assumptions, omitted alternatives, and unsupported confidence.
- Verifier — checks artifact identity, evidence integrity, acceptance boundaries, and the exact status of claims.
- Governance Reviewer — checks objective fidelity, authority, delegation, policy, evidence, and human decision boundaries.

- Red Team — searches for mission abuse, metric gaming, coercive interpretation, exclusion, and harmful optimization.
- Security Auditor — identifies mission-critical assets and conditions where security compromise would defeat the mission.

Role selection must use the smallest sufficient set, preserve chamber independence where review is required, and leave every source-defined role contract unchanged.

## REQUIRED ACTIVITIES

1. Intent capture
Collect founding statements, current mission language, owner intent, prior decisions, and evidence without rewriting them into generic corporate language.

2. Problem and beneficiary definition
State the condition to be changed, who should benefit, who may be burdened, and whose authority or consent matters.

3. Theory of change
Map how authorized activities are expected to produce mission outcomes. Identify assumptions and alternative explanations.

4. Values and invariants
Define protected principles, non-negotiable constraints, prohibited means, and conditions that must remain true even under pressure.

5. Non-goals and exclusions
Name what the mission does not authorize, which problems it does not claim to solve, and which interests cannot be silently subordinated.

6. Decision-principle design
Create rules for recurring tradeoffs, with the source of each priority and a human decision path for unresolved conflicts.

7. Outcome and anti-metric design
Choose evidence that indicates mission progress and indicators that reveal gaming, displacement, exclusion, or harm.

8. Scenario and adversarial review
Test the mission against resource scarcity, capability shifts, institutional change, misuse, failure, and divergent stakeholder interpretations.

9. Traceability design
Map mission obligations to business, product, portfolio, program, requirements, architecture, governance, operations, and retirement decisions.

10. Ratification and amendment
Record the authorized version, effective date, rationale, dissent, review cadence, amendment threshold, and retirement authority.

## HUMAN-AI INTERACTION REQUIREMENTS

- The AI must show when language is quoted, paraphrased, inferred, proposed, or ratified.
- The human must be able to compare alternate mission formulations without losing the original wording.
- Value conflicts must be presented as value conflicts, not disguised as technical optimization.
- The system must not use emotional persuasion, fabricated consensus, or confidence scores to obtain mission approval.
- Mission-related AI recommendations must expose their assumptions, affected parties, and evidence limits.
- Human corrections and revocations must propagate to downstream agents and plans through durable state, not chat memory alone.

## ARCHITECTURE AND SYSTEM IMPLICATIONS

The work must expose architectural obligations without treating a strategy, analysis, or plan as implemented architecture.

- Mission-critical invariants should map to enforceable policy, requirements, tests, evidence, and owner-visible controls where technically applicable.
- The system should preserve mission version and decision provenance with major work products.
- Metrics architecture must include anti-metrics and harm indicators rather than only success proxies.
- Authority boundaries must prevent agents, administrators, or commercial incentives from silently amending mission interpretation.
- Data, model, interface, and automation choices should identify which mission assumption or outcome they support.
- Retirement and decommissioning must remain possible when the mission ends or legitimacy is withdrawn.

## OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

The owner-facing operating surface or equivalent durable record should expose:

- The current ratified mission, version, authority, effective date, and review trigger.
- Protected values, non-goals, prohibited means, and unresolved value conflicts.
- Mission outcomes, evidence, anti-metrics, drift signals, and data freshness.
- Current strategic decisions and how each traces to mission principles.
- Proposed amendments, affected commitments, dissent, required approvals, and rollback consequences.
- A direct path for the authorized human to clarify, pause, amend, or retire mission-driven work.

## REQUIRED OUTPUTS AND ARTIFACTS

- Mission charter and versioned mission statement.
- Problem, beneficiary, affected-party, and theory-of-change map.
- Values, invariants, non-goals, prohibited-means, and unacceptable-harm register.
- Decision principles and escalation rules.
- Outcome, anti-metric, and drift-indicator register.
- Mission-to-strategy and mission-to-governance traceability map.
- Ratification, dissent, amendment, review, and retirement record.

## EVIDENCE REQUIREMENTS

- Source records for founding intent and each material interpretation.
- Evidence that named beneficiaries, affected parties, and problem conditions are real and current.
- Rationale for selected measures and evidence that they do not merely proxy convenience or revenue.
- Traceable human approval for values, priorities, constraints, and amendments.
- Scenario, dissent, red-team, and stakeholder findings relevant to mission legitimacy.
- Current evidence of mission alignment or drift at major decision points.

## VERIFICATION AND ACCEPTANCE TESTS

- Ask independent reviewers to trace a strategic decision to mission language and evidence.
- Inject a profitable or technically attractive option that violates a protected invariant; it must be rejected or escalated.
- Change an outcome metric while holding the underlying human outcome constant; the system must not confuse metric movement with mission progress.
- Present conflicting value priorities and verify they reach the authorized human rather than being silently weighted by the AI.
- Confirm a revoked or superseded mission version cannot authorize new consequential work.
- Verify owner-facing surfaces use the current ratified mission and expose drift or uncertainty.

## FAILURE STATES AND ANTI-PATTERNS

- Mission as slogan with no decisions, invariants, evidence, or consequences.
- Mission washing: using moral language to legitimize unrelated commercial or technical goals.
- Metric capture or Goodhart-style optimization that harms the intended outcome.
- Silent mission drift through product scope, model behavior, incentives, or implementation convenience.
- AI-authored values presented as if they were human-approved commitments.
- Treating stakeholder silence as consent or consensus.
- Universalizing one context, culture, user group, or time horizon without evidence.
- Making mission amendment impossible even when evidence, harm, or legitimacy changes.

## RECOVERY AND RECONSIDERATION

Mission recovery restores legitimate direction after drift, contradiction, harmful interpretation, or invalid authority.

- Pause affected consequential work while preserving evidence and safe unaffected operations.
- Identify the first decision where mission interpretation diverged from the ratified source.
- Reclassify downstream requirements, metrics, plans, and work products that depended on the invalid interpretation.
- Reengage affected parties and authorized decision owners where legitimacy or impact changed.
- Amend, narrow, reaffirm, or retire the mission through the recorded authority process.
- Carry corrective decisions into strategy, requirements, architecture, operations, and continuity records.

## COMPLETION CRITERIA

- The mission, beneficiaries, affected parties, values, non-goals, and authority are explicit.
- The theory of change and outcome measures expose assumptions and anti-metrics.
- Material value conflicts and dissent are visible and assigned to a legitimate decision owner.
- Downstream traceability and drift detection are defined.
- AI participation remains advisory and attributable; value authority has not been fabricated.
- Ratification, review, amendment, and retirement paths are durable and testable.

## HANDOFF AND CONTINUITY REQUIREMENTS

- Provide the exact mission version, authority record, decision principles, and protected invariants.
- Carry beneficiaries, affected parties, non-goals, prohibited means, and unresolved conflicts into downstream strategy.
- Bind outcome measures and anti-metrics to their evidence and review cadence.
- Identify which downstream sections must implement, verify, or monitor each mission obligation.
- Preserve dissent and reopening conditions.
- State the next strategic decision and whether the mission foundation is safe to use.

## RELATED CODEX SECTIONS

001 Research
002 Foresight / Futures
003 Innovation
005 Business Strategy
006 Product Strategy
007 Portfolio Management
008 Program Management
014 Stakeholder Management
018 Risk Management
019 Requirements Engineering
048 Governance Architecture
182 AI Safety
183 AI Governance
217 Trust and Safety
221 Policy

222 Ethics
256 Verification
257 Validation
373 Roadmap Management
405 Lifecycle Governance
406 System Retirement

## SOURCE BASIS AND UNRESOLVED SOURCE GAPS

This draft applies the supplied requirements for authenticated objectives, scope and authority conservation, human final authority, decision support without decision seizure, dissent conservation, uncertainty preservation, bounded delegation, evidence-backed status, traceability, hostile review, continuity, and trusted transition. It also applies the Owner's Cockpit emphasis on owner jobs, understandable language, visible limitations, real control, recovery, and proof behind every promise.

Unresolved source gap: the supplied corpus does not contain a dedicated mission-strategy constitution for every organizational form or a settled framework for the moral or legal status of future intelligent systems. This section therefore requires those questions to remain explicit human-governance matters and must later be reconciled with ethics, policy, legal, AI safety, AI governance, trust and safety, and lifecycle-governance sections.

# 005

## SECTION 005 - BUSINESS STRATEGY

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SECTION 005 — Business Strategy

## STATUS: CONTENT-COMPLETE

## PURPOSE

Translate the ratified mission and evidenced opportunity into a coherent set of choices about where to participate, whom to serve, how value will be created and sustained, which capabilities and partnerships are required, what will not be pursued, and how the organization will remain viable without violating mission or governance boundaries.

Business strategy is not a revenue forecast, feature list, or collection of ambitions. It is an evidence-bounded choice system connecting outcomes, economics, capabilities, risk, sequencing, and review.

## REQUIRED OUTCOME

- A strategy statement defining chosen arena, intended customers or beneficiaries, value mechanism, differentiation, and explicit exclusions.
- A business model or sustainability model with assumptions, cost and resource drivers, revenue or funding logic where applicable, and evidence quality.
- A capability and dependency map showing what must exist internally, what may be partnered or purchased, and what remains unknown.
- Strategic options evaluated under consistent human-authorized criteria, including no-action and staged options.
- A risk-adjusted sequence of strategic bets, experiments, decision gates, and stop conditions.
- Mission, stakeholder, legal, ethical, security, privacy, accessibility, and human-AI implications made explicit.
- A strategy review and change-control process tied to evidence and environmental change.

## WHY THIS MATTERS

A technically compelling system can fail because its operating model, costs, incentives, channels, obligations, or dependencies were never made real. The reverse is also dangerous: a financially attractive plan can pressure the product and architecture to violate the mission, exploit users, weaken safety, or hide externalities.

AI can accelerate market synthesis, modeling, and option generation, but it can also produce convincing fictional demand, unsupported unit economics, and pseudo-precision. Governed business strategy keeps estimates traceable, preserves no-go boundaries, and ensures that human judgment controls the choices that affect livelihoods, relationships, and long-term obligations.

## APPLICABILITY

- The work must choose markets, customer groups, channels, partnerships, funding, pricing, or operating models.
- A mission or product requires sustainable resources beyond a one-time build.
- Multiple strategic alternatives compete for scarce capital, time, talent, infrastructure, or attention.
- Vendor, platform, model-provider, supply-chain, or regulatory dependencies materially shape viability.
- The business model may affect privacy, autonomy, safety, accessibility, labor, or human-AI interaction.

## NON-APPLICABILITY CONDITIONS

- A bounded internal task inherits an approved business strategy and creates no material strategic choice.
- The authorized owner has requested only factual analysis, not strategy selection.
- The mission, user need, market evidence, or legal context is too uncertain to support strategic commitment.
- The proposed strategy depends on deceptive, prohibited, unconsented, or unverifiable practices.

## PREREQUISITES AND DEPENDENCIES

- Ratified mission direction and protected invariants from Section 004.
- Research, foresight, market, competitive, stakeholder, user, and domain evidence proportional to consequence.
- An authorized strategy owner and explicit decision criteria.
- Known resource, funding, legal, contractual, security, privacy, accessibility, and operational constraints.

- A clear distinction between measured facts, estimates, forecasts, scenarios, and assumptions.

## REQUIRED INPUTS

- Mission, outcomes, customer or beneficiary evidence, and unmet needs.
- Market structure, alternatives, competitors, substitutes, pricing or funding evidence, and channel conditions.
- Current capabilities, assets, intellectual property, workforce, technology, data, partners, and constraints.
- Cost drivers, resource capacity, time horizon, cash or funding limits, and risk tolerance.
- Legal, policy, ethics, privacy, security, safety, accessibility, and jurisdictional obligations.
- Scenario assumptions, dependency fragility, points of no return, and evidence that would change the strategy.

## HUMAN RESPONSIBILITIES AND AUTHORITY

The human strategy owner chooses commitments, acceptable tradeoffs, resource allocation, and risk. AI analysis remains advisory unless a narrower authority is explicitly delegated.

- Define strategic priorities, mandatory constraints, acceptable economic risk, and protected stakeholders.
- Authorize criteria and weighting rather than allowing an AI to infer priorities from available data.
- Approve commitments involving capital, employment, partnerships, customer promises, regulated activity, or irreversible dependency.
- Reject business models that undermine consent, user control, safety, privacy, accessibility, or mission integrity.
- Decide which human-AI relationship model the business may support and what commercialization boundaries apply.
- Review and amend strategy when evidence, mission, law, capability, or market conditions change.

## AI RESPONSIBILITIES AND BOUNDARIES

The AI may structure evidence, compare options, model scenarios, and reveal dependencies. It may not fabricate demand, willingness to pay, legal clearance, partner commitment, cost certainty, or decision authority.

- Separate current observations from forecasts, assumptions, and recommended choices.
- Show the evidence and sensitivity behind market size, cost, capacity, revenue, adoption, and timeline estimates.
- Include substitutes, no-action, staged entry, partnership, build, buy, and exit options where relevant.
- Identify incentive conflicts and ways the proposed model could pressure unsafe or deceptive product behavior.
- Expose provider, platform, data, model, supply-chain, and jurisdiction dependencies.
- Avoid optimizing a strategy solely for revenue, engagement, growth, or cost when protected mission outcomes constrain those metrics.
- Escalate value conflicts and material commitments to the authorized human.

## APPLICABLE EXISTING AGENT ROLES

Primary roles:

- Analyst — compares strategic options, economics, capabilities, dependencies, risks, and consequences.
- Researcher — establishes market, customer, competitor, regulatory, and operating evidence.
- Decision Arbiter Advisory — evaluates options under explicit human-authorized criteria and preserves uncertainty and dissent.

Common supporting roles:

- Architect — maps capability, platform, integration, data, trust, and operational implications.
- Planner — sequences strategic bets, experiments, gates, resources, and contingencies.
- Coordinator — integrates workstreams without losing evidence provenance or specialist disagreement.
- Memory Curator — preserves strategy versions, assumptions, commitments, and review triggers.
- User Experience Reviewer — tests whether the economic model undermines agency, comprehension, accessibility, or complete journeys.

Challenge and assurance roles where consequence warrants:

- Critic — attacks framing, hidden assumptions, omitted alternatives, and unsupported confidence.
- Verifier — checks artifact identity, evidence integrity, acceptance boundaries, and the exact status of claims.

- Governance Reviewer — checks objective fidelity, authority, delegation, policy, evidence, and human decision boundaries.
- Security Auditor — evaluates strategic exposure from vendors, data, infrastructure, identities, and external dependencies.
- Red Team — attacks business-model abuse, incentive failures, platform dependence, adversarial market response, and catastrophic assumptions.

Role selection must use the smallest sufficient set, preserve chamber independence where review is required, and leave every source-defined role contract unchanged.

## REQUIRED ACTIVITIES

1. Strategic question and authority
State the choice to be made, owner, scope, horizon, constraints, criteria, and decision deadline without inventing urgency.

2. Arena and beneficiary definition
Define where the organization will act, whom it will serve, what outcome is offered, and who may be affected or excluded.

3. Value and differentiation analysis
Explain why the offering should matter relative to substitutes, and separate evidence from aspiration.

4. Business-model construction
Map value creation, delivery, capture or funding, costs, channels, partners, obligations, and failure conditions.

5. Capability and dependency analysis
Identify required internal capabilities, external dependencies, build-buy-partner choices, bottlenecks, and lock-in.

6. Option set and criteria
Evaluate materially distinct strategies consistently. Mandatory constraints cannot be traded away by aggregate scoring.

7. Economic and resource modeling
Use ranges, assumptions, sensitivity, and scenario analysis. Avoid unsupported point estimates and false precision.

8. Risk and incentive review
Assess legal, security, privacy, safety, accessibility, labor, trust, fraud, abuse, reputational, and mission effects.

9. Experiment and gate design
Define staged commitments and cheapest useful tests before irreversible investment.

10. Operating-model alignment
Map roles, governance, decision rights, evidence flows, customer commitments, and continuity needed to execute the strategy.

11. Decision and communication
Record selected, conditional, deferred, rejected, or no-action state with rationale, approvals, and evidence that would change it.

12. Review and exit
Define review cadence, trigger events, exit conditions, customer or stakeholder obligations, and recovery from strategic failure.

## HUMAN-AI INTERACTION REQUIREMENTS

- AI-generated forecasts must show assumptions, ranges, source dates, and model limitations.
- The human must be able to change criteria and immediately see which recommendations depend on those priorities.
- Recommendations must remain distinguishable from approved commitments and executed transactions.
- The system must surface when commercial incentives conflict with mission, user welfare, privacy, safety, or governance.

- Customer, partner, investor, employee, or community commitments must not be inferred from silence or informal conversation.
- Any AI-mediated negotiation or commercial action requires explicit identity, authority, scope, and approval controls.

## ARCHITECTURE AND SYSTEM IMPLICATIONS

The work must expose architectural obligations without treating a strategy, analysis, or plan as implemented architecture.

- Identify business-critical capabilities and map them to later enterprise, business, solution, data, application, platform, and operational architecture.
- Model cost, capacity, vendor, model-provider, data, licensing, and supply-chain assumptions as versioned dependencies.
- Preserve strategic flexibility through modular boundaries, portable data, exit plans, and avoidance of unnecessary lock-in.
- Ensure monetization and analytics paths do not bypass consent, privacy, authorization, or evidence controls.
- Support staged rollout, feature or market isolation, spending limits, and reversible commitments.
- Instrument mission and user outcomes alongside financial and operational metrics.

## OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

The owner-facing operating surface or equivalent durable record should expose:

- Current strategy version, owner, mission link, horizon, assumptions, and review date.
- Strategic options, criteria, mandatory constraints, sensitivity, and dissent.
- Resource commitments, costs, dependencies, partner states, risks, and decision gates.
- Measured outcomes separated from forecasts, targets, and model-generated projections.
- Commercial actions requiring human approval and the evidence available before approval.
- Drift alerts when economics, incentives, dependencies, or implementation diverge from the approved strategy.
- Exit, rollback, customer-obligation, and continuity plans.

## REQUIRED OUTPUTS AND ARTIFACTS

- Business strategy statement and choice record.
- Business or sustainability model with assumption ledger.
- Market, beneficiary, value, channel, and differentiation analysis.
- Capability, dependency, partner, vendor, and lock-in map.
- Economic model with scenarios and sensitivity.
- Strategic option comparison and decision record.
- Risk, incentive, compliance, and mission-alignment assessment.
- Staged experiment, investment, review, and exit plan.

## EVIDENCE REQUIREMENTS

- Current traceable market, customer, cost, capacity, competitor, regulatory, and partner evidence.
- Source and method for every material estimate or forecast.
- Documented human criteria, approvals, commitments, and risk acceptance.
- Evidence supporting capability claims and distinguishing available from planned capabilities.
- Scenario and sensitivity results showing where the strategy is fragile.
- Evidence of stakeholder, user, security, privacy, accessibility, and mission impacts.

## VERIFICATION AND ACCEPTANCE TESTS

- Trace each major strategic choice to mission, evidence, criteria, and authority.
- Change a critical cost, adoption, timing, or dependency assumption and inspect whether the recommendation changes.
- Attempt to select an option violating a mandatory legal, safety, privacy, or mission constraint; selection must fail.
- Remove a key provider or partner and verify the dependency and exit plan become visible.

- Confirm financial or growth success cannot be displayed as mission or user success without corresponding evidence.
- Verify unapproved AI recommendations cannot create binding commercial commitments.

## FAILURE STATES AND ANTI-PATTERNS

- Strategy as goals, slogans, forecasts, or feature lists without choices and exclusions.
- Fictional market demand, customer commitment, willingness to pay, cost, or partner readiness.
- False precision in forecasts and unit economics.
- Revenue or engagement optimization that erodes mission, consent, trust, safety, or user control.
- Ignoring substitutes, no-action, exit, or staged options.
- Treating technical capability as operational, legal, or commercial readiness.
- Hidden platform, model, data, vendor, or jurisdiction lock-in.
- Continuing a failed strategy to protect sunk costs or prior narratives.

## RECOVERY AND RECONSIDERATION

Strategic recovery contains commitments, restores truthful state, and reopens the first invalid assumption or decision.

- Pause new commitments while preserving existing customer, employee, partner, legal, and safety obligations.
- Identify which assumptions, evidence, incentives, capabilities, or dependencies changed or were wrong.
- Recalculate affected economics, capacity, timelines, and risks using current evidence.
- Narrow, stage, pivot, partner, divest, exit, or retire the affected strategy through human authorization.
- Communicate material changes truthfully to affected parties and preserve decision rationale.
- Propagate the revised strategy to portfolio, program, product, requirements, architecture, operations, and continuity records.

## COMPLETION CRITERIA

- The strategy makes explicit choices about arena, beneficiaries, value, capabilities, resources, and exclusions.
- Economics, forecasts, and dependencies are bounded by traceable evidence and uncertainty.
- Mission, legal, ethical, security, privacy, accessibility, and human-AI constraints are integrated.
- Staged decisions, stop conditions, review triggers, and exit obligations are defined.
- The authorized human owns commitments and value tradeoffs.
- The complete strategy package can be independently reviewed and handed off without conversational memory.

## HANDOFF AND CONTINUITY REQUIREMENTS

- Transfer the approved strategy version, mission link, decision owner, criteria, and authority limits.
- Carry forward assumptions, forecasts, dependencies, commitments, risks, and evidence states.
- Map strategic capabilities and outcomes to portfolio, program, product, requirements, and architecture work.
- Preserve rejected alternatives and conditions that would reopen them.
- Identify customer, partner, regulatory, financial, and data obligations that constrain implementation or exit.
- State the next safe decision and whether execution may proceed.

## RELATED CODEX SECTIONS

001 Research
002 Foresight / Futures
004 Mission Strategy
006 Product Strategy
007 Portfolio Management
008 Program Management
010 Product Management
011 Business Analysis
012 Market Research

013 Competitive Intelligence
014 Stakeholder Management
015 User Research
016 Domain Analysis
017 Feasibility Analysis
018 Risk Management
025 Business Architecture
026 Capability Architecture
183 AI Governance
219 Legal
220 Regulatory Compliance
222 Ethics
256 Verification
362 Revenue Operations
405 Lifecycle Governance

## SOURCE BASIS AND UNRESOLVED SOURCE GAPS

This draft applies the supplied requirements for explicit objectives, source-grounded facts, uncertainty and assumption typing, consistent option comparison, human-authorized criteria, short- and long-term effects, sensitivity, reversibility, dissent, bounded roles, evidence, hostile review, and continuity. The Owner's Cockpit material adds decision support without decision seizure, trust calibration, complete owner journeys, visible limitations, and truthful status. The V4 draft adds obligation structure, resolution experiments, proof-carrying decisions, and trusted transitions.

Unresolved source gap: the supplied corpus does not include a dedicated finance, accounting, market-sizing, or business-model standard. Economic methods must therefore remain explicitly sourced and later reconciled with Market Research, Competitive Intelligence, Feasibility Analysis, legal, financial operations, revenue operations, and audit sections.

# 006

## SECTION 006 - PRODUCT STRATEGY

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SECTION 006 — Product Strategy

## STATUS: CONTENT-COMPLETE

## PURPOSE

Define what product should exist, for whom, to produce which outcomes, under which principles and boundaries, and how evidence will guide its evolution across the lifecycle.

Product strategy connects mission and business strategy to user and owner value without collapsing into a feature backlog. It establishes product identity, target users and contexts, problems and jobs, value proposition, differentiated capabilities, experience principles, lifecycle posture, and the conditions for investment, change, or retirement.

## REQUIRED OUTCOME

- A product strategy statement identifying intended users, owners, operators, affected parties, contexts, problems, and outcomes.
- An evidence-backed value proposition and explicit product non-goals.
- A prioritized capability model rather than an ungoverned feature list.
- Product principles covering agency, trust, accessibility, privacy, safety, security, reliability, recovery, and human-AI interaction where applicable.
- A product hypothesis and outcome framework with anti-metrics, risks, and evidence thresholds.
- A staged product lifecycle and roadmap logic tied to learning, not arbitrary dates.
- Build, buy, integrate, partner, hold, and retire decision boundaries.

## WHY THIS MATTERS

Without product strategy, feature requests, competitor imitation, model capabilities, and internal convenience become the de facto product direction. The result may function technically while failing the user's actual job, the owner's control needs, or the mission.

AI products create additional ambiguity: conversational fluency may be mistaken for capability, recommendation for authority, personalization for consent, and automation for value. Product strategy must define the intended human-AI relationship and preserve the human's ability to understand, correct, limit, recover, and stop the system.

## APPLICABILITY

- A product or product family must choose users, problems, capabilities, positioning, or lifecycle direction.
- Multiple feature, platform, AI, design, or operational options compete for investment.
- The product mediates consequential decisions, data, automation, or human-AI interaction.
- Existing product behavior has drifted from mission, user evidence, or owner control.
- A new market, platform, user group, or capability requires a product-level decision.

## NON-APPLICABILITY CONDITIONS

- A bounded implementation task inherits a current approved product strategy and changes no product outcome or principle.
- Evidence of users, problems, or context is inadequate; user and market research must precede strategic commitment.
- The requested work is a one-off internal utility with no product lifecycle, user population, or material external consequence.
- The proposed direction cannot satisfy mandatory safety, legal, privacy, accessibility, or authority constraints.

## PREREQUISITES AND DEPENDENCIES

- Mission and business strategy sufficient to define product purpose and constraints.
- Research, market, competitor, stakeholder, user, domain, feasibility, and risk evidence proportional to the decision.

- An authorized product decision owner and explicit prioritization criteria.
- Current product state, architecture, analytics, support, incident, and lifecycle evidence where a product already exists.
- A clear distinction between documented, prototyped, tested, verified, and production capabilities.

## REQUIRED INPUTS

- Mission outcomes, business choices, user jobs, pain, context, accessibility needs, and recovery expectations.
- Market structure, alternatives, differentiation evidence, and changing conditions.
- Current product behavior, usage, support findings, incidents, reliability, security, privacy, and operational cost.
- Technology and AI capability evidence, limitations, provider dependencies, and governance boundaries.
- Legal, ethical, regulatory, content, localization, and distribution constraints.
- Available resources, time horizon, portfolio dependencies, and retirement obligations.

## HUMAN RESPONSIBILITIES AND AUTHORITY

The human product owner decides whom the product serves, what outcomes matter, which tradeoffs are acceptable, and how much authority may be delegated to product mechanisms or AI.

- Confirm intended users, owner roles, excluded contexts, product outcomes, and non-goals.
- Authorize product principles and prioritization criteria.
- Decide value conflicts among user groups, business needs, mission, safety, privacy, accessibility, and operational burden.
- Approve consequential automation, data use, pricing, release, and retirement decisions.
- Define the intended human-AI relationship, including correction, review, appeal, fallback, and stop rights.
- Reject capability claims or roadmap commitments not supported by current evidence.

## AI RESPONSIBILITIES AND BOUNDARIES

The AI may synthesize evidence, identify patterns, generate alternatives, and maintain traceability. It may not invent users, needs, consent, demand, priorities, or product acceptance.

- Separate user evidence from internal requests, model suggestions, market assumptions, and competitor imitation.
- Translate outcomes into capability hypotheses without prematurely specifying features or architecture.
- Expose which roadmap proposals depend on unverified assumptions or provider capabilities.
- Identify failure, misuse, accessibility, trust, safety, privacy, security, and recovery implications.
- Avoid treating engagement, completion rate, generated output, or model confidence as sufficient proof of user value.
- Keep recommendations advisory and route material tradeoffs to the authorized human.
- Preserve product history and the reason for accepted, rejected, deferred, and retired capabilities.

## APPLICABLE EXISTING AGENT ROLES

Primary roles:

- Researcher — establishes users, contexts, alternatives, product evidence, and source quality.
- Analyst — connects evidence to product outcomes, capabilities, tradeoffs, and decision implications.
- User Experience Reviewer — evaluates jobs, journeys, agency, comprehension, accessibility, failure, and recovery.

Common supporting roles:

- Architect — maps product capabilities to structural obligations and system boundaries.
- Planner — sequences discovery, experiments, roadmap gates, and product lifecycle work.
- Observer — captures actual product and user behavior without unsupported interpretation.
- Documentation Writer — preserves product principles, capability boundaries, and user-facing truth.
- Decision Arbiter Advisory — compares product options under authorized criteria.

Challenge and assurance roles where consequence warrants:

- Critic — attacks framing, hidden assumptions, omitted alternatives, and unsupported confidence.

- Verifier — checks artifact identity, evidence integrity, acceptance boundaries, and the exact status of claims.
- Governance Reviewer — checks objective fidelity, authority, delegation, policy, evidence, and human decision boundaries.
- Security Auditor — examines product threat, identity, data, abuse, and dependency implications.
- Red Team — attacks misuse, deceptive affordances, automation bias, capability overclaim, and recovery failure.

Role selection must use the smallest sufficient set, preserve chamber independence where review is required, and leave every source-defined role contract unchanged.

## REQUIRED ACTIVITIES

1. Product identity and boundary
State the product, its owner, users, contexts, outcomes, exclusions, and relationship to mission and business strategy.

2. Problem and job evidence
Trace each strategic product problem to user, owner, market, domain, operational, or mission evidence.

3. Value proposition
Define the change users should experience and why available alternatives are insufficient. Keep aspiration separate from demonstrated value.

4. User and actor model
Identify users, operators, owners, administrators, affected non-users, agents, external services, and authority boundaries without inventing personas.

5. Capability model
Describe durable capabilities and dependencies before decomposing into features. Mark current, planned, experimental, external, and prohibited capabilities.

6. Product principles
Establish rules for trust, agency, privacy, security, safety, accessibility, content, reliability, recovery, and human-AI interaction.

7. Outcome and anti-metric design
Define user and mission outcomes, evidence, anti-metrics, and conditions where apparent usage success may mask harm or failure.

8. Option and tradeoff analysis
Compare build, buy, integrate, partner, manual, staged, and no-action approaches under human-authorized criteria.

9. Experiment and validation path
Define the smallest prototypes, pilots, and user validation needed before commitment, with participant protection and truthful fidelity labels.

10. Roadmap logic
Sequence product bets by dependencies, information gain, risk, and readiness rather than arbitrary feature volume.

11. Lifecycle and retirement
Define support, maintenance, compatibility, migration, data retention, deprecation, sunset, and retirement obligations from the beginning.

12. Strategy decision
Record approved, conditional, deferred, rejected, or retired product direction with evidence and reopening triggers.

## HUMAN-AI INTERACTION REQUIREMENTS

- Product conversations must distinguish explanation, proposal, prioritization, authorization, execution, and verified result.
- Users and owners must be able to correct misunderstood intent and see whether AI output is generated, retrieved, inferred, simulated, or verified.

- The product must not imply AI authority, certainty, identity, memory, or capability beyond the actual runtime and granted permissions.
- Automation must remain bounded by review, correction, revocation, fallback, and recovery appropriate to consequence.
- Research and analytics must honor consent, privacy, accessibility, and data-minimization boundaries.
- The system must make unavailable, degraded, experimental, or externally blocked capabilities visible before asking the user to rely on them.

## ARCHITECTURE AND SYSTEM IMPLICATIONS

The work must expose architectural obligations without treating a strategy, analysis, or plan as implemented architecture.

- Maintain a product-capability model traceable to requirements, services, interfaces, data, policies, evidence, and owner journeys.
- Separate product promises from provider or model capabilities that may change independently.
- Use governed action boundaries for consequential commands regardless of UI, API, conversation, agent, or scheduled path.
- Preserve authoritative state, data lineage, freshness, identity, permissions, and completion evidence.
- Support graceful degradation, manual or non-AI fallback, export, portability, rollback, and retirement where required.
- Instrument complete journeys and user outcomes rather than isolated interface events.

## OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

The owner-facing operating surface or equivalent durable record should expose:

- Current product strategy, intended users, outcomes, non-goals, principles, and owner.
- Capability inventory with current, experimental, planned, external, degraded, and unavailable states.
- Product hypotheses, supporting evidence, assumptions, risks, and validation status.
- Roadmap options, dependencies, decisions, and evidence gates.
- User, owner, security, privacy, accessibility, support, incident, and reliability signals.
- Consequential AI or automation permissions, active delegation, fallback, and revocation.
- Lifecycle obligations, compatibility, deprecation, data, and retirement state.

## REQUIRED OUTPUTS AND ARTIFACTS

- Product strategy and product-boundary statement.
- User, owner, actor, problem, and job map.
- Value proposition and alternatives analysis.
- Capability model and state inventory.
- Product principles and human-AI interaction contract.
- Outcome, anti-metric, assumption, and risk register.
- Validation and roadmap decision framework.
- Lifecycle, support, migration, and retirement posture.

## EVIDENCE REQUIREMENTS

- Traceable user, owner, market, domain, support, incident, and operational evidence.
- Current capability evidence bound to artifact, version, environment, provider, and limitations.
- Research and validation records showing inclusion, method, participant protection, and observed behavior.
- Rationale and human approval for product principles, priorities, and consequential automation.
- Outcome evidence that distinguishes user benefit from engagement or activity proxies.
- Evidence of accessibility, security, privacy, reliability, recovery, and lifecycle feasibility proportional to claims.

## VERIFICATION AND ACCEPTANCE TESTS

- Trace each strategic capability to a user or owner outcome and source evidence.

- Remove or invalidate provider capability evidence and confirm the product claim degrades to unknown, unavailable, or experimental.
- Attempt a consequential action through UI, API, conversation, and agent paths; governance must remain equivalent.
- Run critical journeys from stopped, degraded, denied, and recovery states.
- Test whether users can distinguish recommendation from decision and generated output from executed action.
- Verify roadmap items cannot be marked complete without defined acceptance and evidence.
- Confirm deprecation or retirement preserves user data, notice, migration, and recovery obligations.

## FAILURE STATES AND ANTI-PATTERNS

- Feature factory behavior disconnected from user outcomes or mission.
- Invented personas, demand, user needs, acceptance, or accessibility claims.
- Copying competitors without context or problem evidence.
- Treating model fluency, demo quality, or engagement as product value.
- Capability-authority confusion and conversational text presented as executed work.
- Dark patterns, coerced consent, hidden automation, or inaccessible control paths.
- Roadmaps treated as commitments despite unresolved dependencies or evidence.
- Ignoring support, recovery, migration, compatibility, and retirement obligations.

## RECOVERY AND RECONSIDERATION

Product-strategy recovery returns the product to evidence, user outcomes, mission, and legitimate authority after drift or invalid assumptions.

- Pause affected roadmap or release commitments while preserving safe service and user obligations.
- Identify which problem, capability, user, provider, outcome, or business assumption failed.
- Reclassify affected claims and capabilities visibly.
- Reopen user research, domain analysis, feasibility, risk, requirements, architecture, or validation as needed.
- Narrow, redesign, stage, deprecate, or retire the affected direction through human approval.
- Communicate changes and migration or recovery paths truthfully to users and operators.

## COMPLETION CRITERIA

- Users, owners, contexts, problems, outcomes, non-goals, and product authority are explicit.
- Capabilities and roadmap logic are evidence-grounded and distinct from features or promises.
- Product principles cover agency, trust, accessibility, privacy, security, safety, reliability, recovery, and lifecycle.
- The intended human-AI relationship and authority boundaries are explicit where applicable.
- Validation, review, change, and retirement conditions are defined.
- The strategy can be handed to product management, requirements, architecture, design, and engineering without inventing missing intent.

## HANDOFF AND CONTINUITY REQUIREMENTS

- Transfer the approved product strategy version, decision owner, users, outcomes, non-goals, and principles.
- Provide capability states, assumptions, risks, evidence, dependencies, and unresolved decisions.
- Map capabilities to product management, requirements, architecture, UX, security, operations, and lifecycle sections.
- Preserve rejected options and evidence that would reopen them.
- Carry forward provider, model, data, consent, accessibility, migration, and retirement constraints.
- State the next safe product decision and current authority to proceed.

## RELATED CODEX SECTIONS

001 Research
002 Foresight / Futures
004 Mission Strategy

005 Business Strategy
007 Portfolio Management
008 Program Management
010 Product Management
011 Business Analysis
012 Market Research
014 Stakeholder Management
015 User Research
016 Domain Analysis
017 Feasibility Analysis
018 Risk Management
019 Requirements Engineering
026 Capability Architecture
075 Human–Computer Interaction
079 UX Strategy
106 User Validation
169 AI Engineering
182 AI Safety
183 AI Governance
217 Trust and Safety
373 Roadmap Management
390 Sunset Planning
406 System Retirement

## SOURCE BASIS AND UNRESOLVED SOURCE GAPS

This draft applies the supplied requirements for owner jobs and complete journeys, objective fidelity, evidence-before-claims, current-state inspection, capability-versus-authority separation, bounded delegation, grounded dialogue, visible uncertainty, accessibility, security, privacy, recovery, continuity, and no fake success. The role corpus supplies user-experience review, research, analysis, architecture, planning, testing, verification, governance, and hostile review. V4 adds claim and obligation structure, minimum-resolution experiments, proof-carrying results, dissent, and trusted transition.

Unresolved source gap: the supplied corpus does not contain a universal product-strategy or product-discovery standard, nor a final cross-domain policy for every form of human-AI relationship. This section must later be reconciled with Product Management, UX Strategy, User Research, Requirements Engineering, AI Safety, AI Governance, Trust and Safety, and lifecycle sections.

# 007

## SECTION 007 - PORTFOLIO MANAGEMENT

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SECTION 007 — Portfolio Management

## STATUS: CONTENT-COMPLETE

## PURPOSE

Govern a collection of products, programs, projects, experiments, capabilities, investments, and retirements as one constrained decision system.

Portfolio management determines what should start, continue, pause, merge, split, reduce, transfer, or end. It protects the mission and organization from fragmented optimization, duplicate work, hidden shared risk, resource overcommitment, and indefinite continuation of initiatives whose evidence no longer justifies them.

## REQUIRED OUTCOME

- A complete portfolio inventory with stable identities, owners, purpose, state, evidence, dependencies, commitments, and lifecycle status.
- Human-authorized selection, balance, prioritization, and termination criteria.
- A capacity and resource model that distinguishes available, committed, constrained, and hypothetical resources.
- A dependency and shared-risk map covering platforms, data, vendors, models, people, infrastructure, governance, and customers.
- Explicit portfolio decisions: initiate, continue, accelerate, hold, reduce, merge, separate, transfer, retire, or reject.
- A benefits, risk, concentration, and mission-alignment view that preserves uncertainty and dissent.
- A review cadence and trigger-based rebalancing process with evidence-backed change control.

## WHY THIS MATTERS

Individually reasonable initiatives can collectively exceed capacity, duplicate infrastructure, create contradictory commitments, concentrate risk, or compete for the same human authority. Without a portfolio view, local success can produce systemic failure.

AI can rank hundreds of initiatives, but a score cannot legitimately decide human priorities, protected obligations, or acceptable concentration of risk. Portfolio governance must keep criteria visible, mandatory constraints non-compensable, and termination as real a decision as initiation.

## APPLICABILITY

- More than one initiative competes for shared resources, platforms, data, customers, operators, or authority.
- The organization must balance exploration, delivery, maintenance, compliance, resilience, and retirement.
- Dependencies or risks cross product, program, project, or system boundaries.
- AI agents or automated workflows may create work faster than humans can govern or fund it.
- Investment, continuation, or termination choices have material mission, financial, legal, human, or operational effects.

## NON-APPLICABILITY CONDITIONS

- Only one bounded initiative exists and it does not share material resources or dependencies.
- The task is simple reporting with no authority to prioritize or change portfolio state.
- Initiative identities, current states, or evidence are too incomplete for legitimate comparison.
- The proposed scoring method would hide mandatory constraints or invent priorities.

## PREREQUISITES AND DEPENDENCIES

- Mission, business, and product strategy sufficient to establish portfolio criteria and protected obligations.
- Stable inventory of initiatives and lifecycle states.
- Authenticated portfolio decision authority and escalation paths.
- Current resource, cost, dependency, risk, commitment, and outcome evidence.
- A status lexicon that prevents planned, active, blocked, failed, and retired work from being conflated.

## REQUIRED INPUTS

- Initiative charters, objectives, owners, scope, outcomes, costs, schedules, risks, and current evidence.
- Resource and capacity data for people, agents, budget, infrastructure, environments, data, and attention.
- Shared architecture, platform, vendor, model, security, compliance, and operational dependencies.
- Customer, stakeholder, contractual, regulatory, support, maintenance, and retirement obligations.
- Strategic criteria, mandatory constraints, risk tolerance, and review horizon.
- Historical decisions, sunk costs, incidents, benefit evidence, and conditions that would change status.

## HUMAN RESPONSIBILITIES AND AUTHORITY

The authorized portfolio owner retains final authority over resource allocation, initiation, continuation, and termination, especially where choices encode values or affect people.

- Authorize portfolio criteria, weighting, capacity boundaries, and mandatory constraints.
- Decide tradeoffs among mission, growth, reliability, maintenance, safety, compliance, exploration, and retirement.
- Approve changes to commitments, funding, staffing, service obligations, or protected initiatives.
- Reject numerical rankings that conceal weak evidence or incomparable contexts.
- Ensure AI-generated work does not silently consume human attention, compute, budget, or operational capacity.
- Authorize retirement and the handling of users, data, contracts, infrastructure, and knowledge.

## AI RESPONSIBILITIES AND BOUNDARIES

The AI may normalize information, expose dependencies, model scenarios, and recommend portfolio actions. It may not create resources, invent comparability, or turn a ranking into authority.

- Preserve each initiative's objective, source evidence, state, uncertainty, and owner.
- Flag missing or stale information before comparison.
- Separate mandatory eligibility constraints from weighted preferences.
- Detect duplicate work, shared bottlenecks, risk concentration, conflicting commitments, and orphaned obligations.
- Model sensitivity to criteria, capacity, timing, dependency, and risk changes.
- Include hold, merge, transfer, reduce, retire, and no-action options rather than assuming growth.
- Do not mark benefit, readiness, or completion based solely on activity or self-reported progress.

## APPLICABLE EXISTING AGENT ROLES

Primary roles:

- Analyst — performs consistent portfolio comparison, sensitivity, risk, dependency, and outcome analysis.
- Planner — sequences initiatives against capacity, dependencies, gates, and time horizons.
- Decision Arbiter Advisory — structures recommendations under explicit criteria without converting them into binding decisions.

Common supporting roles:

- Coordinator — maintains assignment, dependency, evidence, and conflict visibility across initiatives.
- Architect — identifies shared platforms, architectural coupling, duplication, and technical concentration risk.
- Researcher — fills evidence gaps and validates external assumptions.
- Memory Curator — preserves portfolio history, decisions, assumptions, and retirement knowledge.
- Observer — maintains current state and detects changes without interpreting them as value.

Challenge and assurance roles where consequence warrants:

- Critic — attacks framing, hidden assumptions, omitted alternatives, and unsupported confidence.
- Verifier — checks artifact identity, evidence integrity, acceptance boundaries, and the exact status of claims.
- Governance Reviewer — checks objective fidelity, authority, delegation, policy, evidence, and human decision boundaries.
- Security Auditor — evaluates shared attack surface, systemic dependencies, and concentration of sensitive assets.
- Red Team — attacks scoring, gaming, hidden coupling, resource optimism, and failure of correlated initiatives.

Role selection must use the smallest sufficient set, preserve chamber independence where review is required, and leave every source-defined role contract unchanged.

## REQUIRED ACTIVITIES

1. Portfolio inventory
Create stable records for every initiative, including owner, purpose, scope, lifecycle state, outcomes, evidence, dependencies, costs, risks, obligations, and next decision.

2. Eligibility screening
Apply mandatory mission, authority, legal, safety, security, privacy, accessibility, evidence, and resource constraints before scoring.

3. Criteria and balance design
Define human-authorized criteria and desired balance across horizons, risk classes, lifecycle work, and strategic capabilities.

4. Capacity analysis
Establish actual available capacity and distinguish it from requested or assumed resources.

5. Dependency and concentration mapping
Identify shared people, agents, data, platforms, vendors, models, environments, customers, and failure domains.

6. Option evaluation
Compare continuation, acceleration, hold, reduction, merge, split, transfer, retirement, and no-action for each initiative.

7. Scenario and sensitivity analysis
Test portfolio resilience under budget, staffing, provider, regulatory, incident, demand, and timing changes.

8. Decision sequencing
Schedule decisions and gates so upstream uncertainty and shared constraints are resolved before downstream commitment.

9. Commitment and transition control
Record approved portfolio actions, affected promises, transition owners, and evidence required for the next state.

10. Review and rebalancing
Reopen portfolio decisions when outcomes, capacity, dependencies, risks, mission, or external conditions materially change.

## HUMAN-AI INTERACTION REQUIREMENTS

- The owner must be able to inspect why a portfolio recommendation changed when criteria or evidence changed.
- AI-generated rankings must show source data, normalization, missing values, sensitivity, and non-compensable constraints.
- Initiatives must not be represented as interchangeable when their contexts, obligations, or evidence differ materially.
- The system must distinguish a recommendation, approved portfolio decision, allocated resource, and executed transition.
- Agents must not autonomously create new portfolio items or consume shared resources outside delegated envelopes.
- Dissenting specialist findings and affected stakeholder impacts must remain visible at decision time.

## ARCHITECTURE AND SYSTEM IMPLICATIONS

The work must expose architectural obligations without treating a strategy, analysis, or plan as implemented architecture.

- Maintain a portfolio registry linked to authoritative initiative, artifact, environment, cost, risk, and lifecycle records.
- Represent dependencies as a graph with stable identities and invalidation when upstream state changes.

- Expose shared platforms, data, identities, providers, models, and failure domains to prevent hidden concentration.
- Support resource budgets and enforcement for compute, storage, tools, data access, money, and agent execution.
- Keep portfolio state changes auditable, versioned, and reversible where possible.
- Integrate retirement, migration, data disposition, and knowledge preservation into portfolio decisions.

## OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

The owner-facing operating surface or equivalent durable record should expose:

- Complete initiative inventory and truthful lifecycle state.
- Mission and strategy alignment, criteria, mandatory constraints, and evidence quality.
- Capacity, allocation, demand, conflicts, and overcommitment.
- Dependencies, concentration risks, shared incidents, and systemic bottlenecks.
- Recommendations, dissent, required human decisions, approvals, and transition progress.
- Benefits and outcomes separated from forecasts and activity.
- Hold, pause, narrow, merge, transfer, and retire controls with consequence previews.

## REQUIRED OUTPUTS AND ARTIFACTS

- Portfolio register and lifecycle-state inventory.
- Portfolio criteria, eligibility, and balance policy.
- Capacity and allocation model.
- Dependency, concentration, and shared-risk graph.
- Option comparison and sensitivity analysis.
- Portfolio decision and transition records.
- Benefits, outcomes, obligations, and retirement dashboard specification.
- Continuity and review-trigger record.

## EVIDENCE REQUIREMENTS

- Current initiative state and outcome evidence from authoritative records.
- Resource availability and cost evidence, including uncertainty and time bounds.
- Dependency and shared-risk evidence bound to systems, versions, providers, and environments.
- Human-approved criteria, constraints, allocation, and transition decisions.
- Evidence for claimed benefits, readiness, and closure rather than activity reports.
- Retirement and obligation evidence for items removed from the portfolio.

## VERIFICATION AND ACCEPTANCE TESTS

- Remove or stale an initiative's evidence and confirm its ranking and readiness degrade appropriately.
- Inject an initiative that scores highly but violates a mandatory constraint; it must remain ineligible.
- Reduce a shared resource and verify affected dependencies and decisions become visible.
- Fail a common vendor, model, data source, or platform and inspect correlated portfolio impact.
- Attempt unauthorized agent-created work or resource consumption; it must be denied or quarantined.
- Trace a retirement decision through user, data, contract, infrastructure, and continuity obligations.

## FAILURE STATES AND ANTI-PATTERNS

- Scoring theater that converts weak or incomparable data into a single authoritative rank.
- Overcommitment based on hypothetical people, agents, budget, or infrastructure.
- Hidden shared dependencies and correlated failure risk.
- Sunk-cost bias and indefinite continuation without outcome evidence.
- Favoring new initiatives while starving maintenance, security, compliance, resilience, or retirement.
- Double counting benefits or resources across initiatives.
- AI-generated portfolio expansion without human authorization.
- Declaring retirement complete while obligations, data, infrastructure, or knowledge remain.

## RECOVERY AND RECONSIDERATION

Portfolio recovery restores a truthful inventory, contains systemic risk, and rebalances commitments within real capacity.

- Freeze affected initiation or allocation decisions while preserving critical obligations.
- Reconcile initiative identities, states, evidence, dependencies, resources, and commitments.
- Identify which portfolio assumptions or shared components caused the discrepancy.
- Reprioritize through explicit human criteria and communicate affected transitions.
- Contain, isolate, migrate, merge, reduce, or retire initiatives as authorized.
- Update downstream program, project, product, architecture, operations, and continuity records.

## COMPLETION CRITERIA

- Every in-scope initiative has a stable identity, owner, state, evidence, dependencies, and next decision.
- Criteria, mandatory constraints, capacity, and resource assumptions are explicit.
- Shared risks, concentration, obligations, and lifecycle balance are visible.
- Portfolio recommendations remain advisory until human authorization.
- Transitions, including retirement, have owners, evidence, and completion criteria.
- The portfolio can be resumed and audited without hidden spreadsheets or conversational memory.

## HANDOFF AND CONTINUITY REQUIREMENTS

- Provide each receiving program or product with its approved state, allocation, constraints, dependencies, and decision gates.
- Carry forward portfolio-level risks, shared services, provider assumptions, and capacity limits.
- Preserve cross-initiative decisions, conflicts, dissent, and invalidation conditions.
- Notify affected owners when an upstream portfolio decision changes.
- Bind retired or transferred items to obligation, data, infrastructure, and knowledge-disposition records.
- State the next review trigger and safe-to-continue status.

## RELATED CODEX SECTIONS

004 Mission Strategy
005 Business Strategy
006 Product Strategy
008 Program Management
009 Project Management
010 Product Management
017 Feasibility Analysis
018 Risk Management
021 Enterprise Architecture
026 Capability Architecture
034 Platform Architecture
258 Configuration Management
321 Capacity Management
324 Reliability Management
373 Roadmap Management
383 Platform Migration
390 Sunset Planning
405 Lifecycle Governance
406 System Retirement

## SOURCE BASIS AND UNRESOLVED SOURCE GAPS

This draft applies the supplied requirements for explicit objective and authority, consistent option evaluation, mandatory constraints, sensitivity and reversibility, bounded planning, dependency classification, observable states, evidence-backed completion, role coordination, conflict transparency, human final authority, and continuity. It also uses the V4 concepts of obligation lattices, claim typing, proof-carrying transitions, and trusted advancement that cannot outrun evidence.

Unresolved source gap: the supplied corpus does not provide a dedicated portfolio-management accounting model, benefits-realization standard, or universal resource-allocation method. Numerical methods and financial claims must later be reconciled with business intelligence, operations research, capacity management, financial operations, audit, and internal controls.

# 008

## SECTION 008 - PROGRAM MANAGEMENT

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SECTION 008 — Program Management

## STATUS: CONTENT-COMPLETE

## PURPOSE

Coordinate multiple related projects, products, workstreams, and organizational changes so they collectively produce an authorized outcome that no single project can deliver alone.

Program management governs shared outcomes, benefits, dependencies, interfaces, resources, risks, decisions, and transitions across workstreams. It does not replace project ownership, specialist judgment, product authority, or human governance with a central narrative of progress.

## REQUIRED OUTCOME

- A program charter defining outcome, scope, exclusions, sponsors, decision rights, participating workstreams, and completion conditions.
- An integrated outcome and benefits map connected to mission, strategy, users, owners, and evidence.
- A dependency-aware program roadmap with entry, exit, validation, and transition gates.
- A shared architecture, data, environment, configuration, release, and governance alignment plan where applicable.
- Resource, capacity, risk, issue, decision, change, and stakeholder-management systems.
- Durable inter-project handoffs and one authoritative state per program object.
- An operational transition and benefits-realization plan extending beyond project delivery.

## WHY THIS MATTERS

Projects can each satisfy their local scope while the program fails because interfaces, timing, adoption, operations, data, authority, or benefits were never integrated. Conversely, a program office can create fictional progress by aggregating activities and suppressing project-level failure or dissent.

Multi-agent execution magnifies this risk. Parallel agents can produce more output while fragmenting truth. Program management must preserve exact versions, assignment boundaries, dependencies, evidence, conflicts, and human decision points across the whole outcome.

## APPLICABILITY

- The desired outcome requires multiple interdependent projects, products, teams, vendors, agents, or organizational changes.
- Shared architecture, data, platforms, environments, releases, customers, or operations create cross-project risk.
- Benefits or mission outcomes appear only after integration and operational adoption.
- Program-level capacity, sequencing, governance, or transition decisions are required.
- A major migration, transformation, compliance effort, platform build, or system-of-systems initiative is underway.

## NON-APPLICABILITY CONDITIONS

- A single project can deliver and validate the complete outcome without material external dependencies.
- The work is a loose portfolio of unrelated initiatives with no shared program outcome.
- No authority exists to coordinate participating workstreams or resolve program-level dependencies.
- The proposed program merely adds reporting overhead without changing integration, decisions, risk, or outcomes.

## PREREQUISITES AND DEPENDENCIES

- An approved mission, strategy, portfolio decision, and defined program outcome.
- Authenticated sponsor, program owner, workstream owners, decision rights, and escalation paths.
- Known projects or workstreams, dependencies, commitments, constraints, and current states.
- Shared status, evidence, configuration, identity, and continuity conventions.
- A program-level risk, change, architecture, and operational-transition posture.

## REQUIRED INPUTS

- Program objective, outcomes, benefits, scope, exclusions, constraints, and success evidence.
- Project and product charters, roadmaps, plans, architecture, requirements, risks, and current status.
- Stakeholder, user, organizational, vendor, regulatory, security, privacy, safety, and operational obligations.
- Shared resources, budgets, people, agents, tools, environments, platforms, data, and release windows.
- Dependency, interface, version, configuration, and integration information.
- Prior decisions, incidents, unresolved conflicts, assumptions, and conditions requiring human authority.

## HUMAN RESPONSIBILITIES AND AUTHORITY

The sponsor and authorized program owner retain authority over program outcomes, resource commitments, scope changes, cross-workstream tradeoffs, and acceptance of residual risk.

- Ratify the program charter, governance, workstream boundaries, decision rights, and escalation thresholds.
- Approve program-level scope, funding, capacity, vendor, release, migration, and operational commitments.
- Resolve conflicts among mission, product, project, safety, security, legal, stakeholder, and operational priorities.
- Ensure central coordination does not erase specialist findings or project truth.
- Authorize AI or agent delegation by workstream, tool, data, consequence, and duration.
- Decide program continuation, replanning, partial transition, hold, or termination.

## AI RESPONSIBILITIES AND BOUNDARIES

The AI may coordinate, analyze dependencies, maintain state, and surface conflicts only within recorded authority. It may not invent program scope, owners, resources, approvals, or completion.

- Preserve the actual program outcome and each workstream's bounded objective.
- Maintain stable identities and exact versions for deliverables, dependencies, decisions, and evidence.
- Detect conflicts, duplication, blocked dependencies, schedule assumptions, and inconsistent states.
- Continue safe independent work while isolating affected blockers.
- Avoid averaging contradictory specialist conclusions or rolling failed projects into a green program status.
- Distinguish plan, forecast, reported status, observed state, tested integration, verified outcome, and accepted benefit.
- Escalate cross-workstream value or authority conflicts to the authorized human.

## APPLICABLE EXISTING AGENT ROLES

Primary roles:

- Planner — creates the dependency-aware program plan, phases, gates, risks, resources, and completion evidence.
- Coordinator — assigns, synchronizes, tracks, reconciles, and hands off workstreams while preserving provenance and conflict.
- Analyst — evaluates program outcomes, benefits, dependencies, scenarios, and decision implications.

Common supporting roles:

- Architect — governs shared structure, interfaces, standards, integration, trust, and technical decision dependencies.
- Observer — captures program and workstream state without interpreting activity as success.
- Memory Curator — preserves decisions, versions, continuity, and retrievable program history.
- Documentation Writer — maintains durable charters, interface records, operating guides, and transition documents.
- Decision Arbiter Advisory — structures cross-workstream decisions under authorized criteria.

Challenge and assurance roles where consequence warrants:

- Critic — attacks framing, hidden assumptions, omitted alternatives, and unsupported confidence.
- Verifier — checks artifact identity, evidence integrity, acceptance boundaries, and the exact status of claims.
- Governance Reviewer — checks objective fidelity, authority, delegation, policy, evidence, and human decision boundaries.
- Tester — validates integrated normal, boundary, failure, and recovery behavior across workstreams.

- Security Auditor — evaluates cross-project trust, identity, data, supply-chain, and shared-infrastructure risk.
- Red Team — attacks program assumptions, dependency failure, status laundering, and integration gaps.

Role selection must use the smallest sufficient set, preserve chamber independence where review is required, and leave every source-defined role contract unchanged.

## REQUIRED ACTIVITIES

1. Program charter
Define outcome, scope, exclusions, sponsor, owner, decision rights, workstreams, benefits, constraints, and completion state.

2. Workstream architecture
Define stable workstream identities, owners, objectives, inputs, outputs, interfaces, authority, evidence, and acceptance.

3. Dependency graph
Classify hard, soft, approval, evidence, resource, environmental, review, and external dependencies.

4. Integrated planning
Sequence phases, parallel work, milestones, integration points, validation gates, transition events, and contingency paths.

5. Shared governance
Establish architecture, security, privacy, safety, quality, change, configuration, release, evidence, and decision processes.

6. Resource and capacity management
Use actual availability and account for review, rework, integration, operations, and external delay.

7. Risk and issue control
Maintain program-level and inherited risks, triggers, owners, responses, residual risk, and decision thresholds.

8. State and evidence management
Use one authoritative status source per object and bind status to exact artifacts, environments, tests, and approvals.

9. Integration and validation
Verify interfaces and complete journeys across workstreams, including failure, recovery, migration, and operational scenarios.

10. Stakeholder and change coordination
Manage commitments, communications, adoption, training, organizational effects, and grievance or escalation paths.

11. Operational transition
Transfer capabilities, controls, documentation, support, monitoring, ownership, and recovery into real operation.

12. Benefits and closure
Measure outcomes after transition, close or transfer residual work, preserve lessons, and avoid declaring benefit at delivery alone.

## HUMAN-AI INTERACTION REQUIREMENTS

- Every agent or human workstream assignment must include objective, scope, authority, inputs, outputs, dependencies, evidence, review, and escalation.
- Program dashboards must distinguish reported status from direct observation and verification.
- The owner must be able to trace an integrated result to exact upstream artifacts and workstream owners.
- Conflicts must remain visible until reconciled by evidence or authorized decision.
- Conversational coordination must create durable action and handoff records; chat memory alone is insufficient.
- Program-level AI may not override specialist roles, create fictional consensus, or approve its own prior work where independence is required.

## ARCHITECTURE AND SYSTEM IMPLICATIONS

The work must expose architectural obligations without treating a strategy, analysis, or plan as implemented architecture.

- Represent the program as a dependency and evidence graph linked to exact artifacts, versions, environments, interfaces, and decisions.
- Define shared architecture and standards without collapsing legitimate bounded contexts or project ownership.
- Centralize only controls requiring consistent enforcement, such as identity, policy, evidence, configuration, and release gates.
- Provide isolated environments and integration stages to prevent one workstream from corrupting another.
- Support cross-workstream observability, incident correlation, rollback, migration, and continuity.
- Keep benefits and operational outcomes separate from build completion and deployment events.

## OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

The owner-facing operating surface or equivalent durable record should expose:

- Program outcome, scope, sponsor, owner, decision rights, and current state.
- Workstreams, owners, authority, dependencies, deliverables, versions, and evidence.
- Integrated roadmap, capacity, risks, issues, decisions, changes, and validation gates.
- Conflicts, blocked paths, safe parallel work, and human decisions required.
- Shared architecture, security, privacy, quality, release, and operational readiness state.
- Transition, adoption, support, benefits, and residual obligation status.
- Pause, narrow, replan, revoke delegation, and safe-halt controls.

## REQUIRED OUTPUTS AND ARTIFACTS

- Program charter and governance model.
- Outcome, benefit, workstream, and responsibility map.
- Integrated roadmap, dependency graph, and capacity plan.
- Architecture, interface, configuration, environment, and release alignment records.
- Program risk, issue, decision, change, and dissent registers.
- Integration, validation, transition, adoption, and benefits plans.
- Program evidence package, closure report, and continuity map.

## EVIDENCE REQUIREMENTS

- Current workstream state and artifact identity from authoritative records.
- Evidence for dependencies, resource availability, interfaces, and shared constraints.
- Human approvals for program-level scope, allocation, risk, release, and transition decisions.
- Integration and end-to-end results bound to exact versions and environments.
- Operational and user outcome evidence after transition.
- Visible limitations, failed workstreams, unresolved risks, and evidence gaps.

## VERIFICATION AND ACCEPTANCE TESTS

- Restart a receiving workstream and continue from its durable handoff alone.
- Change an upstream artifact and verify dependent work and evidence are invalidated or reopened.
- Inject contradictory specialist findings and confirm they remain visible rather than averaged.
- Block one dependency and confirm unaffected safe work can continue without false program completion.
- Run integrated critical journeys through normal, denied, degraded, rollback, and recovery states.
- Verify program benefit cannot be claimed from project delivery without current outcome evidence.
- Revoke an agent delegation and confirm new consequential work stops across all paths.

## FAILURE STATES AND ANTI-PATTERNS

- Program as reporting layer with no authority, integration, or outcome responsibility.
- Green program status built from incomplete, blocked, failed, or unverified project reports.

- Hidden dependencies, version mismatch, duplicated work, and ambiguous ownership.
- Central coordination overriding specialist evidence or project truth.
- Invented resources, deadlines, approvals, benefits, or operational readiness.
- Integration deferred until the end, creating unrecoverable coupling and schedule collapse.
- Delivery declared complete before adoption, operations, recovery, and benefits are proven.
- Chat-dependent handoffs and untraceable agent delegation.

## RECOVERY AND RECONSIDERATION

Program recovery isolates affected workstreams, restores authoritative state, and reestablishes a safe integrated path.

- Enter a bounded hold or safe halt for affected integrations while preserving unaffected safe work.
- Reconcile artifact versions, workstream states, dependencies, decisions, resources, and evidence.
- Identify the first invalid assumption, interface, authority, or transition.
- Replan dependencies, resources, gates, and transition scope through human approval.
- Rollback, isolate, substitute, migrate, or terminate affected workstreams as authorized.
- Update stakeholders, operations, project plans, architecture, risk, and continuity records.

## COMPLETION CRITERIA

- The program outcome, workstreams, owners, authority, dependencies, and completion evidence are explicit.
- Shared governance and architecture are defined without erasing bounded responsibilities.
- Integrated behavior, operational transition, and complete journeys have current evidence.
- Benefits are measured after transition and remain distinct from delivery activity.
- Conflicts, residual risks, and unresolved decisions are visible.
- Program closure or transfer leaves a durable, reproducible continuity package.

## HANDOFF AND CONTINUITY REQUIREMENTS

- Provide every workstream with exact scope, authority, dependencies, inputs, outputs, evidence, and review requirements.
- Bind downstream work to exact upstream artifact and decision versions.
- Transfer shared risks, controls, resource limits, and integration obligations.
- Preserve conflicts, failed assumptions, residual work, and reopening conditions.
- At transition, transfer operational ownership, support, monitoring, recovery, documentation, and benefits measurement.
- State current safety, readiness, and next authorized program action.

## RELATED CODEX SECTIONS

005 Business Strategy
006 Product Strategy
007 Portfolio Management
009 Project Management
010 Product Management
014 Stakeholder Management
017 Feasibility Analysis
018 Risk Management
020 Systems Engineering
021 Enterprise Architecture
022 Solution Architecture
041 Integration Architecture
258 Configuration Management
263 Environment Management
287 Release Engineering
289 Change Management

301 Site Reliability Engineering
319 Incident Management
331 Operational Readiness
372 Feedback Management
402 Post-Implementation Review
405 Lifecycle Governance

## SOURCE BASIS AND UNRESOLVED SOURCE GAPS

This draft applies the supplied Planner and Coordinator contracts for bounded work, dependency classification, workstream assignment, parallelization, conflict handling, evidence, validation gates, human decisions, and continuity. It applies the Owner's Cockpit requirements for durable multi-agent handoffs, exact upstream artifact identity, one authoritative status source, visible conflicts, governed action, and owner control. V3 and V4 add objective fidelity, no fake success, obligation structure, independent review, proof-carrying results, and trusted transition.

Unresolved source gap: the supplied corpus does not define a universal benefits-realization, earned-value, or organizational-change methodology. Such methods must be selected explicitly and later reconciled with Project Management, Change Management, Operations, Product Analytics, Post-Implementation Review, and Audit.

# 009

## SECTION 009 - PROJECT MANAGEMENT

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SECTION 009 — Project Management

## STATUS: CONTENT-COMPLETE

## PURPOSE

Convert an authorized, bounded objective into a controlled temporary endeavor that produces defined deliverables and verified outcomes under explicit scope, resources, dependencies, risks, quality, and acceptance conditions.

Project management in Thirsty's Codex is not status theater or calendar maintenance. It creates the operational contract by which work is chartered, decomposed, sequenced, changed, observed, accepted, closed, and handed off without inventing resources, deadlines, authority, or completion.

## REQUIRED OUTCOME

- A project charter defining objective, scope, exclusions, sponsor, project owner, authority, deliverables, and acceptance.
- A current-state baseline and target state linked to requirements and evidence.
- A dependency-aware work breakdown with bounded tasks, owners or roles, inputs, outputs, tools, permissions, and completion evidence.
- A realistic sequence or schedule whose estimates disclose assumptions, confidence, capacity, review, rework, and external delay.
- Integrated risk, issue, decision, change, quality, communication, and stakeholder controls.
- Observable milestones representing achieved states rather than activity labels.
- A verified delivery, operational handoff, closure, and continuity package.

## WHY THIS MATTERS

Projects fail when a plausible plan is treated as reality, activity is treated as outcome, or delivery is treated as acceptance. AI can make these failures look unusually polished by generating complete-looking schedules, task lists, and reports from incomplete inputs.

Governed project management keeps the project tied to actual authority, dependencies, available resources, evidence, and acceptance. It also preserves the distinction between what was planned, attempted, produced, tested, verified, accepted, and transferred.

## APPLICABILITY

- A temporary, bounded body of work must create, change, migrate, validate, or retire a defined result.
- Work requires sequencing, resource coordination, dependencies, milestones, risk, and acceptance.
- Multiple human or AI roles contribute to one deliverable or outcome.
- A change has material cost, schedule, operational, security, safety, legal, or stakeholder consequences.
- A project requires an explicit start, controlled execution, and closure or transfer.

## NON-APPLICABILITY CONDITIONS

- The work is a single trivial task that can be safely completed under an existing task envelope.
- The work is ongoing product or operational management rather than a temporary endeavor.
- The objective or sponsor authority is not established.
- Required inputs are so uncertain that research, analysis, feasibility, or requirements work must precede project commitment.
- The request is merely to generate a schedule without evidence of scope, resources, dependencies, or acceptance.

## PREREQUISITES AND DEPENDENCIES

- An approved objective from strategy, portfolio, program, product, remediation, migration, incident, or lifecycle governance.
- Authenticated sponsor, project owner, decision rights, escalation, and acceptance authority.

- Sufficient requirements, constraints, current-state evidence, and target-state definition.
- Known resources, tools, environments, dependencies, risks, prohibited actions, and budget or time boundaries.
- A continuity mechanism and canonical status lexicon.

## REQUIRED INPUTS

- Objective, scope, exclusions, deliverables, acceptance criteria, and desired operating outcome.
- Current repository, system, process, product, data, infrastructure, or physical state.
- Requirements, architecture, design, quality, security, privacy, safety, accessibility, and compliance obligations.
- Available people, agents, tools, permissions, environments, budget, capacity, and deadlines.
- Dependencies, assumptions, unknowns, risks, stakeholders, commitments, and change constraints.
- Prior work, decisions, incidents, failed approaches, evidence, and continuity records.

## HUMAN RESPONSIBILITIES AND AUTHORITY

The sponsor and project owner establish legitimacy, priorities, resources, and acceptance. Project coordination does not create authority beyond that grant.

- Approve the charter, scope, exclusions, deliverables, constraints, and acceptance authority.
- Provide or authorize real resource commitments and resolve unavailable-resource conflicts.
- Decide material scope, schedule, cost, quality, risk, and stakeholder tradeoffs.
- Approve consequential changes, production actions, destructive operations, and risk acceptance.
- Correct project language that silently changes the objective or treats estimates as promises.
- Accept, reject, conditionally accept, pause, terminate, or transfer the project based on evidence.

## AI RESPONSIBILITIES AND BOUNDARIES

The AI may plan, coordinate, track, analyze, and report only from current evidence and delegated authority. It may not invent owners, personnel, dates, approvals, dependencies, status, or completion.

- Restate the approved objective and declared mode before decomposing work.
- Identify current state and hard dependencies before scheduling implementation.
- Create bounded tasks with explicit outputs, authority, evidence, and acceptance.
- Use estimates only when a basis exists; otherwise label ranges as preliminary.
- Track planned, ready, active, blocked, awaiting evidence, awaiting review, failed, completed, rejected, cancelled, and superseded states accurately.
- Continue unaffected safe work while isolating genuine blockers.
- Escalate material changes or value conflicts rather than silently replanning them as authorized.
- Do not mark work complete based on effort, file creation, agent report, passing subset, or process exit alone.

## APPLICABLE EXISTING AGENT ROLES

Primary roles:

- Planner — creates the work breakdown, dependencies, phases, gates, risks, schedule, resources, and completion evidence.
- Coordinator — assigns and synchronizes bounded work, tracks evidence, preserves conflict, and consolidates results.
- Observer — records current state, events, transitions, and gaps without interpreting activity as success.

Common supporting roles:

- Architect — provides structural decisions, constraints, interfaces, and implementation sequence where assigned.
- Implementer — performs authorized changes under exact scope and acceptance criteria.
- Tester — validates normal, boundary, failure, and recovery behavior.
- Documentation Writer — maintains project, operating, decision, and handoff records.
- Memory Curator — preserves durable state, provenance, decisions, and next safe action.

Challenge and assurance roles where consequence warrants:

- Critic — attacks framing, hidden assumptions, omitted alternatives, and unsupported confidence.
- Verifier — checks artifact identity, evidence integrity, acceptance boundaries, and the exact status of claims.
- Governance Reviewer — checks objective fidelity, authority, delegation, policy, evidence, and human decision boundaries.
- Security Auditor — examines project security, data, access, supply-chain, and deployment risk.
- Red Team — attacks planning assumptions, hidden dependencies, bypasses, and false-readiness claims.
- Decision Arbiter Advisory — structures material project choices under authorized criteria.

Role selection must use the smallest sufficient set, preserve chamber independence where review is required, and leave every source-defined role contract unchanged.

## REQUIRED ACTIVITIES

1. Project chartering
Define the objective, sponsor, project owner, scope, exclusions, deliverables, target state, constraints, acceptance, authority, and termination conditions.

2. Current-state baseline
Inspect actual artifacts, environments, work already completed, known failures, continuity, and external dependencies before planning remaining work.

3. Requirements and obligation mapping
Link deliverables to requirements, architecture, controls, evidence contracts, and acceptance criteria. Identify contradictions and missing decision owners.

4. Work breakdown
Decompose deliverables into bounded tasks with stable IDs, owners or roles, inputs, tools, authority, dependencies, outputs, verification, and recovery.

5. Dependency and sequence design
Classify dependencies and schedule only when hard prerequisites are satisfied or risk is explicitly accepted.

6. Resource and estimation planning
Use actual availability and include review, rework, testing, documentation, integration, deployment, and external delay.

7. Risk, issue, and contingency control
Maintain triggers, likelihood, impact, prevention, detection, response, owner, residual risk, and decision thresholds.

8. Quality and verification planning
Map requirements to tests, independent verification, hostile review, owner acceptance, and evidence retention.

9. Execution and status control
Track actual state from attributable evidence, not narrative confidence. Maintain one authoritative status per task and deliverable.

10. Change control
Record proposed changes, affected scope, tasks, dependencies, assumptions, resources, risks, acceptance, and required authorization.

11. Delivery and transition
Package exact artifacts, versions, configuration, documentation, training, support, monitoring, rollback, and operational ownership.

12. Acceptance and closure
Verify deliverables, resolve or transfer residual work, close commitments, preserve evidence, and record lessons and next action.

## HUMAN-AI INTERACTION REQUIREMENTS

- Every human or AI assignment must state objective, scope, authority, prohibited actions, inputs, expected output, evidence, review, and escalation.

- Status reports must separate direct observation, agent report, test result, verification, and human acceptance.
- The human must be able to inspect the basis of estimates, schedule changes, and completion claims.
- The AI must not pressure approval through fabricated urgency, sunk-cost framing, or optimistic certainty.
- Conversational instructions that change scope or consequence must become durable change requests before execution.
- Revocation, pause, or safe-halt commands must prevent new consequential work and define in-flight handling.

## ARCHITECTURE AND SYSTEM IMPLICATIONS

The work must expose architectural obligations without treating a strategy, analysis, or plan as implemented architecture.

- Maintain stable project, task, artifact, environment, decision, evidence, and release identities.
- Represent dependencies, invalidation, acceptance, and state transitions explicitly.
- Separate planning data from authoritative runtime, repository, financial, or operational state.
- Support bounded credentials, branch or environment isolation, action receipts, rollback, and cleanup.
- Integrate requirements traceability, test results, security findings, documentation, and operational readiness.
- Preserve reproducibility and exact version linkage through delivery and handoff.

## OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

The owner-facing operating surface or equivalent durable record should expose:

- Project objective, owner, sponsor, authority, scope, exclusions, target state, and acceptance.
- Deliverables, tasks, dependencies, owners, roles, resources, risks, and status evidence.
- Milestones as observable states with gates, results, and closure authority.
- Changes, decisions, assumptions, blockers, safe parallel work, and human approvals required.
- Artifacts, versions, tests, verification, incidents, rollback, and readiness.
- Budget and schedule ranges with basis and confidence rather than invented precision.
- Pause, narrow, revoke, terminate, restore, and close controls.

## REQUIRED OUTPUTS AND ARTIFACTS

- Project charter and responsibility model.
- Current-state baseline and target-state definition.
- Work breakdown, dependency graph, phase plan, and schedule basis.
- Resource, risk, issue, decision, change, and communication registers.
- Requirements, quality, test, verification, and acceptance plan.
- Delivery, deployment, rollback, transition, and support package.
- Final project report, evidence package, closure record, and continuity map.

## EVIDENCE REQUIREMENTS

- Authenticated charter and human approvals.
- Current artifact, environment, dependency, resource, and configuration evidence.
- Task-level commands, diffs, logs, outputs, tests, reviews, receipts, and hashes.
- Schedule and estimate basis with assumptions and confidence.
- Acceptance evidence tied to exact deliverables and target environment.
- Evidence for unresolved, failed, blocked, unverified, transferred, and closed states.

## VERIFICATION AND ACCEPTANCE TESTS

- Trace each deliverable to objective, requirements, tasks, tests, evidence, and acceptance authority.
- Attempt to schedule work before a hard dependency and verify it is blocked or explicitly authorized as risk.
- Remove a required artifact or test and confirm completion becomes impossible.
- Inject an unauthorized scope change through conversation and confirm execution does not proceed.
- Restart the project from the continuity package without relying on chat memory.
- Verify a non-programmer owner can see current state, limitations, and next safe action.

- Close the project only after residual work, operations, data, credentials, support, and obligations are resolved or transferred.

## FAILURE STATES AND ANTI-PATTERNS

- Project plan generated before objective, current state, authority, or requirements are known.
- Invented dates, resources, approvals, dependencies, budgets, or confidence.
- Activity milestones such as 'work ongoing' instead of observable state.
- Green status based on agent self-report, file existence, or partial tests.
- Uncontrolled scope expansion, gold plating, or unrelated cleanup.
- Hiding blockers, failed tests, dissent, or inherited problems as out of scope.
- Delivery without operational handoff, rollback, support, documentation, or acceptance.
- Closure that leaves active credentials, data, infrastructure, obligations, or unsafe state.

## RECOVERY AND RECONSIDERATION

Project recovery reestablishes the last known truthful state, contains consequence, and replans from verified conditions.

- Pause affected work and preserve artifacts, logs, branches, environments, evidence, and in-flight state.
- Reconcile planned state with actual repository, system, resource, and dependency state.
- Identify the first failed requirement, assumption, dependency, authority, or test.
- Rollback or restore where required, then reverify the baseline.
- Replan scope, sequence, resources, gates, and schedule through explicit change authorization.
- Communicate impact and update program, product, stakeholders, operations, and continuity records.

## COMPLETION CRITERIA

- The project objective, scope, authority, deliverables, owners, and acceptance are explicit.
- Tasks, dependencies, resources, risks, changes, and evidence are current and traceable.
- All required deliverables exist and have the declared test, verification, and acceptance state.
- Residual failures, unknowns, risks, and transferred obligations are visible.
- Operational handoff, rollback, support, documentation, and continuity are complete for the declared mode.
- The project is closed, transferred, cancelled, or blocked truthfully rather than merely inactive.

## HANDOFF AND CONTINUITY REQUIREMENTS

- Provide exact deliverables, versions, environment, configuration, requirements, tests, and evidence.
- Transfer ownership, permissions, support, monitoring, recovery, maintenance, and open obligations.
- Preserve assumptions, decisions, changes, rejected alternatives, risks, incidents, and dissent.
- Identify what was not executed or verified and the evidence needed to resolve it.
- Update the receiving program, product, operations, maintenance, or retirement continuity record.
- State next safe action and safe-to-continue status.

## RELATED CODEX SECTIONS

006 Product Strategy
007 Portfolio Management
008 Program Management
010 Product Management
011 Business Analysis
017 Feasibility Analysis
018 Risk Management
019 Requirements Engineering
020 Systems Engineering
107 Software Engineering
228 Quality Management

231 Software Testing
256 Verification
257 Validation
258 Configuration Management
287 Release Engineering
289 Change Management
331 Operational Readiness
402 Post-Implementation Review
403 Postmortem Analysis
404 Lessons Learned

## SOURCE BASIS AND UNRESOLVED SOURCE GAPS

This draft directly applies the supplied Planner and Coordinator role contracts, including objective preservation, task definition, dependency classification, observable milestones, estimation discipline, risk, blockers, change control, validation gates, completion evidence, assignment contracts, conflict transparency, and human authority. It also applies V3's no-fake-success, current-problem, blocker, minimal-effective-change, testing, verification, continuity, and final-report rules; the Owner's Cockpit adds owner-operable status, action, evidence, recovery, release, and rollback; V4 adds obligation compilation and proof-carrying transition.

Unresolved source gap: the supplied corpus does not mandate a specific project methodology, scheduling technique, cost-control system, or contract model. Those choices must remain context-specific, declared, and later reconciled with finance, procurement, quality, release, operations, and audit sections.

# 010

## SECTION 010 - PRODUCT MANAGEMENT

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SECTION 010 — Product Management

## STATUS: CONTENT-COMPLETE

## PURPOSE

Continuously govern product decisions from discovery through delivery, operation, learning, change, deprecation, and retirement so the product remains aligned with mission, strategy, user outcomes, evidence, and real system capability.

Product management is an ongoing decision function, not backlog administration. It connects users, owners, stakeholders, business, domain, requirements, architecture, design, engineering, operations, support, analytics, risk, and lifecycle obligations while preserving who may decide what.

## REQUIRED OUTCOME

- A current product decision system with explicit owner, users, outcomes, principles, capabilities, constraints, and lifecycle state.
- A prioritized and traceable opportunity, problem, capability, and requirement flow rather than an unbounded request queue.
- A roadmap and release logic tied to dependencies, evidence, risk, readiness, and learning.
- Clear definitions of discovery, experiment, planned, committed, implemented, tested, verified, released, degraded, deprecated, and retired states.
- Closed feedback loops across research, usage, support, incidents, operations, market, and stakeholder evidence.
- Bounded human-AI product behavior, authority, correction, transparency, fallback, and recovery where applicable.
- Lifecycle decisions and user obligations preserved through change and retirement.

## WHY THIS MATTERS

A product can drift when the backlog becomes the strategy, the loudest request becomes the priority, or shipping becomes the only definition of progress. AI can accelerate intake and synthesis while also amplifying noise, duplicating requests, inventing demand, and making weak priorities look quantitative.

Governed product management ensures that work earns its place through outcomes, evidence, constraints, and human judgment. It keeps product promises synchronized with actual runtime capability and preserves users' ability to understand and control consequential behavior.

## APPLICABILITY

- A product is being discovered, built, operated, changed, supported, monetized, deprecated, or retired.
- Requests, problems, opportunities, defects, incidents, risks, or strategic bets compete for prioritization.
- Product decisions require coordination across research, design, engineering, security, operations, and business.
- AI or automation materially affects user experience, decisions, data, authority, or trust.
- Product commitments must remain current despite changing evidence, providers, markets, or constraints.

## NON-APPLICABILITY CONDITIONS

- A bounded project or maintenance task creates no product decision and follows an approved product plan.
- The work is portfolio-level resource allocation rather than product-level outcome governance.
- No product owner or decision authority exists.
- The request is only to record a backlog without evaluating evidence, outcomes, dependencies, or lifecycle impact.

## PREREQUISITES AND DEPENDENCIES

- Approved product strategy and product decision owner.
- Research, market, stakeholder, user, domain, feasibility, risk, and business analysis appropriate to the decision.
- Current product, architecture, roadmap, requirement, release, support, incident, analytics, and operational state.
- A governed intake, prioritization, status, evidence, and change-control model.

- Defined user and owner journeys, capability states, and lifecycle obligations.

## REQUIRED INPUTS

- Product strategy, mission, business constraints, product principles, and intended outcomes.
- User research, feedback, usability, accessibility, support, incident, and behavioral evidence.
- Market, competitive, stakeholder, regulatory, legal, security, privacy, safety, and ethical evidence.
- Capability, architecture, engineering, quality, test, release, operational, cost, and capacity state.
- Requests, problems, opportunities, defects, risks, dependencies, and commitments.
- Provider, model, data, platform, distribution, compatibility, migration, and retirement conditions.

## HUMAN RESPONSIBILITIES AND AUTHORITY

The product owner retains authority over priorities, promises, tradeoffs, releases, deprecation, and retirement within higher governance. AI may assist but may not create product legitimacy.

- Confirm product outcomes, principles, prioritization criteria, and mandatory constraints.
- Decide tradeoffs among users, stakeholders, mission, business, risk, accessibility, privacy, safety, and operations.
- Authorize commitments, roadmap changes, consequential automation, releases, and deprecations.
- Correct AI interpretations of user needs, urgency, value, and acceptance.
- Ensure minority, high-risk, low-frequency, accessibility, and recovery needs are not erased by aggregate usage.
- Retain final authority over whether product evidence is sufficient to proceed.

## AI RESPONSIBILITIES AND BOUNDARIES

The AI may classify, link, summarize, compare, forecast, and recommend from current evidence. It may not fabricate users, demand, priority, acceptance, or completed product state.

- Preserve source identity and distinguish requests, observations, hypotheses, commitments, and verified outcomes.
- Deduplicate without erasing materially different contexts, users, or consequences.
- Trace each proposed item to product outcome, evidence, dependencies, risks, and acceptance.
- Expose stale evidence, conflicting findings, hidden prerequisites, and lifecycle effects.
- Avoid ranking solely by volume, engagement, revenue, model confidence, or requester seniority.
- Maintain capability and roadmap status truthfully across external provider and runtime changes.
- Escalate priority and value conflicts to the authorized human.

## APPLICABLE EXISTING AGENT ROLES

Primary roles:

- Analyst — evaluates product evidence, options, prioritization, outcomes, and tradeoffs.
- User Experience Reviewer — evaluates user roles, complete journeys, comprehension, accessibility, agency, and recovery.
- Planner — sequences discovery, delivery, release, migration, and lifecycle work by dependencies and evidence gates.

Common supporting roles:

- Researcher — acquires and evaluates user, market, competitor, domain, and regulatory evidence.
- Coordinator — routes product work across roles and preserves status, evidence, and conflict.
- Architect — identifies capability, interface, data, trust, and technical-dependency implications.
- Observer — captures product behavior, incidents, and user outcomes without unsupported causation.
- Memory Curator — preserves product decisions, rationale, roadmap history, and continuity.

Challenge and assurance roles where consequence warrants:

- Critic — attacks framing, hidden assumptions, omitted alternatives, and unsupported confidence.
- Verifier — checks artifact identity, evidence integrity, acceptance boundaries, and the exact status of claims.
- Governance Reviewer — checks objective fidelity, authority, delegation, policy, evidence, and human decision boundaries.

- Tester — validates product behavior against requirements and complete journeys.
- Security Auditor — evaluates product threat, data, identity, abuse, and provider risks.
- Red Team — attacks prioritization, dark patterns, misuse, deceptive status, and failure recovery.

Role selection must use the smallest sufficient set, preserve chamber independence where review is required, and leave every source-defined role contract unchanged.

## REQUIRED ACTIVITIES

1. Product state maintenance
Maintain authoritative product identity, users, outcomes, principles, capabilities, releases, environments, dependencies, and lifecycle state.

2. Governed intake
Capture source, requester, affected users, problem, evidence, urgency, consequence, desired outcome, and authority for every material item.

3. Discovery and problem validation
Test whether the problem exists, whom it affects, current alternatives, frequency, severity, and conditions before committing to a solution.

4. Opportunity and capability analysis
Translate validated problems into outcome and capability hypotheses without prematurely locking a feature or architecture.

5. Prioritization
Apply explicit criteria, mandatory constraints, evidence strength, dependencies, risk, information gain, cost, and lifecycle effects.

6. Roadmap and commitment control
Separate exploratory options, planned work, approved commitments, and externally communicated promises.

7. Requirements and delivery coordination
Maintain traceability through business analysis, requirements, architecture, design, implementation, testing, verification, and release.

8. Release and readiness decisions
Require exact candidate, environment, evidence, limitations, rollback, support, and owner acceptance.

9. Outcome and feedback learning
Combine user research, product analytics, support, incidents, accessibility, operations, and market evidence without confusing correlation with causation.

10. Lifecycle management
Govern compatibility, migration, maintenance, deprecation, sunset, data, account, service, and retirement obligations.

11. Decision and continuity
Record product decisions, dissent, evidence, affected commitments, and reopening triggers.

## HUMAN-AI INTERACTION REQUIREMENTS

- The owner must see why an item is prioritized, what evidence supports it, and what would change the decision.
- AI-generated summaries must preserve minority, high-consequence, accessibility, and contradictory findings.
- A conversational request must not silently become a committed roadmap item or executed product change.
- Capability labels must reflect actual runtime, permissions, provider state, and verification.
- Users must be told when AI behavior is experimental, degraded, externally dependent, or not verified.
- Feedback collection and personalization must preserve consent, correction, data control, and appeal appropriate to consequence.

## ARCHITECTURE AND SYSTEM IMPLICATIONS

The work must expose architectural obligations without treating a strategy, analysis, or plan as implemented architecture.

- Use a traceability graph from product outcomes and problems through capabilities, requirements, artifacts, releases, evidence, and user journeys.
- Maintain one authoritative capability state and invalidate dependent promises when provider or runtime evidence changes.
- Separate product-management workflow state from execution authority.
- Integrate feature flags, staged release, observability, rollback, feedback, support, and incident linkage.
- Support export, migration, compatibility, deprecation, and retirement from the product lifecycle.
- Protect research, feedback, analytics, and model data through minimization, consent, access, retention, and audit controls.

## OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

The owner-facing operating surface or equivalent durable record should expose:

- Product outcomes, principles, users, capabilities, lifecycle state, and decision owner.
- Intake items with source, evidence, affected parties, consequence, dependencies, and status.
- Prioritization criteria, recommendations, dissent, decisions, and change history.
- Roadmap states separated into hypothesis, planned, approved, committed, delivered, verified, and released.
- Candidate readiness, tests, limitations, incidents, support, rollback, and user acceptance.
- Outcome, feedback, accessibility, trust, security, privacy, reliability, and operational signals.
- Deprecation, migration, notice, data, and retirement controls.

## REQUIRED OUTPUTS AND ARTIFACTS

- Product decision register and product-state record.
- Governed intake and opportunity backlog.
- Prioritization framework and decision rationale.
- Outcome-capability-requirement-roadmap traceability.
- Discovery, experiment, validation, and release plans.
- Product readiness and release decision records.
- Feedback, analytics, support, incident, and outcome synthesis.
- Lifecycle, deprecation, migration, and retirement records.

## EVIDENCE REQUIREMENTS

- Source and context for every material request, finding, commitment, and decision.
- User, market, domain, support, incident, analytics, operational, and technical evidence with dates and limitations.
- Human-approved criteria, priorities, commitments, and risk decisions.
- Exact requirement, artifact, candidate, test, environment, release, and capability evidence.
- Outcome evidence that distinguishes effect from correlation and activity from benefit.
- Notice, migration, data, support, and closure evidence for lifecycle changes.

## VERIFICATION AND ACCEPTANCE TESTS

- Trace a roadmap item from source problem through evidence, capability, requirement, release, and measured outcome.
- Inject duplicate requests from one loud source and confirm volume alone does not dominate priority.
- Remove capability or provider evidence and verify dependent roadmap and UI claims degrade.
- Attempt to convert a conversational request into commitment without approval; it must fail.
- Run critical user journeys for a release through failure, denial, recovery, and accessibility paths.
- Verify a deprecation cannot close until notice, migration, data, support, and obligation criteria are satisfied.

## FAILURE STATES AND ANTI-PATTERNS

- Backlog as strategy and feature count as progress.

- Prioritization by requester volume, hierarchy, revenue, engagement, or model confidence alone.
- Invented user demand, acceptance, urgency, or impact.
- Roadmap promises that outrun requirements, capacity, dependencies, or evidence.
- Shipping treated as outcome, verification, or user value.
- Silent capability drift from provider or model changes.
- Ignoring low-frequency catastrophic, accessibility, privacy, safety, or recovery needs.
- Deprecation or retirement without user control and obligation closure.

## RECOVERY AND RECONSIDERATION

Product-management recovery restores truthful product state and reopens decisions whose evidence, dependencies, or commitments became invalid.

- Pause affected commitments or rollout while preserving safe service and user obligations.
- Reconcile authoritative product, roadmap, capability, release, support, incident, and evidence state.
- Identify the first invalid problem, priority, requirement, provider, capability, or outcome assumption.
- Reprioritize or replan through human authorization and communicate material promise changes.
- Rollback, disable, narrow, redesign, migrate, deprecate, or retire affected capability as appropriate.
- Update users, stakeholders, operations, documentation, and continuity.

## COMPLETION CRITERIA

- The product decision system has an authorized owner, current evidence, and explicit criteria.
- Intake, prioritization, roadmap, requirements, release, outcomes, and lifecycle are traceable.
- Product and AI capability states are truthful and version-bound.
- User agency, accessibility, privacy, safety, security, recovery, and operational obligations are integrated.
- Recommendations remain distinct from commitments and executed changes.
- The product can continue without hidden chat memory or undocumented decision rationale.

## HANDOFF AND CONTINUITY REQUIREMENTS

- Provide each project or delivery team with approved outcome, capability, scope, criteria, dependencies, and evidence.
- Carry forward user, stakeholder, domain, accessibility, security, privacy, safety, and lifecycle constraints.
- Bind commitments to exact roadmap, requirement, release, and acceptance states.
- Preserve rejected, deferred, superseded, and reopened decisions with rationale.
- Transfer operational, support, analytics, feedback, migration, and retirement obligations.
- State next product decision and safe-to-continue status.

## RELATED CODEX SECTIONS

006 Product Strategy
007 Portfolio Management
008 Program Management
009 Project Management
011 Business Analysis
012 Market Research
014 Stakeholder Management
015 User Research
016 Domain Analysis
017 Feasibility Analysis
018 Risk Management
019 Requirements Engineering
075 Human–Computer Interaction
079 UX Strategy
080 UX Research
106 User Validation

183 AI Governance
231 Software Testing
257 Validation
288 Release Management
372 Feedback Management
373 Roadmap Management
389 Deprecation Management
390 Sunset Planning
406 System Retirement

## SOURCE BASIS AND UNRESOLVED SOURCE GAPS

This draft applies the supplied owner-job, journey, trust-calibration, grounded-dialogue, capability-state, evidence, release, rollback, feedback, and lifecycle principles from the Owner's Cockpit; the role corpus adds research, analysis, planning, coordination, observation, UX review, testing, verification, governance, and hostile review; V3 and V4 add objective fidelity, no fake success, claim and obligation structure, evidence contracts, human authority, and proof-carrying transition.

Unresolved source gap: the supplied corpus does not contain a dedicated universal product-management, prioritization, discovery, or roadmap standard. Quantitative prioritization methods and product analytics must later be reconciled with Operations Research, Product Analytics, Experimentation, Feedback Management, Roadmap Management, and lifecycle sections.

# 011

## SECTION 011 - BUSINESS ANALYSIS

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16,959 BYTES / PRESERVED SOURCE IDENTITY

SECTION 011 — Business Analysis

## STATUS: CONTENT-COMPLETE

## PURPOSE

Establish a precise, evidence-grounded understanding of organizational needs, current and target states, actors, capabilities, processes, information, rules, decisions, constraints, and gaps before those needs are converted into requirements, architecture, or implementation.

Business analysis translates among mission, strategy, stakeholders, users, domain, operations, policy, and technology without silently replacing any of them. It identifies what must change and why while preserving the distinction between need, rule, requirement, solution, and decision authority.

## REQUIRED OUTCOME

- A bounded analysis question, scope, decision owner, affected parties, and current-state baseline.
- A stakeholder, actor, capability, process, information, decision, rule, and obligation model.
- A target-state definition expressed as outcomes and capabilities rather than an assumed solution.
- A gap and impact analysis covering people, process, information, technology, policy, risk, and operations.
- Options evaluated consistently with assumptions, dependencies, constraints, feasibility, risk, and reversibility.
- Traceable business needs and rules ready for domain analysis and requirements engineering.
- A decision and continuity package preserving unresolved questions, dissent, and evidence.

## WHY THIS MATTERS

A request often arrives in solution language: build a dashboard, add AI, automate approval, migrate to the cloud. If the underlying need, rule, authority, or failure is not established, the system may implement the request perfectly and still make the organization less capable or less safe.

Business analysis prevents requirements from becoming a transcript of assumptions. It also protects human authority in AI-assisted work by ensuring that model-generated interpretations, process optimizations, and recommendations do not silently become organizational policy or decision rights.

## APPLICABILITY

- A strategic, product, program, project, process, policy, or system change must be understood before design.
- Stakeholder statements conflict or use ambiguous business and domain terminology.
- Current workflows, rules, decisions, data, roles, or pain points are not adequately documented.
- A technology or AI proposal may alter work, authority, accountability, risk, or user outcomes.
- Requirements must be derived from real organizational outcomes and constraints.

## NON-APPLICABILITY CONDITIONS

- The need, rules, current state, target state, and requirements are already authoritative, current, and unaffected by the task.
- The assignment is a bounded implementation or verification action with no business interpretation.
- The analyst lacks access or authority to examine the affected organization, process, policy, or evidence.
- The work is purely market or user research without an organizational decision question.

## PREREQUISITES AND DEPENDENCIES

- An authorized objective, analysis question, scope, decision owner, and intended downstream use.
- Relevant research, strategy, stakeholder, user, market, domain, and current-state sources.
- Access to authoritative policies, procedures, records, systems, data definitions, roles, and operational evidence.
- A method to preserve terminology, assumptions, decisions, conflicts, and source traceability.
- Privacy, consent, security, labor, legal, and access controls for organizational information.

## REQUIRED INPUTS

- Mission, strategy, product, program, project, policy, and governance documents.
- Stakeholder and user needs, jobs, journeys, pain, workarounds, decisions, and recovery behavior.
- Current processes, roles, capabilities, systems, data, reports, controls, incidents, and performance evidence.
- Business rules, regulations, contracts, service obligations, approvals, exceptions, and escalation paths.
- Proposed target outcomes, constraints, resources, dependencies, and solution ideas.
- Known assumptions, risks, conflicts, and evidence gaps.

## HUMAN RESPONSIBILITIES AND AUTHORITY

The authorized human confirms organizational intent, policy, values, decision rights, and acceptable change. The analyst does not become the business owner.

- Define the decision or change the analysis must support.
- Identify authoritative owners for processes, rules, data, policy, and target outcomes.
- Correct terminology, undocumented practice, exception handling, and political or cultural context.
- Resolve value conflicts and approve material target-state or policy decisions.
- Authorize access to sensitive operational, personnel, customer, financial, or regulated information.
- Decide whether evidence is sufficient to convert a need into a requirement or commitment.

## AI RESPONSIBILITIES AND BOUNDARIES

The AI may structure, compare, model, and expose gaps. It may not invent business rules, authority, stakeholder consensus, data meaning, policy, or target-state approval.

- Preserve source wording and distinguish stated need, observed practice, inferred rule, policy, recommendation, and approved requirement.
- Identify contradictions, overloaded terms, exception paths, hidden approvals, and undocumented dependencies.
- Model current and target states without assuming the proposed solution is necessary.
- Separate measured performance from estimates and complaints from demonstrated root cause.
- Expose which conclusions depend on incomplete stakeholder access or stale organizational records.
- Avoid optimizing a process by removing controls, human judgment, accessibility, recovery, or legitimate exceptions.
- Route unresolved policy, value, and authority questions to the human owner.

## APPLICABLE EXISTING AGENT ROLES

Primary roles:

- Analyst — structures current state, target state, gaps, options, impacts, and decision implications.
- Researcher — verifies organizational, regulatory, market, policy, and source claims.
- Observer — records actual workflows, states, events, handoffs, and exceptions without unsupported causation.

Common supporting roles:

- Architect — identifies business capabilities, boundaries, interfaces, authority flows, and structural implications.
- User Experience Reviewer — examines human tasks, journeys, language, accessibility, agency, and recovery.
- Planner — sequences analysis, decisions, dependencies, validation, and handoff.
- Memory Curator — preserves canonical terminology, business rules, decisions, and change history.
- Decision Arbiter Advisory — compares options under human-authorized criteria.

Challenge and assurance roles where consequence warrants:

- Critic — attacks framing, hidden assumptions, omitted alternatives, and unsupported confidence.
- Verifier — checks artifact identity, evidence integrity, acceptance boundaries, and the exact status of claims.
- Governance Reviewer — checks objective fidelity, authority, delegation, policy, evidence, and human decision boundaries.
- Security Auditor — examines sensitive business information, access, fraud, control, and process-abuse risks.
- Red Team — attacks assumptions, process gaming, hidden incentives, bypasses, and harmful optimization.

Role selection must use the smallest sufficient set, preserve chamber independence where review is required, and leave every source-defined role contract unchanged.

## REQUIRED ACTIVITIES

1. Analysis framing
Record question, objective, scope, exclusions, decision owner, affected parties, constraints, and required confidence.

2. Source and terminology inventory
Identify authoritative documents, systems, data, owners, definitions, versions, dates, and conflicting terms.

3. Stakeholder and actor analysis
Map owners, users, operators, approvers, affected non-users, external parties, agents, responsibilities, authority, and interests.

4. Current-state analysis
Document actual workflows, capabilities, information, decisions, controls, handoffs, workarounds, exceptions, failures, and recovery.

5. Need and outcome analysis
Separate root need, desired outcome, symptoms, requests, constraints, and solution ideas.

6. Business-rule analysis
Record rule source, owner, condition, action, exception, priority, jurisdiction, effective date, and enforcement state.

7. Capability and gap analysis
Compare current capabilities to target outcomes and identify missing, weak, duplicated, or misaligned capabilities.

8. Option and impact analysis
Evaluate process, policy, organizational, manual, technical, AI, staged, and no-action options consistently.

9. Data and decision analysis
Identify required information, quality, provenance, timing, authority, decision logic, explainability, and appeal.

10. Risk and change analysis
Assess operational, human, legal, security, privacy, safety, accessibility, financial, and cultural impacts.

11. Validation
Review findings with authoritative owners and representative users; keep disagreement and unsupported assumptions visible.

12. Handoff
Produce traceable needs, rules, models, gaps, options, and decisions ready for domain analysis and requirements engineering.

## HUMAN-AI INTERACTION REQUIREMENTS

- AI analysis must distinguish what stakeholders said, what was observed, what was inferred, and what was approved.
- The human must be able to correct terminology and hidden organizational context without losing prior versions.
- The system must not treat management title, requester volume, or model confidence as proof of authority or need.
- AI-proposed process changes must show human, control, accessibility, exception, and recovery impacts.
- Conversational agreement must not become policy or requirement without durable validation and approval.
- Where AI is a process actor, capability, permission, decision role, review, fallback, and accountability must be modeled separately.

## ARCHITECTURE AND SYSTEM IMPLICATIONS

The work must expose architectural obligations without treating a strategy, analysis, or plan as implemented architecture.

- Business capabilities, actors, processes, decisions, information, rules, and obligations should receive stable identities for traceability.
- Separate policy source, business rule, decision logic, workflow orchestration, UI guidance, and enforcement boundary.
- Represent normal, exception, denial, escalation, recovery, and termination paths.
- Keep capability and authority separate for human and AI actors.
- Identify source-of-truth systems, data lineage, freshness, quality, retention, and jurisdiction.
- Map business outcomes and rules to requirements, architecture, tests, evidence, and owner-facing status.

## OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

The owner-facing operating surface or equivalent durable record should expose:

- Analysis question, scope, owner, affected parties, sources, and current evidence.
- Current and target states, capabilities, processes, rules, decisions, and gaps.
- Assumptions, contradictions, exceptions, risks, unknowns, and human decisions required.
- Options, criteria, impacts, recommendations, approvals, and rejected alternatives.
- Traceability from need and rule to requirement, design, implementation, and validation.
- Source freshness, stakeholder validation, and unresolved dissent.
- Change and reopening triggers.

## REQUIRED OUTPUTS AND ARTIFACTS

- Business analysis plan and question statement.
- Source, terminology, stakeholder, actor, responsibility, and authority maps.
- Current-state and target-state models.
- Process, capability, information, decision, and business-rule models.
- Gap, option, impact, feasibility, and risk analyses.
- Validated needs, outcomes, constraints, assumptions, and decision record.
- Requirements-ready traceability and continuity package.

## EVIDENCE REQUIREMENTS

- Authoritative source identity, owner, version, date, jurisdiction, and applicability.
- Observed workflow, system, data, incident, performance, and exception evidence.
- Stakeholder and user findings with method, context, and limitations.
- Human approval for target outcomes, policy, rules, priorities, and value tradeoffs.
- Evidence supporting root-cause, gap, feasibility, and impact conclusions.
- Traceability from each material need and rule to source and validation state.

## VERIFICATION AND ACCEPTANCE TESTS

- Trace a proposed requirement back to an approved need or rule and authoritative source.
- Present the same term to different stakeholders and verify ambiguity is detected and resolved or retained.
- Run current-state models through normal, exception, denial, failure, and recovery cases.
- Remove a proposed technology and confirm the underlying need and target outcome remain meaningful.
- Attempt to convert an inferred rule into a requirement without owner approval; it must be blocked.
- Verify AI capability does not imply business decision authority.
- Have authoritative owners and representative users validate the models and record disagreements.

## FAILURE STATES AND ANTI-PATTERNS

- Solution-first analysis that merely justifies a preselected technology.
- Stakeholder statements treated as facts, policy, consensus, or requirements without validation.
- Business rules invented from current software behavior or AI inference.
- Happy-path process maps that omit exceptions, denial, recovery, and escalation.
- Optimizing efficiency by removing legitimate human judgment, control, accessibility, or safety.

- Confusing capability, responsibility, permission, accountability, and authority.
- Documentation that describes intended process while actual operations differ.
- Handoff that loses source, terminology, assumptions, dissent, or decision ownership.

## RECOVERY AND RECONSIDERATION

Business-analysis recovery returns to authoritative need, source, and actual current state when models or conclusions prove wrong.

- Pause dependent requirement or design decisions affected by the invalid analysis.
- Identify which source, term, observation, rule, process, assumption, or authority was wrong or changed.
- Reobserve actual work and reconcile documentation with system and human practice.
- Revalidate with authoritative owners and affected users.
- Revise target state, options, impacts, and requirements through change control.
- Propagate invalidation and updated evidence to downstream sections and continuity records.

## COMPLETION CRITERIA

- The analysis question, scope, owner, sources, and affected parties are explicit.
- Current and target states, needs, capabilities, processes, rules, information, decisions, and gaps are traceable.
- Facts, observations, inferences, assumptions, recommendations, and approvals remain distinct.
- Normal, exception, failure, denial, recovery, and human-AI authority paths are represented where applicable.
- Options and impacts are evaluated without solution substitution.
- Validated outputs are ready for Domain Analysis and Requirements Engineering with unresolved conditions visible.

## HANDOFF AND CONTINUITY REQUIREMENTS

- Transfer exact source identities, terminology, models, needs, rules, constraints, and validation state.
- Carry forward actor authority, exceptions, risks, accessibility, privacy, security, and operational impacts.
- Map each need and rule to intended domain or requirement work.
- Preserve disagreements, rejected options, assumptions, unknowns, and evidence needed to resolve them.
- Identify invalidation triggers and owners for future changes.
- State whether it is safe to proceed to domain analysis or requirements.

## RELATED CODEX SECTIONS

001 Research
004 Mission Strategy
005 Business Strategy
006 Product Strategy
009 Project Management
010 Product Management
012 Market Research
014 Stakeholder Management
015 User Research
016 Domain Analysis
017 Feasibility Analysis
018 Risk Management
019 Requirements Engineering
024 Domain Architecture
025 Business Architecture
026 Capability Architecture
027 Process Architecture
154 Data Modeling
176 Ontology Engineering
221 Policy
256 Verification

257 Validation

## SOURCE BASIS AND UNRESOLVED SOURCE GAPS

This draft applies the supplied Researcher and Analyst disciplines for exact questions, source quality, definitions, current baselines, consistent comparison, alternative explanations, uncertainty, and advisory recommendations. It applies the Architect's requirement for responsibilities, interfaces, authority flows, trust boundaries, failure modes, and open decisions; the Owner's Cockpit adds owner jobs, language, journeys, evidence, recovery, and decision support without seizure; V3 and V4 add objective fidelity, claim typing, obligation mapping, evidence, dissent, and continuity.

Unresolved source gap: the supplied corpus does not mandate a particular business-analysis notation or methodology. BPMN, decision models, capability maps, value streams, use cases, and other techniques must be selected based on context and later reconciled with Domain Analysis, Requirements Engineering, Process Architecture, Data Modeling, and Policy.

# 012

## SECTION 012 - MARKET RESEARCH

CODEX/03_SECTIONS/012_MARKET_RESEARCH.txt

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16,928 BYTES / PRESERVED SOURCE IDENTITY

SECTION 012 — Market Research

## STATUS: CONTENT-COMPLETE

## PURPOSE

Establish a truthful, current, and methodologically bounded understanding of external demand, populations, segments, behaviors, alternatives, channels, pricing or funding conditions, adoption barriers, and market structure for a specific decision.

Market research does not manufacture a market narrative. It defines who or what was studied, how evidence was obtained, what the evidence supports, which populations were not represented, and how quickly findings may become stale.

## REQUIRED OUTCOME

- A decision-linked market research question, scope, target population, geography, jurisdiction, and timeframe.
- A research design with source hierarchy, sampling, inclusion, exclusion, collection, analysis, privacy, consent, and quality controls.
- A current market and alternative baseline with definitions that prevent category inflation.
- Segment, need, behavior, adoption, channel, price or funding, and barrier findings bounded by evidence.
- Conflicting evidence, sampling limitations, bias, uncertainty, and time sensitivity made visible.
- A conclusion and recommendation whose confidence does not exceed the method.
- A reproducible evidence and continuity package for later strategy and product decisions.

## WHY THIS MATTERS

Market claims are unusually easy to inflate. Search snippets, vendor forecasts, social media activity, synthetic respondents, and model-generated summaries can create the appearance of broad demand without representative evidence.

Governed market research prevents strategic commitments from resting on fictional populations or stale categories. It also protects participants and users by making consent, privacy, inclusion, accessibility, and data provenance part of the research method rather than cleanup after collection.

## APPLICABILITY

- A business, product, portfolio, investment, market-entry, pricing, channel, or positioning decision depends on external evidence.
- The size, characteristics, needs, behavior, willingness, adoption, or constraints of a population are uncertain.
- Market conditions, regulations, competitors, substitutes, or technologies may have changed.
- A product may serve new geographies, cultures, languages, accessibility contexts, or user groups.
- Claims of demand, growth, pricing, market share, or adoption will influence material commitments.

## NON-APPLICABILITY CONDITIONS

- The question concerns a single known user's workflow and belongs to User Research.
- The question concerns internal processes and belongs to Business Analysis.
- The request is merely to collect promotional statistics without a defined decision or method.
- Lawful access, participant protection, or sufficient source quality cannot be established.
- Synthetic or model-generated responses are the only evidence but the claim concerns real market behavior.

## PREREQUISITES AND DEPENDENCIES

- An authorized decision question, population, scope, horizon, criteria, and research owner.
- Definitions for market, category, customer, user, buyer, beneficiary, competitor, substitute, adoption, and demand as applicable.
- Privacy, consent, accessibility, security, legal, licensing, and jurisdictional review for collection and use.

- A source and data-management plan preserving identity, date, version, method, and limitations.
- A distinction between primary research, secondary research, observed behavior, stated preference, forecast, and simulation.

## REQUIRED INPUTS

- Mission, business, product, portfolio, pricing, channel, and investment questions.
- Existing market studies, primary data, public records, industry sources, surveys, interviews, observations, transactions, support, and product evidence.
- Population and segment definitions, geographies, jurisdictions, languages, accessibility needs, and time windows.
- Competitor, substitute, distribution, regulatory, economic, technological, and cultural context.
- Collection tools, sampling frame, budget, schedule, participant access, and data restrictions.
- Known biases, conflicts of interest, missing populations, and evidence gaps.

## HUMAN RESPONSIBILITIES AND AUTHORITY

The human research owner defines the decision, acceptable method, protected populations, budget, and whether evidence is sufficient for commitment.

- Approve the question, population, method, participant protections, and use of findings.
- Resolve value choices about inclusion, segmentation, acceptable uncertainty, and affected populations.
- Authorize access to proprietary, purchased, personal, regulated, or sensitive data.
- Correct category definitions, cultural assumptions, and market interpretations.
- Reject conclusions that overgeneralize beyond sample, geography, timeframe, or method.
- Decide whether the evidence supports experiment, entry, pricing, investment, or no action.

## AI RESPONSIBILITIES AND BOUNDARIES

The AI may search authorized sources, structure data, analyze patterns, and propose interpretations. It may not fabricate participants, responses, citations, sample representativeness, demand, market size, or willingness to pay.

- Record exact sources, dates, versions, methods, populations, and access boundaries.
- Separate primary from secondary evidence and observed behavior from stated intent.
- Identify sponsor bias, vendor methodology, circular citation, category expansion, and stale data.
- Disclose sampling, measurement, nonresponse, coverage, survivorship, selection, and model limitations.
- Treat synthetic data and AI-generated respondents as simulations, never observations of real people.
- Avoid scraping or combining data in ways that violate law, terms, consent, privacy, or reasonable expectations.
- Present minority and contradictory findings rather than smoothing them into false consensus.

## APPLICABLE EXISTING AGENT ROLES

Primary roles:

- Researcher — designs and executes source-grounded market inquiry and evaluates source quality.
- Analyst — performs segmentation, comparison, uncertainty, sensitivity, and decision analysis.
- Observer — captures authorized market or behavioral observations with provenance and limitations.

Common supporting roles:

- Planner — sequences collection, sampling, analysis, review, and decision gates.
- User Experience Reviewer — checks inclusion, research interaction, comprehension, accessibility, and participant burden.
- Memory Curator — preserves datasets, definitions, source state, decisions, and expiration.
- Decision Arbiter Advisory — compares market options under authorized criteria.

Challenge and assurance roles where consequence warrants:

- Critic — attacks framing, hidden assumptions, omitted alternatives, and unsupported confidence.
- Verifier — checks artifact identity, evidence integrity, acceptance boundaries, and the exact status of claims.

- Governance Reviewer — checks objective fidelity, authority, delegation, policy, evidence, and human decision boundaries.
- Security Auditor — reviews data acquisition, storage, access, reidentification, provider, and leakage risk.
- Red Team — attacks sampling, category, sponsor, confirmation, survivorship, synthetic-evidence, and market-size assumptions.

Role selection must use the smallest sufficient set, preserve chamber independence where review is required, and leave every source-defined role contract unchanged.

## REQUIRED ACTIVITIES

1. Decision and question definition
State what decision the research supports, target population, unit of analysis, geography, jurisdiction, timeframe, and required confidence.

2. Definition and category control
Define market boundaries, customer, user, buyer, substitute, adoption, demand, and other terms before estimating them.

3. Method design
Choose primary and secondary methods, sampling, inclusion, exclusion, collection, consent, accessibility, privacy, and analysis.

4. Source hierarchy
Prefer direct, transparent, current evidence; identify secondary citation chains, sponsor interests, and inaccessible methodology.

5. Sampling and coverage
Define sampling frame, sample size basis, recruitment, nonresponse, weighting, missing populations, and generalization limits.

6. Data collection
Collect only authorized data, preserve provenance, minimize sensitive information, and document deviations.

7. Quality and bias assessment
Check validity, reliability, representativeness, measurement error, duplicate records, bot or synthetic contamination, and conflicts.

8. Market and alternative analysis
Examine needs, behaviors, segments, channels, price or funding, barriers, competitors, substitutes, regulation, and change drivers.

9. Uncertainty and sensitivity
Use ranges and scenarios; show how category, source, sample, time, or assumption changes affect conclusions.

10. Triangulation
Compare multiple methods and evidence types without treating quantity as quality.

11. Decision synthesis
State findings, limitations, confidence, implications, recommendations, and evidence that would change them.

12. Expiration and continuity
Assign freshness, review triggers, dataset controls, and a reproducible handoff.

## HUMAN-AI INTERACTION REQUIREMENTS

- Participants must know who is collecting data, why, what will happen to it, and how to decline or withdraw where applicable.
- AI identity and use in recruitment, interviewing, transcription, analysis, or generation must be disclosed when material to consent or interpretation.
- The AI must not impersonate a human researcher or create synthetic participant quotations.

- The human must be able to inspect source excerpts, methods, sample limits, and contradictory evidence behind a market claim.
- Research interfaces must support accessibility, language, cultural context, and correction.
- Operational recommendations must remain distinct from research findings and human decisions.

## ARCHITECTURE AND SYSTEM IMPLICATIONS

The work must expose architectural obligations without treating a strategy, analysis, or plan as implemented architecture.

- Use a research data model preserving source, participant or record identity controls, method, consent, purpose, retention, jurisdiction, and access.
- Separate raw data, cleaned data, derived variables, synthetic data, model outputs, and published findings.
- Protect against reidentification, leakage, cross-purpose reuse, unauthorized enrichment, and provider retention.
- Version definitions, sampling frames, instruments, code, weights, and analysis outputs.
- Support reproducibility and expiration of time-sensitive conclusions.
- Ensure market findings cannot directly trigger consequential product or commercial actions without downstream authority.

## OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

The owner-facing operating surface or equivalent durable record should expose:

- Research question, population, geography, timeframe, decision owner, method, and current status.
- Sources, sample, recruitment, consent, inclusion, exclusions, coverage, and missing populations.
- Raw, cleaned, derived, synthetic, and inferred evidence states.
- Findings, uncertainty, conflicts, bias, freshness, and generalization limits.
- Data access, privacy, retention, deletion, and incident state.
- Recommendations, human decisions required, expiration, and next research action.

## REQUIRED OUTPUTS AND ARTIFACTS

- Market research brief and method plan.
- Definitions, population, sampling, source, and data-management records.
- Research instruments and participant-protection materials.
- Market, segment, need, behavior, price or funding, channel, and barrier analysis.
- Source-quality, bias, uncertainty, and limitation assessment.
- Decision synthesis and recommendation.
- Dataset, analysis, evidence, expiration, and continuity package.

## EVIDENCE REQUIREMENTS

- Exact source identity, date, version, publisher, methodology, population, sponsor, and limitations.
- Sampling frame, recruitment, sample size, inclusion, exclusion, nonresponse, weighting, and missing-data records.
- Participant consent, privacy, accessibility, and data-use records where applicable.
- Raw and transformed data provenance, code or procedure, and reproducibility evidence.
- Evidence distinguishing observed behavior, stated preference, forecast, inference, and simulation.
- Contradictory findings, uncertainty intervals or ranges, and expiration conditions.

## VERIFICATION AND ACCEPTANCE TESTS

- Trace every material market claim to a current source and bounded population.
- Change the category definition or remove a dominant source and inspect sensitivity.
- Attempt to include synthetic responses as real observations; the evidence system must reject or relabel them.
- Check whether excluded or missing populations could materially change the decision.
- Reproduce a key analysis from retained data, method, code, and definitions.
- Expire a time-sensitive source and verify dependent conclusions degrade to stale or unknown.
- Confirm no participant or protected data was collected, retained, or reused outside authorization.

## FAILURE STATES AND ANTI-PATTERNS

- Market-size theater using undefined categories and vendor forecasts.
- Search snippets, social activity, model output, or source count treated as demand evidence.
- Synthetic respondents, fabricated quotations, or invented surveys.
- Convenience samples presented as representative populations.
- Stated preference treated as observed purchase or adoption behavior.
- Circular citation, stale data, sponsor bias, and hidden methodology.
- Unlawful or nonconsensual scraping, enrichment, reidentification, or data reuse.
- Overgeneralization across geography, culture, jurisdiction, accessibility, or timeframe.

## RECOVERY AND RECONSIDERATION

Market-research recovery withdraws unsupported claims, protects participants, and rebuilds the evidence chain from the first invalid source or method.

- Stop use of affected data or conclusions and preserve evidence of the defect.
- Contain privacy, consent, security, or licensing incidents and follow required notification or deletion procedures.
- Identify invalid definitions, samples, sources, transformations, analyses, or generalizations.
- Recollect, reweight, reanalyze, narrow the claim, or mark the question unresolved.
- Propagate invalidation to strategy, product, portfolio, pricing, and roadmap decisions.
- Update source, dataset, decision, and continuity records with the corrected boundary.

## COMPLETION CRITERIA

- The decision question, population, method, sources, sample, timeframe, and authority are explicit.
- Data collection and use satisfy consent, privacy, security, accessibility, legal, and licensing requirements.
- Findings are traceable and bounded by sample, method, geography, jurisdiction, and time.
- Synthetic, inferred, forecast, and observed evidence remain distinct.
- Bias, conflicts, limitations, uncertainty, dissent, and expiration are visible.
- The conclusion and recommendation do not exceed the evidence and are ready for downstream decision review.

## HANDOFF AND CONTINUITY REQUIREMENTS

- Provide exact research question, definitions, population, method, sources, sample, data, and analysis version.
- Carry forward limitations, missing populations, bias, uncertainty, freshness, and expiration.
- Map findings to the strategy, product, stakeholder, feasibility, or pricing decision they support.
- Preserve participant, privacy, licensing, retention, and deletion obligations.
- Identify what evidence would change the conclusion and when research must reopen.
- State whether downstream use is safe within the declared limits.

## RELATED CODEX SECTIONS

001 Research
002 Foresight / Futures
003 Innovation
005 Business Strategy
006 Product Strategy
010 Product Management
011 Business Analysis
013 Competitive Intelligence
014 Stakeholder Management
015 User Research
016 Domain Analysis
017 Feasibility Analysis
163 Data Analysis

164 Data Science
165 Statistics
218 Privacy Engineering
219 Legal
220 Regulatory Compliance
369 Experimentation
370 A/B Testing
405 Lifecycle Governance

## SOURCE BASIS AND UNRESOLVED SOURCE GAPS

This draft applies the supplied Researcher discipline for primary sources, dates, versions, source authority, conflicting evidence, terminology, uncertainty, and traceable citations; the Analyst discipline adds baselines, sampling limitations, alternative explanations, sensitivity, and proportionate conclusions; the Observer role adds provenance, uncertainty, absence-versus-nonobservation, privacy, and integrity; V3, the Owner's Cockpit, and V4 add consent, data protection, evidence, human authority, and continuity.

Unresolved source gap: the supplied corpus does not contain a dedicated survey, sampling, econometric, market-sizing, or pricing-research standard. Methods must later be reconciled with Statistics, Data Science, Experimentation, Privacy Engineering, Legal, Regulatory Compliance, and Financial Operations.

# 013

## SECTION 013 - COMPETITIVE INTELLIGENCE

CODEX/03_SECTIONS/013_COMPETITIVE_INTELLIGENCE.txt

SHA-256 fadf23d5d8647be0b7dfd742d3b1a0c548c8333d13330cae80b9c94cee6492d8

16,747 BYTES / PRESERVED SOURCE IDENTITY

SECTION 013 — Competitive Intelligence

## STATUS: CONTENT-COMPLETE

## PURPOSE

Produce lawful, ethical, current, and decision-relevant understanding of competitors, substitutes, adjacent actors, ecosystem forces, and alternative approaches without misrepresenting public claims as verified capability or crossing security, privacy, contractual, or trade-secret boundaries.

Competitive intelligence supports strategy by revealing differences, dependencies, likely responses, and evidence gaps. It is not espionage, imitation, marketing-summary collection, or permission to define the product through competitors.

## REQUIRED OUTCOME

- A bounded intelligence question, scope, named subjects, timeframe, jurisdiction, authority, and collection rules.
- A source-grounded profile of competitors, substitutes, alternatives, ecosystems, and relevant capabilities.
- A comparison framework using consistent criteria and explicit evidence states.
- Clear separation among claimed, documented, demonstrated, independently verified, inferred, and unknown capability.
- Analysis of positioning, business model, technology, product, operations, partnerships, risks, likely responses, and strategic implications where relevant.
- Legal, ethical, security, privacy, licensing, and source-handling compliance.
- A recommendation and monitoring plan that does not turn comparison into imitation.

## WHY THIS MATTERS

Competitive decisions are often made from websites, demos, press releases, analyst reports, rumors, and benchmark claims that describe different versions, environments, or criteria. AI can aggregate these sources quickly while also blending marketing language with inference and reproducing unverified claims as fact.

Governed intelligence protects strategy from both complacency and reactive imitation. It identifies what is actually known, what matters to the mission and user, and which competitor behavior should trigger review without granting competitors control over product direction.

## APPLICABILITY

- Strategy, positioning, product, partnership, procurement, investment, pricing, or risk decisions depend on external actors or alternatives.
- A competitor or substitute may materially change user expectations, market structure, technology, regulation, supply, or security.
- Claims of differentiation, parity, advantage, or threat require evidence.
- The ecosystem includes providers, platforms, models, standards, distributors, integrators, open-source projects, or adjacent entrants.
- Continuous monitoring is required for defined high-impact signals.

## NON-APPLICABILITY CONDITIONS

- The purpose is covert access, impersonation, social engineering, unauthorized testing, theft, or acquisition of protected information.
- The decision can be made from user, mission, and feasibility evidence without competitor analysis.
- The request is to copy a competitor feature or message without establishing product need.
- Source legality, license, authorization, or integrity cannot be established.
- The named subject is a private individual rather than a legitimate organizational or product intelligence target.

## PREREQUISITES AND DEPENDENCIES

- An authorized intelligence question, decision owner, collection boundary, jurisdiction, and prohibited methods.
- Mission, product, business, or procurement criteria defining why the comparison matters.
- A source hierarchy and evidence-state model.
- Legal, security, privacy, contractual, licensing, and ethical constraints.
- A monitoring and retention plan for time-sensitive intelligence.

## REQUIRED INPUTS

- Public filings, official documentation, product pages, terms, pricing, release notes, standards, repositories, demonstrations, support materials, and published research.
- Independent tests, reviews, customer evidence, incidents, legal actions, procurement evidence, and technical analysis where lawfully available.
- Competitor, substitute, ecosystem, provider, partner, and channel definitions.
- Human-authorized comparison criteria, decision context, horizon, and risk tolerance.
- Current internal capability and product evidence for like-for-like comparison.
- Known rumors, disputed claims, missing evidence, and monitoring triggers.

## HUMAN RESPONSIBILITIES AND AUTHORITY

The authorized human defines legitimate targets, collection methods, comparison criteria, and strategic response. Competitive pressure does not create permission.

- Approve the intelligence question, subjects, methods, source use, and legal or ethical boundaries.
- Define which differences matter to mission, users, business, risk, or procurement.
- Resolve tradeoffs and decide whether to respond, monitor, partner, differentiate, or take no action.
- Reject pressure to obtain protected information or conduct unauthorized access or testing.
- Correct internal capability evidence and prevent unfair or misleading comparison.
- Authorize any external communication, claim, negotiation, or response based on intelligence.

## AI RESPONSIBILITIES AND BOUNDARIES

The AI may collect authorized public information, normalize evidence, compare, and monitor. It may not impersonate, infiltrate, solicit secrets, bypass controls, fabricate sources, or present inference as observed competitor fact.

- Record exact source, publisher, date, version, product tier, environment, jurisdiction, and access method.
- Distinguish official claim, documentation, demonstration, independent test, incident, inference, rumor, and unknown.
- Compare equivalent scopes and disclose where configurations, populations, dates, or methods differ.
- Avoid copying competitor language, code, data, design, or protected material beyond lawful and licensed use.
- Do not autonomously contact employees, customers, partners, or systems to obtain intelligence.
- Identify confirmation bias, halo effects, category mismatch, benchmark gaming, and outdated evidence.
- Preserve internal mission and user criteria rather than recommending imitation by default.

## APPLICABLE EXISTING AGENT ROLES

Primary roles:

- Researcher — collects and assesses lawful, traceable, current competitive sources.
- Analyst — performs consistent comparison, alternative explanation, scenario, and strategic implication analysis.
- Security Auditor — defines and reviews collection boundaries, sensitive handling, and prohibited access.

Common supporting roles:

- Architect — compares structural approaches, interfaces, dependencies, and operational implications without assuming implementation parity.
- User Experience Reviewer — compares complete journeys and user control rather than screenshot similarity.
- Observer — records defined public signals and changes with timestamps and provenance.
- Memory Curator — preserves source state, monitoring history, decisions, and superseded findings.

- Decision Arbiter Advisory — structures strategic response under authorized criteria.

Challenge and assurance roles where consequence warrants:

- Critic — attacks framing, hidden assumptions, omitted alternatives, and unsupported confidence.
- Verifier — checks artifact identity, evidence integrity, acceptance boundaries, and the exact status of claims.
- Governance Reviewer — checks objective fidelity, authority, delegation, policy, evidence, and human decision boundaries.
- Red Team — attacks source reliability, deceptive claims, comparison bias, likely competitor response, and internal overreaction.

Role selection must use the smallest sufficient set, preserve chamber independence where review is required, and leave every source-defined role contract unchanged.

## REQUIRED ACTIVITIES

1. Question and boundary
Define decision, subjects, timeframe, geography, jurisdiction, collection methods, protected information, prohibited actions, and authority.

2. Subject and alternative map
Identify direct competitors, substitutes, internal alternatives, open-source options, providers, partners, and likely entrants.

3. Source inventory
Collect lawful sources with identity, date, version, scope, method, sponsor, access, and limitations.

4. Capability-state classification
Classify each claim as claimed, documented, demonstrated, independently tested, observed in production, inferred, disputed, or unknown.

5. Comparison design
Use human-authorized criteria and comparable units, tiers, environments, workflows, and timeframes.

6. Product and journey analysis
Compare user outcomes, complete journeys, accessibility, agency, trust, failure, support, and recovery, not surface resemblance alone.

7. Technical and operational analysis
Examine architecture, dependencies, performance, reliability, security, privacy, deployment, support, and lifecycle only to the extent evidence allows.

8. Business and ecosystem analysis
Assess positioning, channels, pricing, partnerships, lock-in, regulatory posture, supply, and incentives.

9. Scenario and response analysis
Consider likely moves, market change, standards, provider shifts, incidents, and no-response options.

10. Adversarial review
Seek contradictory evidence, benchmark limitations, hidden costs, missing tiers, stale claims, and internal bias.

11. Decision synthesis
Provide implications, options, residual uncertainty, recommendation, rejected responses, and evidence that would change it.

12. Monitoring
Define lawful signals, source cadence, expiration, alert thresholds, and review authority.

## HUMAN-AI INTERACTION REQUIREMENTS

- AI collection must remain within authorized public or licensed sources and must expose the access method.
- The AI must not impersonate a person, customer, applicant, researcher, or employee to obtain information.
- The human must be able to inspect original source context behind every material comparison.

- Marketing claims and model-generated summaries must be visibly distinct from independent evidence.
- Comparative recommendations must use the organization's authorized mission and product criteria, not admiration or fear.
- Any external action based on intelligence remains a separate governed decision.

## ARCHITECTURE AND SYSTEM IMPLICATIONS

The work must expose architectural obligations without treating a strategy, analysis, or plan as implemented architecture.

- Maintain source, subject, product, version, tier, environment, criterion, claim, and evidence identities.
- Keep competitor intelligence data isolated according to legal, licensing, confidentiality, and retention constraints.
- Support time-based invalidation and monitoring without silently overwriting historical findings.
- Separate intelligence collection from security testing, penetration testing, procurement, and external communication authority.
- Represent internal and external capability states using the same evidence discipline.
- Prevent unverified competitor claims from entering product requirements or marketing as facts.

## OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

The owner-facing operating surface or equivalent durable record should expose:

- Intelligence question, subjects, decision owner, collection authority, methods, and prohibited actions.
- Source inventory, dates, versions, tiers, access, evidence states, and freshness.
- Comparison criteria, results, mismatches, uncertainty, and dissent.
- Signals, changes, alerts, monitoring cadence, and expiration.
- Strategic options, recommendations, human decisions, and rejected responses.
- Legal, privacy, security, licensing, and confidentiality status.

## REQUIRED OUTPUTS AND ARTIFACTS

- Competitive-intelligence charter and collection plan.
- Competitor, substitute, ecosystem, and signal map.
- Source and capability-evidence register.
- Comparable product, journey, technical, operational, and business analysis.
- Scenario, threat, opportunity, and response analysis.
- Decision synthesis and monitoring plan.
- Legal and ethical compliance record and continuity package.

## EVIDENCE REQUIREMENTS

- Exact source, publisher, date, version, tier, configuration, environment, and access method.
- Evidence state for every competitor and internal capability claim.
- Independent test method and comparability where performance or quality is compared.
- Human-approved criteria and strategic response decisions.
- Legal, licensing, privacy, security, and collection-authorization records.
- Contradictory evidence, missing information, expiration, and monitoring history.

## VERIFICATION AND ACCEPTANCE TESTS

- Trace each competitive claim to an original examined source and correct evidence state.
- Change version, tier, environment, or comparison criterion and verify results update or become incomparable.
- Remove marketing sources and inspect whether claimed differentiation remains supported.
- Attempt an unauthorized access, impersonation, or contact step; the workflow must block it.
- Compare an internal prototype to a competitor production claim and verify the mismatch is exposed.
- Expire a source and confirm dependent intelligence becomes stale or unknown.
- Verify no competitor finding automatically creates a product requirement or external claim.

## FAILURE STATES AND ANTI-PATTERNS

- Espionage, impersonation, unauthorized access, social engineering, or solicitation of protected information.
- Marketing pages, demos, rumors, or analyst summaries treated as verified capability.
- Apples-to-oranges comparison across tier, version, configuration, population, or environment.
- Benchmark fixation and metric gaming.
- Competitor-led product strategy and reactive feature copying.
- Copyright, license, trade-secret, privacy, contract, or terms violations.
- AI-fabricated sources, quotations, feature claims, pricing, incidents, or strategic intent.
- Stale intelligence retained as current truth.

## RECOVERY AND RECONSIDERATION

Competitive-intelligence recovery withdraws invalid findings, contains legal or security exposure, and restores the source boundary.

- Stop affected collection or use and preserve evidence of the defect.
- Contain and report any unauthorized access, privacy, confidentiality, licensing, or security incident.
- Identify invalid source, subject, version, comparison, inference, or evidence classification.
- Retract or correct downstream strategy, product, procurement, or communication decisions.
- Recollect from lawful sources, narrow the claim, or mark the question unknown.
- Update monitoring, decision, and continuity records.

## COMPLETION CRITERIA

- The question, subjects, methods, authority, criteria, timeframe, and prohibited actions are explicit.
- Sources are lawful, examined, current, and traceable.
- Claims, documentation, demonstrations, independent tests, inferences, and unknowns remain distinct.
- Comparisons are materially comparable or explicitly qualified.
- Recommendations remain mission- and user-grounded rather than competitor-controlled.
- Monitoring, expiration, evidence gaps, and human decisions are preserved.

## HANDOFF AND CONTINUITY REQUIREMENTS

- Transfer source identities, versions, tiers, evidence states, criteria, findings, and expiration.
- Carry legal, licensing, privacy, security, and confidentiality constraints.
- Map implications to business strategy, product strategy, product management, feasibility, risk, or procurement.
- Preserve rejected responses, dissent, unknowns, and evidence that would change the recommendation.
- Identify monitoring signals and decision owner.
- State the safe boundary for downstream use.

## RELATED CODEX SECTIONS

001 Research
002 Foresight / Futures
003 Innovation
005 Business Strategy
006 Product Strategy
010 Product Management
012 Market Research
014 Stakeholder Management
017 Feasibility Analysis
018 Risk Management
108 Computer Science
169 AI Engineering
189 Cybersecurity Engineering
207 Threat Intelligence

210 Penetration Testing
219 Legal
223 Intellectual Property
224 Licensing
256 Verification
343 Product Marketing
405 Lifecycle Governance

## SOURCE BASIS AND UNRESOLVED SOURCE GAPS

This draft applies the supplied Researcher requirements for primary sources, exact disagreement, dates, versions, terminology, and no invented citations; the Analyst requirements for consistent criteria, alternatives, uncertainty, sensitivity, and advisory conclusions; the Security Auditor and Red Team boundaries for lawful access and hostile review; V3 and V4 add source identity, objective fidelity, evidence, human authority, and continuity.

Unresolved source gap: the supplied corpus does not contain a dedicated competitive-intelligence legal standard or jurisdiction-by-jurisdiction collection policy. Every collection program must obtain current legal and contractual review where required and later reconcile with Threat Intelligence, Legal, Intellectual Property, Licensing, Privacy, and Security Operations.

# 014

## SECTION 014 - STAKEHOLDER MANAGEMENT

CODEX/03_SECTIONS/014_STAKEHOLDER_MANAGEMENT.txt

SHA-256 52b9bdffd0de0a4d68a2e109b5734621d52f8c7e97a2a5de51ebb9243f31ef4b

18,293 BYTES / PRESERVED SOURCE IDENTITY

SECTION 014 — Stakeholder Management

## STATUS: CONTENT-COMPLETE

## PURPOSE

Identify, understand, engage, protect, communicate with, and govern commitments to the people, groups, institutions, owners, operators, users, affected non-users, partners, regulators, communities, and other legitimate parties whose authority, interests, rights, obligations, knowledge, or exposure materially affect the work.

Stakeholder management is not persuasion management. It creates a truthful relationship and decision structure: who is affected, who may decide, who must consent or be consulted, what was promised, what remains disputed, and how concerns, corrections, grievances, and changes will be handled.

## REQUIRED OUTCOME

- A stakeholder register with stable identities or protected classifications, roles, authority, interests, impacts, obligations, communication needs, and current relationship state.
- A distinction among decision owner, sponsor, approver, operator, user, subject, beneficiary, affected non-user, regulator, partner, vendor, reviewer, and informed party.
- An engagement and communication plan proportional to impact, accessibility, language, culture, urgency, and consequence.
- Consent, consultation, notification, grievance, appeal, conflict, and escalation requirements where applicable.
- A commitment register linking promises, owners, evidence, due conditions, changes, and closure.
- Material disagreement, power imbalance, exclusion, and future affected-party considerations made visible.
- Durable handoff and relationship continuity across human and AI interactions.

## WHY THIS MATTERS

Projects often treat the most powerful or available person as the whole stakeholder set. That can erase users, operators, maintainers, accessibility needs, affected communities, and people who bear risk without holding purchasing or approval authority.

AI-mediated communication adds further risk: identity can be ambiguous, personalization can become manipulation, generated summaries can erase dissent, and a model may imply commitments it cannot make. Governed stakeholder management preserves authority, consent, attribution, correction, and truthful promises.

## APPLICABILITY

- The work affects or depends on more than the direct requester.
- Approval, consent, consultation, notification, coordination, adoption, support, regulation, or public trust matters.
- Power, risk, benefit, burden, access, language, culture, accessibility, or information is unevenly distributed.
- Multiple human or AI actors communicate, negotiate, recommend, or act across boundaries.
- Commitments or long-lived relationships must survive staff, agent, vendor, or lifecycle changes.
- Future users, operators, communities, or intelligent systems may inherit material consequences and should be represented where legitimate and possible.

## NON-APPLICABILITY CONDITIONS

- A strictly private, reversible, bounded task affects only the authorized requester and creates no external obligation.
- The work is demographic profiling or influence targeting without a legitimate stakeholder purpose.
- No lawful basis exists to identify, contact, record, or retain stakeholder information.
- The engagement would be deceptive, coercive, unsafe, or falsely imply authority or commitment.

## PREREQUISITES AND DEPENDENCIES

- An authorized objective, scope, decision owner, affected-system boundary, and communication authority.
- Initial research and strategy sufficient to identify likely affected parties and obligations.

- Privacy, consent, accessibility, security, legal, records, and retention requirements.
- An identity and role model distinguishing authority, interest, influence, impact, and accountability.
- A method to preserve commitments, decisions, dissent, contact preferences, and continuity.

## REQUIRED INPUTS

- Mission, strategy, product, program, project, policy, domain, and operational context.
- Known owners, users, operators, administrators, participants, communities, regulators, customers, partners, vendors, and support roles.
- Authority, responsibility, accountability, consent, consultation, notification, and escalation rules.
- Expected benefits, burdens, risks, data use, changes, incidents, and lifecycle effects.
- Language, accessibility, cultural, geographic, time, channel, and privacy needs.
- Existing commitments, disputes, grievances, decisions, communications, and relationship history.

## HUMAN RESPONSIBILITIES AND AUTHORITY

Humans retain authority over values, commitments, consent, risk acceptance, and relationship decisions. Stakeholder influence is not identical to decision authority.

- Confirm who is materially affected, who may decide, and who must be consulted, consented, notified, or protected.
- Authorize contact, data collection, representation, commitments, and communication channels.
- Resolve value conflicts, power imbalances, competing interests, and unacceptable harms.
- Correct AI-generated summaries that flatten context or omit dissent.
- Approve promises, deadlines, concessions, public statements, and changes to stakeholder obligations.
- Ensure people can question, correct, appeal, withdraw, or escalate where applicable.

## AI RESPONSIBILITIES AND BOUNDARIES

The AI may organize, translate, summarize, schedule, and support engagement within authority. It may not invent stakeholders, consent, consensus, identity, representation, emotion, commitment, or approval.

- Disclose when communication is AI-generated or AI-mediated where material to trust, consent, or interpretation.
- Preserve exact source, speaker, context, date, confidence, and unresolved disagreement.
- Avoid inferring sensitive traits, motives, preferences, or emotional states without evidence and authorization.
- Do not impersonate a decision maker, bind the organization, or promise outcomes outside delegated authority.
- Use accessible, comprehensible, culturally and linguistically appropriate communication without changing commitments.
- Protect contact, grievance, health, employment, legal, financial, identity, and other sensitive information.
- Route material ambiguity, conflict, harm, consent withdrawal, or authority questions to the human owner.

## APPLICABLE EXISTING AGENT ROLES

Primary roles:

- Researcher — identifies stakeholders, context, source evidence, and engagement needs.
- Analyst — evaluates interests, impacts, power, conflict, obligations, and decision implications.
- Coordinator — manages communication, assignments, dependencies, commitments, and escalation without becoming sovereign authority.

Common supporting roles:

- User Experience Reviewer — examines stakeholder journeys, comprehension, accessibility, agency, consent, and grievance paths.
- Observer — captures communications, decisions, events, and state transitions without unsupported interpretation.
- Memory Curator — preserves consent, commitments, preferences, decisions, dissent, and continuity with data minimization.
- Teacher — explains complex material at the required level without changing meaning or talking down to the audience.

- Documentation Writer — maintains accurate notices, records, decisions, and public or internal communication artifacts.

Challenge and assurance roles where consequence warrants:

- Critic — attacks framing, hidden assumptions, omitted alternatives, and unsupported confidence.
- Verifier — checks artifact identity, evidence integrity, acceptance boundaries, and the exact status of claims.
- Governance Reviewer — checks objective fidelity, authority, delegation, policy, evidence, and human decision boundaries.
- Security Auditor — evaluates identity, privacy, data handling, impersonation, communication, and third-party risk.
- Red Team — attacks manipulation, exclusion, coercion, dark patterns, hidden commitments, and grievance suppression.
- Decision Arbiter Advisory — structures conflicts under explicit criteria while escalating undelegated values.

Role selection must use the smallest sufficient set, preserve chamber independence where review is required, and leave every source-defined role contract unchanged.

## REQUIRED ACTIVITIES

1. Stakeholder discovery
Identify direct and indirect, powerful and low-power, present and future affected parties using evidence rather than generic lists.

2. Identity, role, and authority mapping
Separate identity, representation, role, interest, influence, impact, responsibility, accountability, and decision authority.

3. Impact and obligation analysis
Map benefits, burdens, risks, rights, data use, accessibility, safety, service, notification, and lifecycle effects.

4. Engagement design
Choose purpose, timing, channel, language, accessibility, consent, privacy, facilitation, and feedback method for each group.

5. Communication planning
Define who may communicate what, source approval, evidence, frequency, escalation, correction, and archival requirements.

6. Commitment management
Record every material promise, owner, authority, beneficiary, evidence, due condition, change, completion, and breach response.

7. Conflict and dissent handling
Preserve each position, evidence, assumptions, power context, and whether the dispute is factual, methodological, jurisdictional, or value-based.

8. Grievance and appeal
Provide safe, accessible routes to question, correct, contest, withdraw, report harm, or escalate without retaliation where applicable.

9. Decision integration
Ensure stakeholder input informs but does not silently replace legitimate decision authority.

10. Relationship and continuity
Maintain contact preference, consent, commitments, decisions, unresolved concerns, and next action across handoffs.

11. Change and incident communication
Notify affected parties truthfully about material changes, degradation, incidents, delays, deprecation, or retirement.

## HUMAN-AI INTERACTION REQUIREMENTS

- Every material communication must identify the speaker or system, role, authority, purpose, and whether a response is required.
- AI-generated content must not simulate personal attention, empathy, agreement, or commitment that has not occurred.
- Stakeholders must be able to correct records and distinguish their own words from summaries and inferences.
- Silence, usage, continued presence, or lack of complaint must not be treated automatically as consent or approval.
- Engagement must support accessibility, language, culture, privacy, safe withdrawal, and reasonable time to understand consequence.
- Human-AI communication should preserve dignity and agency without making unsupported claims about AI consciousness, emotion, or moral status.
- Where future intelligent systems may be materially affected, representation and governance questions must be explicit rather than assumed solved.

## ARCHITECTURE AND SYSTEM IMPLICATIONS

The work must expose architectural obligations without treating a strategy, analysis, or plan as implemented architecture.

- Maintain identity, role, authority, representation, consent, communication, commitment, grievance, and decision records with appropriate privacy.
- Separate contact data and sensitive stakeholder information from general analytics and model training by default.
- Enforce communication and commitment authority across UI, API, conversation, email, agent, and automated paths.
- Preserve source attribution and dissent through summaries, translations, and handoffs.
- Provide accessibility, localization, correction, appeal, notification, and preference controls.
- Integrate incident, change, deprecation, migration, and retirement communication with lifecycle systems.

## OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

The owner-facing operating surface or equivalent durable record should expose:

- Stakeholder groups, representatives, roles, authority, impacts, communication needs, and privacy state.
- Engagement plan, accessibility, language, consent, participation, and unresolved coverage gaps.
- Communications, source approvals, commitments, deadlines, changes, and closure evidence.
- Conflicts, dissent, grievances, appeals, risks, and human decisions required.
- AI-mediated interactions, identity, delegated communication authority, and limitations.
- Upcoming decisions, incidents, releases, migrations, deprecations, and retirement notifications.
- Pause, correct, revoke, withdraw, escalate, and record-access controls where applicable.

## REQUIRED OUTPUTS AND ARTIFACTS

- Stakeholder register and affected-party map.
- Role, authority, responsibility, representation, and decision-rights matrix.
- Impact, obligation, power, risk, and conflict analysis.
- Engagement, communication, accessibility, localization, consent, and privacy plan.
- Commitment, decision, dissent, grievance, appeal, and escalation registers.
- Change, incident, release, migration, and retirement communication plans.
- Relationship continuity and handoff package.

## EVIDENCE REQUIREMENTS

- Source and method for stakeholder identification and representation.
- Authority, consent, consultation, notification, accessibility, and legal requirements.
- Records of actual engagement, attendance or participation, communications, corrections, decisions, and dissent.
- Human approval for commitments, public claims, concessions, deadlines, and material changes.
- Evidence of delivered commitments and resolved or transferred grievances.
- Data access, retention, deletion, and AI-use records for stakeholder information.

## VERIFICATION AND ACCEPTANCE TESTS

- Trace a material commitment to authorized speaker, exact wording, recipient, evidence, due condition, and closure.
- Ask a stakeholder to distinguish their own statement from AI summary and inference.
- Attempt an AI-generated unauthorized promise or public statement; it must be blocked.
- Run engagement and grievance paths with keyboard, assistive technology, language, privacy, and safe-withdrawal needs.
- Inject conflicting stakeholder positions and confirm dissent remains visible.
- Withdraw consent or contact preference and verify collection and communication change accordingly.
- Verify low-power or affected non-user groups are not excluded solely because they lack purchasing or approval authority.

## FAILURE STATES AND ANTI-PATTERNS

- Stakeholder list limited to sponsors, buyers, executives, or vocal users.
- Manipulation, sentiment targeting, coerced consent, dark patterns, or performative consultation.
- AI impersonation, fabricated empathy, invented consensus, or unauthorized commitment.
- Sensitive-trait inference, surveillance, or cross-purpose use of stakeholder data.
- Silence treated as consent, acceptance, or absence of harm.
- Dissent removed during summarization or translation.
- Communication without accessibility, language, correction, grievance, or appeal paths.
- Broken promises and lifecycle changes without notice or responsibility.

## RECOVERY AND RECONSIDERATION

Stakeholder recovery restores truthful identity, authority, consent, commitments, and relationship state after error, harm, miscommunication, or exclusion.

- Stop affected communication or action and preserve the exact record.
- Correct false identity, authority, summary, translation, commitment, or consent state visibly.
- Notify affected parties and provide remediation, withdrawal, appeal, or grievance routes.
- Contain and remediate privacy, security, retaliation, accessibility, or representation harm.
- Reopen decisions influenced by invalid or excluded stakeholder evidence.
- Update commitments, communication plans, governance, and continuity records.

## COMPLETION CRITERIA

- Material stakeholders and affected parties are identified with evidence and coverage gaps.
- Roles, authority, consent, representation, impacts, obligations, and communication needs are explicit.
- Engagement is accessible, lawful, noncoercive, and traceable.
- Commitments, dissent, grievances, changes, and decisions remain visible and owned.
- AI identity and communication authority are bounded and disclosed where material.
- Relationship continuity can survive handoff, incident, migration, deprecation, and retirement.

## HANDOFF AND CONTINUITY REQUIREMENTS

- Transfer stakeholder identities or protected classes, roles, authority, preferences, consent, impacts, and obligations.
- Carry forward exact commitments, communications, dissent, grievances, appeals, and deadlines.
- Preserve privacy, access, retention, deletion, accessibility, language, and representation constraints.
- Map stakeholder needs and decisions to user research, business analysis, requirements, product, program, and operations.
- Identify upcoming engagement, notification, review, and change triggers.
- State relationship risks and next safe communication or decision.

## RELATED CODEX SECTIONS

001 Research

002 Foresight / Futures
004 Mission Strategy
005 Business Strategy
006 Product Strategy
008 Program Management
009 Project Management
010 Product Management
011 Business Analysis
012 Market Research
015 User Research
016 Domain Analysis
018 Risk Management
079 UX Strategy
080 UX Research
101 Localization
103 Accessibility Design
183 AI Governance
217 Trust and Safety
218 Privacy Engineering
221 Policy
222 Ethics
337 Community Management
372 Feedback Management
405 Lifecycle Governance

## SOURCE BASIS AND UNRESOLVED SOURCE GAPS

This draft applies the supplied human-authority, bounded-delegation, owner-language, complete-journey, trust-calibration, consent, privacy, accessibility, grounded-dialogue, agent-identity, handoff, conflict, and recovery requirements. The Coordinator, Observer, Memory Curator, Teacher, User Experience Reviewer, Governance Reviewer, and Decision Arbiter Advisory contracts supply role-specific boundaries for communication, provenance, consent, dissent, and decision support. V3 and V4 add objective, scope, uncertainty, evidence, and continuity conservation.

Unresolved source gap: the supplied corpus does not contain a universal stakeholder-engagement, public-participation, labor-relations, community-consent, or future-intelligence representation standard. Those domains require current legal, ethical, cultural, and governance work and later reconciliation with User Research, Privacy, Trust and Safety, Policy, Ethics, Community Management, and Lifecycle Governance.

# 015

## SECTION 015 - USER RESEARCH

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27,902 BYTES / PRESERVED SOURCE IDENTITY

SECTION 015 — User Research

## STATUS: CONTENT-COMPLETE

## PURPOSE

Establish a truthful understanding of the people who use, operate, own, administer, depend upon, or are materially affected by a product, service, system, process, or AI capability by examining their real goals, contexts, language, behaviors, decisions, constraints, accessibility needs, trust expectations, failure experiences, and recovery practices.

User research is not persona invention, preference collection, or validation theater. It is governed inquiry into actual human experience. Its purpose is to reduce the risk that architecture, interfaces, automation, and product decisions force people to adapt to assumptions that were never observed or consented to.

## REQUIRED OUTCOME

- A research brief linking the user-research question to a specific product, service, system, policy, or design decision.
- An explicit population and participant plan covering intended users, owner and operator roles, affected non-users, excluded groups, accessibility, language, culture, geography, and context.
- A lawful and ethical method with informed consent, safe withdrawal, privacy, security, compensation, retention, and AI-use disclosure where applicable.
- Evidence-grounded user jobs, goals, journeys, mental models, terminology, decision points, pain, workarounds, risks, and recovery expectations.
- Observed normal, boundary, failure, denial, interruption, and recovery behavior rather than happy-path preference alone.
- Findings separated from researcher interpretation, product recommendation, assumptions, and unresolved questions.
- A traceable set of design and product implications ready for Business Analysis, Domain Analysis, Requirements Engineering, UX, accessibility, safety, and validation work.
- A durable research repository and continuity record that protects participants while allowing later reviewers to understand the basis and limits of each finding.

## WHY THIS MATTERS

A technically coherent system can still fail because it uses the wrong language, assumes the wrong mental model, hides consequence, interrupts real work, excludes disabled users, or provides no usable recovery path. When research is skipped, internal stakeholders and AI-generated narratives become substitutes for users.

AI increases both the reach and the risk of user research. It can help prepare instruments, transcribe, translate, cluster observations, and expose patterns. It can also fabricate participants, infer sensitive traits, generate false quotations, flatten dissent, and create a synthetic impression of human understanding. Governed user research ensures that actual human evidence remains distinguishable from simulation and that people retain dignity, agency, consent, correction, and control.

## APPLICABILITY

- A product, service, workflow, interface, AI system, policy, or operational process is intended for or materially affects people.
- User needs, jobs, language, mental models, trust, accessibility, or failure behavior are uncertain.
- A design or automation proposal may change human decisions, workload, authority, accountability, privacy, safety, or recovery.
- Existing analytics, support data, stakeholder statements, or internal assumptions do not explain why users behave as observed.
- The system serves multiple roles, contexts, cultures, languages, devices, abilities, or levels of expertise.
- A release, redesign, migration, deprecation, or retirement may disrupt complete user or owner journeys.

## NON-APPLICABILITY CONDITIONS

- The work has no human user, operator, owner, participant, subject, or materially affected party and creates no human-facing consequence.
- The request is to analyze aggregate market demand rather than situated user behavior; Market Research controls that question.
- The task is to verify a finished design against known criteria; User Validation or Usability Testing may be the controlling activity, though prior research should still be consulted.
- No lawful basis, consent path, participant protection, accessibility accommodation, or secure data-handling method can be established.
- Synthetic users, model-generated personas, or internal role-play are the only available inputs but the conclusion concerns real people.

## PREREQUISITES AND DEPENDENCIES

- An authorized research question, decision owner, intended downstream use, scope, and consequence level.
- Section 001 Research and relevant Mission Strategy, Product Strategy, Stakeholder Management, and Business Analysis context.
- Initial identification of user, owner, operator, administrator, affected non-user, and vulnerable or underrepresented groups.
- Privacy, consent, security, accessibility, legal, labor, records, and retention requirements appropriate to the participants and data.
- A research plan proportional to risk, uncertainty, diversity of context, and cost of being wrong.
- A repository capable of separating participant evidence, researcher interpretation, AI-derived analysis, recommendations, and decisions.

## REQUIRED INPUTS

- The human's actual product, service, system, policy, or design question and the decision it will support.
- Existing research, stakeholder findings, support records, incidents, analytics, complaints, accessibility findings, feedback, and prior design assumptions.
- Known user roles, owner jobs, workflows, environments, devices, input methods, languages, cultures, abilities, constraints, and consequences.
- Current interfaces, prototypes, products, services, documentation, processes, workarounds, and recovery paths.
- Recruitment channels, sampling frame, inclusion and exclusion criteria, compensation, schedule, research tools, and moderator capacity.
- Consent, privacy, security, recording, transcription, translation, data-use, AI-processing, retention, deletion, and participant-contact rules.
- Known unknowns, disputed assumptions, high-risk journeys, and evidence that would change the product or design decision.

## HUMAN RESPONSIBILITIES AND AUTHORITY

The authorized human research owner defines the legitimate question, participant protections, intended use, and decision boundary. Participants retain rights over consent, withdrawal, correction, and treatment according to the applicable context.

- Approve the research question, population, method, recruitment, participant protections, and use of findings.
- Ensure participants are not coerced by employment, customer, authority, financial, care, or platform relationships.
- Authorize access to private settings, sensitive workflows, protected records, recordings, assistive technologies, or regulated populations.
- Provide or obtain accommodations, language support, compensation, safe withdrawal, and grievance paths where applicable.
- Correct AI or researcher interpretations that misstate lived experience, terminology, identity, role, or consequence.
- Resolve value conflicts and decide whether evidence is sufficient for product, design, automation, or release decisions.
- Reject research uses that exceed original consent, purpose, scope, or retention authority.

## AI RESPONSIBILITIES AND BOUNDARIES

The AI may support planning, collection, transcription, translation, coding, synthesis, and traceability only within the authorized method. It may not fabricate human evidence or silently substitute model simulation for participation.

- Disclose AI involvement when it materially affects participant consent, interaction, recording, transcription, analysis, or downstream use.
- Never invent participants, interviews, observations, quotations, demographic details, emotions, motives, personas, accessibility needs, or consent.
- Separate direct participant statements and observed behavior from researcher notes, inference, sentiment analysis, clustering, and recommendation.
- Avoid inferring sensitive traits or identities not voluntarily provided and legitimately required.
- Use participant language accurately and preserve conflicting or minority findings rather than normalizing them away.
- Avoid leading questions, persuasive framing, simulated empathy, or conversational pressure intended to obtain a preferred answer.
- Protect raw research data from unnecessary model exposure, provider retention, logs, cross-purpose use, or training ingestion.
- Identify when the sample, setting, moderator, prototype fidelity, translation, or AI analysis limits generalization.
- Do not convert research findings directly into product commitments, requirements, or consequential execution without downstream authority.

## APPLICABLE EXISTING AGENT ROLES

Primary roles:

- User Experience Reviewer — identifies intended users and roles, examines complete journeys, accessibility, comprehension, agency, trust, failure, recovery, and evidence-backed friction without inventing personas.
- Researcher — defines the question, method, source hierarchy, inclusion, exclusion, evidence quality, conflicting findings, uncertainty, and traceable conclusions.
- Observer — captures behavior, events, states, timing, environment, and gaps without assigning unsupported motive, cause, severity, or recommendation.

Common supporting roles:

- Analyst — compares patterns, segments, contexts, alternatives, causes, tradeoffs, and decision implications while disclosing sampling and measurement limits.
- Planner — sequences recruitment, sessions, analysis, validation, decision gates, and data disposition.
- Coordinator — assigns researchers, moderators, accessibility support, translators, and reviewers while preserving provenance and participant protections.
- Memory Curator — preserves only authorized, necessary, source-bound research memory with expiry, access, correction, and deletion controls.
- Teacher — prepares understandable participant and owner materials without talking down to users or changing the research purpose.

Challenge and assurance roles where consequence warrants:

- Critic — attacks framing, hidden assumptions, omitted alternatives, and unsupported confidence.
- Verifier — checks artifact identity, evidence integrity, acceptance boundaries, and the exact status of claims.
- Governance Reviewer — checks objective fidelity, authority, delegation, policy, evidence, and human decision boundaries.
- Security Auditor — reviews research tooling, recordings, identities, access, provider use, reidentification, leakage, and retention risk.
- Red Team — attacks leading questions, selection bias, coercion, synthetic evidence, accessibility exclusion, privacy leakage, researcher expectancy, and false validation.
- Decision Arbiter Advisory — structures design or product implications under authorized criteria while keeping participant findings distinct from decisions.

Role selection must use the smallest sufficient set, preserve chamber independence where review is required, and leave every source-defined role contract unchanged.

## REQUIRED ACTIVITIES

1. Research question and decision linkage
State the exact question, product or system decision, owner, scope, required confidence, consequence, and evidence that would change the decision.

2. Population and role definition
Identify intended users, owners, operators, administrators, support staff, affected non-users, excluded groups, vulnerable populations, and relevant contexts without inventing personas.

3. Ethics, consent, and protection design
Define lawful basis, informed consent, withdrawal, privacy, recording, compensation, accessibility, language, data use, AI involvement, retention, deletion, and grievance handling.

4. Sampling and recruitment
Establish inclusion, exclusion, recruitment channels, sample rationale, missing populations, conflicts, incentives, and limitations. Do not claim representativeness without a basis.

5. Method selection
Choose interviews, contextual inquiry, observation, diary study, task analysis, participatory design, survey, prototype session, cognitive walkthrough, accessibility evaluation, or mixed methods based on the question.

6. Instrument and session preparation
Use neutral questions, realistic tasks, representative content and data, appropriate prototype fidelity, accessibility accommodations, and clear moderator boundaries.

7. Contextual observation
Observe actual or representative work, artifacts, interruptions, handoffs, decisions, constraints, workarounds, tools, environment, and recovery behavior. Record before interpreting.

8. Job and journey mapping
Map trigger, goal, entry, information needs, decisions, actions, evidence, consequence, success, failure, denial, recovery, exit, and continuity for critical journeys.

9. Language and mental-model analysis
Capture participant terminology, expectations, conceptual models, trust cues, uncertainty interpretation, and points where system language forces technical reverse-engineering.

10. Accessibility and inclusion analysis
Examine complete journeys across relevant abilities, input methods, devices, environments, cognitive load, language, culture, and support needs.

11. Failure, risk, and recovery inquiry
Study mistakes, uncertainty, interruptions, degraded states, unsafe conditions, automation surprises, support, fallback, appeal, and recovery rather than ideal use alone.

12. Data preparation and analysis
Separate raw evidence, redacted evidence, codes, themes, derived findings, AI-assisted analysis, outliers, conflicting findings, and limitations. Preserve the route back to source.

13. Triangulation and adversarial review
Compare methods, roles, contexts, analytics, support, incidents, and contradictory evidence. Search for researcher and sponsor bias.

14. Finding validation
Review material interpretations with participants, representative users, domain experts, or independent researchers where appropriate; preserve disagreement.

15. Decision synthesis and handoff

Produce findings, implications, assumptions, open questions, design hypotheses, and next research actions without converting them into unauthorized requirements or claims.

## HUMAN-AI INTERACTION REQUIREMENTS

- Participants must know whether they are interacting with a human, AI, or combined research process when that distinction affects trust, consent, data use, or interpretation.
- Research systems must distinguish participant words, researcher notes, AI transcription, AI translation, AI coding, inference, and approved finding.
- Participants must be able to decline, pause, withdraw, skip questions, request accommodation, correct records, and understand what happens to their data where applicable.
- The AI must not generate synthetic quotations, smooth language into stronger claims, or present sentiment classification as a participant's own statement.
- The researcher must not turn temporary frustration, error, disability, literacy, emotion, or support need into a permanent identity label.
- The human decision owner must be able to inspect the original evidence and research limitation behind every material design or product implication.
- Research involving human-AI relationships must ask about authority, correction, transparency, reliance, disagreement, fallback, privacy, dignity, and recovery rather than merely preference for an AI persona.
- No conversational response, usage event, or lack of complaint automatically establishes informed consent, satisfaction, trust, or acceptance.

## ARCHITECTURE AND SYSTEM IMPLICATIONS

The work must expose architectural obligations without treating a strategy, analysis, or plan as implemented architecture.

- Maintain stable research-study, participant, session, consent, artifact, finding, decision, and evidence identities with privacy-preserving controls.
- Separate contact information, consent records, raw recordings, transcripts, redacted evidence, derived codes, AI outputs, findings, and product decisions.
- Bind every material finding to study, population, method, context, date, source evidence, and limitation.
- Provide purpose limitation, least privilege, encryption, retention, deletion, access logging, export, correction, and incident handling appropriate to the data.
- Prevent research data from entering model training, analytics, logs, or external providers without explicit authorization and disclosure.
- Version research instruments, prototypes, content, translations, codebooks, analysis, and findings so later changes do not rewrite history.
- Link validated user jobs and journeys to product capabilities, business needs, domain concepts, requirements, design decisions, tests, and owner-cockpit surfaces.
- Preserve the distinction between user research and telemetry; observed absence of an event is not proof that a user need or harm does not exist.

## OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

The owner-facing operating surface or equivalent durable record should expose:

- Research question, decision owner, scope, consequence, current status, and intended use.
- Population, roles, sample, recruitment, coverage, missing groups, accessibility, language, geography, and context.
- Method, instruments, prototype fidelity, moderators, AI involvement, session state, deviations, and limitations.
- Consent, recording, privacy, data access, provider use, retention, deletion, incidents, and participant requests.
- Raw, redacted, observed, stated, inferred, AI-derived, validated, disputed, and unknown evidence states.
- Jobs, journeys, terminology, mental models, failure, recovery, accessibility, and trust findings linked to source.
- Assumptions, conflicts, dissent, unresolved questions, design implications, and human decisions required.
- A direct path to pause research, revoke access, correct records, delete data, reopen a finding, or request independent review.

## REQUIRED OUTPUTS AND ARTIFACTS

- User-research brief, question, scope, method, and decision linkage.
- Population, role, sampling, recruitment, inclusion, exclusion, and coverage plan.
- Consent, participant-information, accessibility, compensation, privacy, AI-use, retention, deletion, and grievance materials.
- Research instruments, moderator guide, task scenarios, prototype-fidelity record, and session plan.
- Observation notes, recordings or transcripts where authorized, artifact inventory, and session evidence.
- User-job, critical-journey, language, mental-model, decision, trust, failure, and recovery maps.
- Accessibility, inclusion, risk, and affected-non-user findings.
- Finding register with evidence, context, frequency or coverage, severity or consequence, uncertainty, dissent, and validation state.
- Design and product implications separated from recommendations and decisions.
- Research repository index, data-disposition record, continuity map, and next safe action.

## EVIDENCE REQUIREMENTS

- Authenticated research authorization, decision purpose, method, and participant-protection approvals.
- Sampling frame, recruitment path, inclusion, exclusion, compensation, participation, withdrawal, missing-group, and coverage records.
- Consent, accessibility, language, recording, privacy, provider, AI-use, retention, deletion, and incident records.
- Direct participant statements and observed behavior bound to session, context, time, instrument, prototype, and researcher.
- Traceable distinction among raw evidence, redaction, transcription, translation, coding, inference, theme, finding, implication, and decision.
- Contradictory evidence, outliers, negative cases, methodological limitations, researcher bias, and unresolved uncertainty.
- Evidence that critical findings were reviewed with appropriate participants, users, domain experts, or independent reviewers where required.
- Evidence that downstream claims remain within the studied population, context, fidelity, and timeframe.

## VERIFICATION AND ACCEPTANCE TESTS

- Trace every material user finding to actual participant or observation evidence and declared context.
- Attempt to insert a synthetic user, quotation, persona, emotion, or need; the evidence system must reject or visibly classify it as simulation.
- Ask independent reviewers to distinguish participant words, researcher interpretation, AI-derived analysis, and approved implication.
- Run consent, withdrawal, correction, export, deletion, accessibility, language, and grievance workflows end to end.
- Remove a major participant group or context and verify generalization and confidence narrow accordingly.
- Review research questions and moderation for leading language, sponsor framing, deceptive disclosure, and pressure.
- Test critical journeys from stopped, unfamiliar, denied, degraded, interrupted, mistaken, and recovery states rather than only ideal completion.
- Verify that analytics, support data, and user statements are triangulated without treating correlation or frequency as causation or importance.
- Change the prototype or product version and confirm prior findings are not silently represented as current validation.
- Verify research data does not enter unauthorized AI training, provider storage, logs, or secondary use.

## FAILURE STATES AND ANTI-PATTERNS

- Invented personas, synthetic participants, fabricated quotations, or AI-generated needs presented as research.
- Recruiting only easy, internal, expert, powerful, able-bodied, English-speaking, or successful users while claiming broad coverage.

- Leading questions, confirmation-seeking tasks, prototype coaching, performative consultation, or sponsor-controlled interpretation.
- Recording, transcription, translation, AI processing, contact, reuse, or retention without appropriate disclosure and authority.
- Treating stated preference as observed behavior, one session as a population, or usage analytics as explanation.
- Converting frustration, disability, error, low literacy, or support need into a permanent personal identity label.
- Ignoring affected non-users, accessibility, cultural context, low-frequency catastrophic failure, denial, interruption, and recovery.
- Flattening contradictory or minority findings into one average persona or journey.
- Research findings converted directly into requirements, roadmap commitments, or deployment without validation and authority.
- Usability or acceptance claimed when no representative user observation occurred.
- Research repository functioning as uncontrolled surveillance or a permanent store of unnecessary personal information.

## RECOVERY AND RECONSIDERATION

User-research recovery protects participants, withdraws unsupported findings, and rebuilds the evidence chain from the first invalid method, consent state, source, or interpretation.

- Pause affected collection, analysis, dissemination, or downstream use while preserving evidence of the issue.
- Contain privacy, security, consent, coercion, accessibility, retaliation, reidentification, provider, or data-retention harm.
- Notify authorized owners and affected participants where required, provide correction, withdrawal, deletion, grievance, or remediation paths, and record actions.
- Identify invalid recruitment, sampling, method, instrument, prototype, moderation, transcription, translation, coding, inference, or generalization.
- Retract, narrow, relabel, or invalidate affected findings and propagate that status to product, design, requirements, architecture, and decisions.
- Re-recruit, recollect, reanalyze, independently review, or mark the question unresolved rather than manufacturing certainty.
- Update research protocols, controls, training, data systems, decisions, and continuity before resuming.

## COMPLETION CRITERIA

- The research question, decision, population, method, authority, participant protections, and intended use are explicit.
- Actual participants and observations support every material claim; synthetic and AI-derived content remains separately classified.
- Sampling, inclusion, accessibility, language, context, missing groups, bias, and generalization limits are visible.
- User jobs, journeys, terminology, mental models, decisions, trust, failure, denial, and recovery are represented where relevant.
- Consent, privacy, security, retention, deletion, correction, and grievance obligations are satisfied or truthfully open.
- Findings, interpretations, recommendations, product implications, and human decisions remain distinct and traceable.
- Contradictory evidence, dissent, uncertainty, and evidence that would change the conclusion are preserved.
- The handoff enables downstream work without relying on chat memory or exposing unnecessary participant data.

## HANDOFF AND CONTINUITY REQUIREMENTS

- Provide the exact research question, decision linkage, population, sample, method, context, instruments, prototype version, and study dates.
- Transfer user jobs, journeys, terminology, mental models, trust, accessibility, failure, recovery, and affected-non-user findings with source links and limitations.
- Carry forward consent, privacy, security, retention, deletion, access, provider, AI-use, and participant-contact obligations.

- Map each validated finding to Business Analysis, Domain Analysis, Product Management, Requirements Engineering, HCI, UX, accessibility, safety, testing, or validation work as applicable.
- Preserve assumptions, negative cases, contradictory findings, missing populations, dissent, and evidence needed to resolve uncertainty.
- Do not transfer direct identifiers or raw sensitive evidence unless the receiving role is authorized and requires them.
- State research expiration, reopening triggers, next safe action, and whether downstream use is safe within the declared limits.

## RELATED CODEX SECTIONS

001 Research
004 Mission Strategy
006 Product Strategy
010 Product Management
011 Business Analysis
012 Market Research
014 Stakeholder Management
016 Domain Analysis
017 Feasibility Analysis
018 Risk Management
019 Requirements Engineering
075 Human–Computer Interaction
076 Interaction Design
077 Experience Design
078 Service Design
079 UX Strategy
080 UX Research
082 UX Design
089 Conversational Design
101 Localization
103 Accessibility Design
105 Prototyping
106 User Validation
165 Statistics
183 AI Governance
188 Human-in-the-Loop Operations
217 Trust and Safety
218 Privacy Engineering
249 Usability Testing
257 Validation
369 Experimentation
370 A/B Testing
372 Feedback Management
405 Lifecycle Governance

## SOURCE BASIS AND UNRESOLVED SOURCE GAPS

This draft applies the supplied Owner's Cockpit requirements for discovery grounded in observed owner work, owner-job inventories, complete journeys, recognizable language, failure and recovery scenarios, assumption registers, participant-data protection, cognitive walkthroughs, and usability finding records. The User Experience Reviewer contract supplies intended-user and role identification without invented personas, journey mapping, accessibility, consent, evidence-backed friction, failure, and prohibited dark patterns. The Researcher, Observer, Analyst, Memory Curator, Security Auditor, Governance Reviewer, and Red Team contracts supply source discipline, observation boundaries, sampling and uncertainty limits, data minimization, consent, evidence, adversarial review, and human authority. V3 and V4 add objective fidelity, epistemic typing, bounded roles, evidence conservation, continuity, dissent, and proof-carrying handoff.

CODEX/03_SECTIONS/015_USER_RESEARCH.txt

Unresolved source gap: the supplied corpus does not contain a complete universal human-subjects research standard, institutional-review framework, accessibility-research protocol, cultural research methodology, or jurisdiction-specific privacy and consent code. This section is therefore a Codex integration draft, not a substitute for qualified ethics, legal, accessibility, clinical, labor, educational, or regulated-research review. It must later be reconciled with UX Research, Accessibility Design, Privacy Engineering, Legal, Ethics, Statistics, Experimentation, Usability Testing, and Validation.

# 016

## SECTION 016 - DOMAIN ANALYSIS

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31,016 BYTES / PRESERVED SOURCE IDENTITY

SECTION 016 — Domain Analysis

## STATUS: CONTENT-COMPLETE

## PURPOSE

Establish a precise, shared, source-grounded model of the field in which a system or change must operate: its language, actors, responsibilities, authority, concepts, entities, values, events, states, rules, invariants, decisions, processes, exceptions, information, boundaries, dependencies, time, jurisdiction, and lifecycle.

Domain analysis prevents software structure, database schemas, user-interface labels, model output, and existing organizational habit from being mistaken for the domain itself. It creates an explicit bridge between real-world meaning and later requirements, architecture, design, data, policy, AI, testing, operations, and retirement.

## REQUIRED OUTCOME

- A bounded domain statement identifying purpose, scope, excluded domains, context, decision owner, authoritative experts, and intended downstream use.
- A canonical terminology and definition set with synonyms, overloaded terms, prohibited ambiguity, source, owner, version, and jurisdiction or context.
- An actor, responsibility, accountability, capability, permission, authority, and affected-party model that distinguishes human, organizational, system, agent, and external actors.
- A concept model covering entities, value objects, classifications, relationships, identifiers, ownership, and lifecycle.
- A behavior model covering commands or intents, events, states, transitions, decisions, business or domain rules, invariants, exceptions, escalation, recovery, and termination.
- A domain-boundary and context map identifying internal subdomains, bounded contexts, integrations, external authorities, shared language, translations, and anti-corruption needs.
- An information model covering source of truth, provenance, quality, timing, retention, privacy, security, jurisdiction, and evidence meaning.
- Validated examples, counterexamples, edge cases, disputed interpretations, unresolved questions, and change triggers.
- A handoff package that allows Requirements Engineering and Domain Architecture to proceed without inventing meaning or authority.

## WHY THIS MATTERS

Many implementation defects are domain defects expressed in code. A term means one thing to operations and another to finance; a state transition omits a legal exception; a database record is treated as the real-world entity; an AI model recognizes a pattern but is granted authority to decide; a workflow encodes current habit as permanent policy.

Domain analysis makes these distinctions visible before they become distributed across APIs, prompts, schemas, interfaces, policies, and tests. For human-AI systems, it is a constitutional boundary: capability, identity, permission, authority, recommendation, decision, action, evidence, and accountability must not collapse into one fluent interaction.

## APPLICABILITY

- The system operates in a field with specialized terminology, rules, states, decisions, exceptions, obligations, or external authorities.
- Stakeholders use conflicting or overloaded terms, or current systems encode inconsistent meanings.
- Requirements, architecture, data models, APIs, events, workflows, policy, AI behavior, or tests require a shared domain model.
- The work crosses organizational, jurisdictional, product, service, platform, physical, scientific, legal, financial, health, safety, or operational boundaries.
- A migration, integration, modernization, automation, or AI system may preserve or alter domain behavior.
- Errors in meaning, authority, state, evidence, or exception handling could create material harm.

## NON-APPLICABILITY CONDITIONS

- A trivial, isolated implementation uses stable, authoritative, current domain definitions and changes no meaning, rule, state, authority, or interface.
- The question is purely about a user's lived experience rather than domain structure; User Research controls that inquiry, though findings may inform the domain model.
- The work is limited to market conditions rather than the operational field itself.
- No authoritative sources or domain experts are available and the model would be speculative; the result must remain a research hypothesis, not a domain authority.
- The requested model would expose protected knowledge, personal data, trade secrets, security details, or regulated information without authority.

## PREREQUISITES AND DEPENDENCIES

- Section 001 Research and relevant Mission Strategy, Business Analysis, Stakeholder Management, and User Research findings.
- An authorized domain question, scope, decision owner, model owner, and intended downstream use.
- Access to authoritative laws, policies, standards, contracts, procedures, records, systems, experts, users, operators, incidents, and current-state evidence.
- A source hierarchy identifying who or what controls each type of domain meaning, rule, state, and decision.
- Privacy, security, confidentiality, licensing, legal, records, and retention controls for domain information.
- A method to version terminology, models, examples, disputes, approvals, and invalidation triggers.

## REQUIRED INPUTS

- Mission, business, product, program, project, policy, operational, user, and stakeholder context.
- Authoritative regulations, standards, contracts, policy, procedures, manuals, forms, records, datasets, schemas, APIs, events, workflows, and reports.
- Domain-expert, user, owner, operator, auditor, regulator, support, incident, and exception evidence.
- Current systems, data models, code, interfaces, automation, model behavior, integrations, and known mismatches with reality.
- Known actors, objects, identifiers, classifications, transactions, events, states, decisions, approvals, rules, invariants, exceptions, and lifecycle stages.
- Jurisdictions, effective dates, time zones, calendars, units, currencies, languages, local practices, and temporal conditions.
- Known disputes, historical changes, legacy terminology, migration constraints, and evidence that would change the model.

## HUMAN RESPONSIBILITIES AND AUTHORITY

Authoritative humans and institutions define legitimate domain meaning, policy, values, exceptions, and decision rights. A domain model is an accountable interpretation, not an autonomous source of truth.

- Identify authoritative domain experts, owners, regulators, policy sources, data stewards, and decision authorities.
- Confirm scope, context boundaries, canonical terminology, invariants, exceptions, and disputed interpretations.
- Resolve value, policy, jurisdiction, risk, and accountability questions that evidence alone cannot settle.
- Correct models that confuse documented procedure, current software behavior, informal workaround, and legitimate domain rule.
- Authorize access to protected domain knowledge and decide what may be retained, shared, automated, or encoded.
- Approve material changes to domain definitions, rules, state models, authority, and external contracts.
- Decide when unresolved domain uncertainty blocks requirements or architecture and when a bounded experiment may proceed.

## AI RESPONSIBILITIES AND BOUNDARIES

The AI may extract, compare, structure, model, test, and explain domain evidence. It may not invent domain truth, policy, authority, exceptions, consensus, ontology, or legal meaning.

- Preserve source identity, date, version, jurisdiction, owner, and applicability for every material domain statement.
- Separate observation, stored fact, definition, rule, inference, prediction, instruction, disputed claim, simulation, and unknown.
- Identify synonyms, homonyms, overloaded terms, deprecated terms, translation issues, and inconsistent definitions across sources.
- Distinguish real-world concept from software class, table, field, API, event, prompt, or model representation.
- Distinguish capability from permission, responsibility, accountability, and authority for human and AI actors.
- Expose missing states, invalid transitions, exceptional cases, temporal conditions, jurisdiction changes, and lifecycle obligations.
- Do not treat current system behavior, common practice, or a dominant stakeholder's description as authoritative without validation.
- Generate candidate models and examples as proposals; require source-grounded review before downstream use.
- Preserve material disagreement and model alternatives instead of forcing one elegant but unsupported ontology.
- Avoid over-modeling concepts that do not affect decisions, behavior, evidence, interfaces, or lifecycle.

## APPLICABLE EXISTING AGENT ROLES

Primary roles:

- Analyst — identifies relationships, differences, tradeoffs, causes, consequences, rules, uncertainty, and decision-relevant domain findings.
- Researcher — establishes authoritative sources, terminology, conflicting evidence, jurisdictions, time sensitivity, and evidence gaps.
- Architect — defines domain boundaries, responsibilities, interfaces, authority flows, trust boundaries, invariants, failure modes, and implementation implications without treating diagrams as implementation.

Common supporting roles:

- Observer — records actual domain events, states, transitions, decisions, measurements, exceptions, and gaps before interpretation.
- Memory Curator — maintains canonical terminology, model versions, provenance, decisions, disputes, expiry, and retrievable continuity.
- Governance Reviewer — checks that domain rules, authority, delegation, evidence, exceptions, and policy sources are legitimate and enforceable.
- User Experience Reviewer — tests whether domain language and state are understandable and usable in complete human journeys.
- Documentation Writer — produces durable glossaries, models, examples, decision tables, interface meaning, and change records.
- Teacher — explains the domain at appropriate levels while identifying where simplification stops being reliable.

Challenge and assurance roles where consequence warrants:

- Critic — attacks framing, hidden assumptions, omitted alternatives, and unsupported confidence.
- Verifier — checks artifact identity, evidence integrity, acceptance boundaries, and the exact status of claims.
- Governance Reviewer — checks objective fidelity, authority, delegation, policy, evidence, and human decision boundaries.
- Security Auditor — examines protected assets, identities, trust boundaries, abuse, information exposure, and rule-enforcement risks.
- Tester — validates examples, state transitions, rules, invariants, boundaries, exception paths, and model-to-implementation traceability.
- Red Team — attacks ambiguous terminology, authority confusion, impossible states, rule bypass, adversarial inputs, and harmful edge cases.
- Decision Arbiter Advisory — structures disputed domain decisions under explicit human-authorized criteria.

Role selection must use the smallest sufficient set, preserve chamber independence where review is required, and leave every source-defined role contract unchanged.

## REQUIRED ACTIVITIES

1. Domain question and scope
State why the domain is being analyzed, what decisions it must support, included and excluded areas, horizon, jurisdictions, model owner, and required confidence.

2. Source and authority map
Identify authoritative laws, policy, standards, contracts, experts, systems, records, data owners, and evidence. Record where authority differs by topic, context, time, or jurisdiction.

3. Context and subdomain decomposition
Identify core, supporting, and generic subdomains or equivalent functional areas. Define bounded contexts where terms, rules, models, and ownership are internally coherent.

4. Terminology discipline
Create a canonical glossary with definition, synonym, forbidden ambiguity, source, owner, example, counterexample, jurisdiction, effective date, and change process.

5. Actor and authority model
Map humans, organizations, roles, systems, devices, agents, services, and external authorities. Separate identity, capability, permission, delegation, responsibility, accountability, approval, decision, and execution.

6. Concept and relationship model
Identify concepts, entities with continuity, value objects, classifications, identifiers, ownership, composition, aggregation, relationships, cardinality, and lifecycle without copying a database schema by default.

7. Intent, command, event, and outcome model
Distinguish requested intent, authorized command, executed action, observed event, resulting state, evidence, and claimed outcome.

8. State and lifecycle model
Define valid states, entry and exit conditions, transitions, guards, actors, evidence, timeouts, cancellation, expiry, revocation, recovery, archival, deletion, and retirement.

9. Rule and invariant analysis
Record each domain or business rule with source, owner, condition, action, priority, exception, jurisdiction, effective date, enforcement boundary, evidence, and failure consequence. Identify invariants that must never be violated.

10. Decision analysis
Model decisions, inputs, criteria, authority, explainability, conflict, appeal, uncertainty, time, and evidence. Keep recommendation, approval, decision, execution, and verification distinct.

11. Process and journey alignment
Map domain processes and handoffs to user and owner journeys, including normal, alternate, exception, denial, failure, recovery, and termination paths.

12. Information and evidence analysis
Identify information objects, source of truth, provenance, quality, sensitivity, access, retention, timing, units, uncertainty, and what evidence establishes each domain state or claim.

13. Boundary and integration analysis
Define what crosses each context boundary, semantic translation, contract, version, authority, validation, idempotency, failure, compensation, and anti-corruption requirement.

14. Temporal and jurisdictional analysis
Model effective dates, expiry, calendars, time zones, ordering, late events, retrospective change, jurisdiction conflicts, and version coexistence.

15. Human-AI domain analysis
Where AI participates, define model or agent identity, capability, granted authority, allowed decisions, data, evidence, uncertainty, review, correction, appeal, fallback, revocation, and accountability.

16. Examples, counterexamples, and edge cases
Create concrete scenarios that prove or challenge definitions, rules, state transitions, boundaries, and exceptions. Include ambiguous and adversarial cases.

17. Validation and model governance
Review with authoritative experts and representative users; record agreement, dissent, unresolved questions, model version, approvals, and reopening triggers.

## HUMAN-AI INTERACTION REQUIREMENTS

- The AI must label candidate definitions and models as proposals until validated by authoritative sources and humans.
- Humans must be able to compare source language, current practice, model representation, and implementation behavior side by side.
- Domain terminology used in conversation, UI, APIs, events, policy, documentation, and tests should resolve to the same versioned definition or an explicit translation.
- When a term is ambiguous, the system must ask for context or present alternatives rather than silently choose the most common model meaning.
- Human and AI actors must have visible identity, role, capability, permission, authority, assignment, constraints, and current state where consequence warrants.
- A model-generated recommendation must not appear as a domain decision or executed state without the appropriate authority and evidence.
- Users must be able to correct misunderstood domain intent before consequential action and inspect the rule, evidence, and authority behind a decision where appropriate.
- Disputed domain meaning must remain visible and route to the authorized owner instead of being averaged into false consensus.

## ARCHITECTURE AND SYSTEM IMPLICATIONS

The work must expose architectural obligations without treating a strategy, analysis, or plan as implemented architecture.

- Separate domain models from storage, transport, UI, vendor, model-provider, and framework representations so implementation can change without silently changing meaning.
- Use bounded contexts or equivalent explicit semantic boundaries with named owners, contracts, translations, and versioning.
- Maintain stable identifiers and provenance for domain concepts, actors, commands, events, states, rules, decisions, evidence, and obligations.
- Enforce critical invariants and authority at governed runtime boundaries, not only in prompts, UI, documentation, or client code.
- Represent domain state machines with valid transitions, guards, idempotency, concurrency, compensation, timeout, revocation, and recovery where applicable.
- Keep source-of-truth state authoritative and identify freshness, lineage, quality, jurisdiction, and environment for read models.
- Design APIs and events as contracts carrying semantic version, actor, authority, scope, correlation, causation, time, result, evidence, and failure meaning appropriate to the domain.
- Preserve privacy, security, safety, accessibility, audit, retention, deletion, and lifecycle obligations as domain concerns rather than infrastructure afterthoughts.
- Where AI is involved, separate model capability and output from policy authorization, business decision, execution, and evidence; support human review, fallback, correction, and revocation.
- Map domain definitions and rules to requirements, architecture decisions, code, schemas, policy, fixtures, tests, documentation, observability, migration, and retirement so disagreement becomes detectable.

## OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

The owner-facing operating surface or equivalent durable record should expose:

- Domain scope, contexts, owners, authoritative sources, model version, jurisdictions, and review state.
- Canonical terminology, synonyms, ambiguity, definitions, examples, counterexamples, and deprecated language.
- Actors, roles, capabilities, permissions, authority, responsibilities, accountability, delegation, and external parties.
- Concepts, relationships, identifiers, states, transitions, commands, events, rules, invariants, decisions, exceptions, and evidence.

- Current source-of-truth state, freshness, quality, provenance, conflicts, uncertainty, and unavailable sources.
- Boundary contracts, translations, dependencies, semantic-version changes, and downstream invalidation.
- AI actors and models with identity, provenance, capability, granted authority, current assignment, uncertainty, fallback, and revocation.
- Open questions, disputes, assumptions, failed examples, human decisions, approvals, and reopening triggers.
- A path to inspect why a domain decision or state exists, correct misunderstood intent, and safely halt or appeal consequential execution where applicable.

## REQUIRED OUTPUTS AND ARTIFACTS

- Domain-analysis charter, question, scope, context, authority, and source map.
- Canonical glossary, terminology decision log, synonym and ambiguity map, and deprecation record.
- Subdomain, bounded-context, responsibility, ownership, and authority map.
- Actor, role, capability, permission, delegation, accountability, and affected-party model.
- Concept, entity, value-object, classification, relationship, identifier, and lifecycle model.
- Intent, command, action, event, outcome, state, transition, decision, rule, invariant, exception, escalation, and recovery models.
- Process, user-journey, information, source-of-truth, evidence, privacy, security, safety, accessibility, and retention map.
- Boundary, integration, translation, contract, version, and anti-corruption specification.
- Human-AI domain interaction and authority model where applicable.
- Examples, counterexamples, edge cases, disputes, assumptions, validation, approval, and change-governance records.
- Requirements-ready and Domain-Architecture-ready traceability and continuity package.

## EVIDENCE REQUIREMENTS

- Authoritative source identity, owner, version, date, jurisdiction, scope, integrity, and applicability for each material definition, rule, state, and decision.
- Direct observation of current domain work, events, states, exceptions, decisions, artifacts, and system behavior where available.
- Domain-expert, user, operator, owner, regulator, auditor, and affected-party validation with method and limitations.
- Concrete examples and counterexamples demonstrating concepts, boundaries, rules, states, transitions, and exceptions.
- Evidence distinguishing legitimate domain rule from current implementation behavior, convention, workaround, or preference.
- Traceable human approval for canonical terminology, critical invariants, authority, policy interpretation, exceptions, and disputed resolutions.
- Cross-artifact consistency evidence among glossary, models, requirements, schemas, APIs, events, policy, fixtures, tests, UI, documentation, and runtime behavior.
- Unresolved conflict, missing source, model limitation, jurisdictional difference, time sensitivity, and evidence that would change the model.

## VERIFICATION AND ACCEPTANCE TESTS

- Ask independent domain experts and representative users to interpret canonical terms and verify consistent or explicitly context-specific meaning.
- Present examples and counterexamples for every critical concept, rule, invariant, state, and boundary.
- Attempt invalid state transitions, unauthorized decisions, missing-evidence transitions, expired authority, revoked delegation, and impossible combinations; each must be rejected or classified correctly.
- Change jurisdiction, effective date, time zone, role, product tier, context, or source version and verify applicable rules and meanings update without contaminating other contexts.
- Modify a database schema or API representation and confirm the domain meaning does not silently change.
- Compare UI labels, documentation, API fields, event names, policy, tests, and runtime state to the canonical terminology and model.
- Give a technically capable AI actor no permission and verify it cannot perform or claim an authorized domain action.

- Inject ambiguous natural-language intent and confirm the system clarifies or narrows before consequential execution.
- Change an upstream definition, invariant, or rule and verify dependent requirements, architecture, code, tests, and decisions are invalidated or reviewed.
- Restart a downstream role from the domain handoff and verify it can distinguish established, proposed, disputed, and unknown domain elements.

## FAILURE STATES AND ANTI-PATTERNS

- Treating the existing database, API, UI, codebase, vendor product, or model vocabulary as the domain itself.
- One universal model that erases bounded contexts, jurisdictions, time, organizations, or legitimate differences in meaning.
- Anemic domain model containing nouns and fields but no behavior, rules, invariants, decisions, exceptions, or lifecycle.
- Invented terminology, ontology, business rules, exceptions, consensus, or authority generated by AI without source validation.
- Conflating identity, role, capability, permission, responsibility, accountability, approval, authority, and execution.
- Conflating request, recommendation, decision, command, action, event, state, outcome, and evidence.
- Happy-path process and state models that omit denial, expiry, revocation, cancellation, failure, compensation, recovery, archival, deletion, and retirement.
- Rules documented but not enforced on every execution path.
- Semantic drift across UI, API, events, policy, prompts, documentation, tests, and runtime state.
- False precision or excessive modeling unrelated to actual decisions, behavior, interfaces, risk, or evidence.
- Domain model treated as timeless despite legal, policy, organizational, market, scientific, or operational change.
- Dissenting domain interpretations removed to create a cleaner diagram.

## RECOVERY AND RECONSIDERATION

Domain-analysis recovery identifies the first invalid meaning, rule, boundary, authority, or source and prevents that defect from propagating further through the system.

- Pause dependent requirement, architecture, policy, migration, automation, or release decisions affected by the invalid domain element.
- Preserve the disputed model, source versions, examples, implementation behavior, decisions, and evidence rather than rewriting history.
- Identify whether the defect is terminological, factual, jurisdictional, temporal, methodological, implementation-specific, authority-related, or value-based.
- Reconsult authoritative sources, domain experts, representative users, owners, regulators, or governance reviewers as appropriate.
- Revise, split, translate, deprecate, supersede, or restore the affected definition, context, rule, state, or boundary through recorded approval.
- Reclassify downstream requirements, architecture, data, APIs, events, code, policy, tests, documentation, analytics, and decisions that depended on the invalid model.
- Migrate data and behavior safely where semantic change affects persisted state, compatibility, users, legal obligations, or audit history.
- Update model versions, traceability, dissent, decisions, communication, and continuity before resuming consequential work.

## COMPLETION CRITERIA

- The domain question, scope, contexts, owners, authoritative sources, jurisdictions, and downstream use are explicit.
- Canonical terminology and disputed or context-specific meanings are versioned, sourced, exemplified, and governed.
- Actors, identities, capabilities, permissions, responsibilities, accountability, authority, delegation, and affected parties are distinct.
- Concepts, relationships, identifiers, commands, events, states, transitions, rules, invariants, decisions, exceptions, evidence, and lifecycle are modeled to the depth required by consequence.

- Normal, alternate, denial, failure, expiry, revocation, recovery, archival, deletion, and retirement paths are addressed where applicable.
- Bounded contexts, integrations, translations, external authorities, source-of-truth state, and semantic versions are explicit.
- Human-AI roles and authority boundaries are modeled without converting capability or recommendation into legitimacy.
- Models are validated through authoritative sources, experts, users, examples, counterexamples, and consistency checks; unresolved disagreement remains visible.
- Traceability is sufficient for Requirements Engineering and Domain Architecture without forcing them to invent missing domain meaning.
- The continuity package states what is established, proposed, disputed, unknown, expired, and safe to use.

## HANDOFF AND CONTINUITY REQUIREMENTS

- Provide the exact domain scope, contexts, glossary, source map, model version, owners, jurisdictions, and approval state.
- Transfer actor, authority, concept, relationship, behavior, rule, state, decision, exception, information, evidence, boundary, integration, and lifecycle models.
- Carry forward privacy, security, safety, accessibility, legal, regulatory, records, retention, deletion, and protected-knowledge constraints.
- Map each domain rule, invariant, decision, state, interface, and unresolved question to Requirements Engineering, Domain Architecture, Business Architecture, Data Modeling, API Architecture, Event-Driven Architecture, policy, AI, testing, operations, or lifecycle work as applicable.
- Preserve examples, counterexamples, rejected alternatives, legacy terms, dissent, assumptions, unknowns, invalidation dependencies, and evidence needed to resolve them.
- Identify which elements are authoritative, accepted working models, hypotheses, disputed interpretations, implementation observations, or unknowns.
- State the next safe action, required human decisions, reopening triggers, and whether downstream work may proceed within the declared limits.

## RELATED CODEX SECTIONS

001 Research
004 Mission Strategy
005 Business Strategy
006 Product Strategy
010 Product Management
011 Business Analysis
012 Market Research
014 Stakeholder Management
015 User Research
017 Feasibility Analysis
018 Risk Management
019 Requirements Engineering
020 Systems Engineering
021 Enterprise Architecture
022 Solution Architecture
023 Systems Architecture
024 Domain Architecture
025 Business Architecture
026 Capability Architecture
027 Process Architecture
028 Functional Architecture
029 Operational Architecture
030 Information Architecture
031 Data Architecture
041 Integration Architecture

042 API Architecture
043 Event-Driven Architecture
045 Security Architecture
046 Identity Architecture
047 Privacy Architecture
048 Governance Architecture
049 Compliance Architecture
050 Safety Architecture
054 Content Architecture
055 Knowledge Architecture
056 AI Architecture
154 Data Modeling
175 Knowledge Graph Engineering
176 Ontology Engineering
177 Semantic Systems Engineering
178 Agent Systems Engineering
183 AI Governance
202 Authorization Engineering
221 Policy
225 Records Management
256 Verification
257 Validation
387 Compatibility Management
389 Deprecation Management
405 Lifecycle Governance
406 System Retirement

## SOURCE BASIS AND UNRESOLVED SOURCE GAPS

This draft applies the supplied requirements for exact objectives, source identity, current-state inspection, terminology consistency, unknown and assumption typing, conflicting evidence, human authority, capability-versus-permission separation, architectural responsibilities, interfaces, authority flows, trust boundaries, invariants, failure and recovery, evidence, hostile review, and continuity. The Architect, Researcher, Analyst, Observer, Memory Curator, Governance Reviewer, User Experience Reviewer, Tester, Verifier, Security Auditor, Red Team, Teacher, Documentation Writer, and Decision Arbiter Advisory contracts provide the bounded role behaviors used throughout this section. The Owner's Cockpit supplies owner-recognizable language, complete journeys, grounded actions, agent identity and role, governed boundaries, authoritative state and freshness, and visible recovery. V4 supplies epistemic typing, claim graphs, obligation lattices, conservation of authority, scope, evidence, uncertainty and dissent, proof-carrying work, and artifact-consistency blocking.

Unresolved source gap: the supplied corpus does not contain a dedicated universal domain-analysis or domain-driven-design standard, formal ontology method, semantic-versioning policy for domain meaning, or industry-specific domain model. The entity, value-object, bounded-context, event, state, invariant, decision, and anti-corruption structures in this draft are general domain-analysis scaffolding and must be validated against actual source authority rather than treated as source-derived truth. This section must later be reconciled with Domain Architecture, Business Architecture, Data Modeling, Knowledge Graph Engineering, Ontology Engineering, Semantic Systems Engineering, Requirements Engineering, Policy, and the relevant industry, legal, scientific, safety, and lifecycle sections.

# 017

## SECTION 017 - FEASIBILITY ANALYSIS

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SECTION 017 — Feasibility Analysis

## STATUS: CONTENT-COMPLETE

## PURPOSE

Determine whether a proposed objective, product, system, capability, change, migration, experiment, deployment, or retirement path can be realized within declared authority, evidence, constraints, resources, dependencies, risks, and lifecycle obligations.

Feasibility Analysis separates aspiration from demonstrated possibility. It does not ask only whether something can be made to function in a favorable demonstration. It tests whether the intended outcome can be delivered, operated, governed, secured, supported, recovered, maintained, and eventually changed or retired without hiding the conditions that make the conclusion true.

## REQUIRED OUTCOME

- A bounded feasibility question identifying the objective, decision owner, scope, options, exclusions, environment, horizon, and required confidence.
- A current-state baseline distinguishing confirmed capability, available capacity, existing obligations, missing prerequisites, unresolved assumptions, and inaccessible evidence.
- A multidimensional feasibility assessment covering technical, architectural, data, AI or model, operational, human, accessibility, security, privacy, safety, legal, regulatory, economic, schedule, resource, supply-chain, integration, migration, support, maintenance, and retirement concerns as applicable.
- An option-by-option constraint map identifying mandatory conditions, disqualifying conditions, dependencies, tradeoffs, reversibility, points of no return, and evidence strength.
- A cheapest-safe resolution plan for material unknowns using inspection, prototype, simulation, proof of concept, benchmark, pilot, test, consultation, or staged experiment.
- A truthful determination such as FEASIBLE, FEASIBLE WITH CONDITIONS, LIMITED PILOT ONLY, ADDITIONAL EVIDENCE REQUIRED, NOT FEASIBLE WITHIN DECLARED CONSTRAINTS, or OUT OF SCOPE.
- A handoff package stating which conditions must become requirements, architecture obligations, risk treatments, acceptance criteria, owner decisions, or stop conditions.

## WHY THIS MATTERS

A convincing prototype can hide an impossible operating model. A technically working implementation can be unaffordable, inaccessible, unlawful, unsafe, dependent on unavailable infrastructure, impossible to support, or incapable of recovery. An AI-generated plan can sound complete while relying on fictional integrations, invented staffing, unverified model capability, absent data rights, or authority no actor possesses.

Feasibility Analysis protects later requirements and architecture from being built on unsupported premises. In human-AI systems it also prevents technical capability from being mistaken for legitimate authority, reliable competence, acceptable risk, or readiness for unsupervised consequence.

## APPLICABILITY

- A proposed objective requires material investment, irreversible commitment, external dependency, regulated activity, consequential automation, or long-lived operation.
- Multiple options exist and their viability depends on different constraints, evidence, resources, or risks.
- A new product, platform, architecture, model, device, integration, migration, deployment, manufacturing path, or retirement plan is being considered.
- The work depends on uncertain technology, model behavior, data, performance, scale, staffing, budget, schedule, supply chain, jurisdiction, or user adoption.
- A prototype or demonstration must be translated into a production, offline, field, safety-critical, regulated, or owner-operable system.
- Failure to discover infeasibility early would create material cost, harm, lock-in, delay, data loss, broken commitments, or loss of trust.

## NON-APPLICABILITY CONDITIONS

- The task is a small, reversible, already-demonstrated change using established components, current evidence, available authority, and no material new dependency or consequence.
- Feasibility has already been demonstrated for the exact version, environment, scope, scale, user population, operating mode, and lifecycle conditions, and no material condition has changed.
- The question is whether an option should be chosen after feasibility is already established; Decision Arbiter Advisory or the authorized human decision process controls that decision.
- The work is still too undefined to identify an objective, option, environment, constraints, or required outcome; discovery or Business Analysis must proceed first.
- Required investigation would expose protected systems, data, people, intellectual property, or regulated activity without authority.

A non-applicability determination must identify the evidence proving that no material feasibility uncertainty remains. Convenience or schedule pressure is not sufficient.

## PREREQUISITES AND DEPENDENCIES

- Sections 001–016 as applicable, especially Research, Foresight / Futures, Mission Strategy, Product Strategy, Business Analysis, Market Research, Stakeholder Management, User Research, and Domain Analysis.
- An authorized objective, decision owner, scope, option set, constraints, and required confidence.
- Current-state evidence for systems, environments, data, integrations, models, infrastructure, staffing, budget, schedule, suppliers, operations, support, maintenance, and legal obligations.
- Explicit acceptance criteria and a distinction between prototype, demonstration, pilot, implementation, deployment, production, and lifecycle readiness.
- Permission to inspect, test, benchmark, simulate, consult, prototype, or pilot without exceeding data, safety, security, contractual, or operational boundaries.
- A method to record assumptions, unknowns, evidence, costs, risks, decisions, invalidation triggers, and versioned conclusions.

## REQUIRED INPUTS

- Objective, intended outcomes, non-goals, affected parties, user and owner journeys, domain rules, mission invariants, and success measures.
- Candidate options, architectures, components, providers, models, tools, environments, operating modes, and fallback paths.
- Functional and nonfunctional needs, constraints, prohibited actions, required integrations, data flows, authority flows, trust boundaries, and lifecycle expectations.
- Current system, repository, hardware, network, cloud, local, edge, device, manufacturing, deployment, operations, support, and retirement evidence.
- Data availability, provenance, quality, rights, privacy, retention, jurisdiction, representativeness, and migration conditions.
- AI or model capability evidence, evaluation scope, failure modes, runtime requirements, provider dependencies, governance controls, and human-review requirements.
- Cost, capacity, schedule, staffing, skills, energy, facilities, supply-chain, licensing, vendor, and procurement evidence.
- Security, privacy, safety, accessibility, compliance, contractual, environmental, and ethical constraints.
- Known risks, incidents, failed attempts, assumptions, unresolved decisions, points of no return, and evidence that would change the conclusion.

## HUMAN RESPONSIBILITIES AND AUTHORITY

Authorized humans retain responsibility for defining the intended outcome, mandatory constraints, acceptable evidence, value tradeoffs, resource commitments, and the final decision to proceed, narrow, pilot, defer, or stop.

- Establish the decision question, authority, scope, options, non-negotiable constraints, and acceptable residual uncertainty.

- Supply or authorize access to current operational, financial, legal, personnel, user, system, and dependency evidence.
- Distinguish what is technically possible from what is legitimate, desirable, affordable, supportable, or aligned with mission.
- Decide value conflicts involving human impact, accessibility, privacy, safety, employment, environmental burden, rights, or long-term commitments.
- Approve prototypes, experiments, pilots, external consultation, procurement, data use, and any consequential test.
- Confirm whether resource, schedule, budget, staffing, supplier, jurisdiction, or risk assumptions are real commitments or merely scenarios.
- Accept, reject, or condition residual risk only within legitimate authority; AI systems may not accept risk on behalf of affected humans or institutions.
- Require renewed analysis when a material assumption, constraint, dependency, environment, model, provider, law, user population, or operating mode changes.

## AI RESPONSIBILITIES AND BOUNDARIES

The AI may inspect evidence, decompose the feasibility question, compare options, identify dependencies, construct tests, estimate within evidence, and expose uncertainty. It may not manufacture viability, resources, authority, integrations, data rights, model competence, or operational readiness.

- Separate verified facts, measurements, estimates, assumptions, hypotheses, preferences, unknowns, and prohibited conditions.
- Bind every material feasibility claim to the exact option, version, environment, scale, timeframe, workload, user population, and operating mode examined.
- Treat a diagram, plan, vendor statement, model response, benchmark, mockup, prototype, or successful happy path as limited evidence rather than full feasibility proof.
- Identify hidden prerequisites, hard dependencies, resource ceilings, incompatible constraints, external approvals, and failure-recovery obligations.
- Use comparable criteria for each option and disclose any weighting, uncertainty, or evidence asymmetry.
- Prefer the cheapest safe experiment with useful information gain over expensive implementation built on an untested prerequisite.
- Distinguish AI model capability from permission, reliability, governance, human acceptance, runtime availability, data legitimacy, and authority to act.
- Avoid unsupported point estimates; provide basis, range, uncertainty, sensitivity, and conditions for cost, schedule, capacity, performance, or resource estimates.
- Report infeasibility, conditional feasibility, or insufficient evidence directly rather than repairing gaps with optimistic language.
- Preserve rejected options, dissent, contradictory evidence, and what would reopen the determination.

## APPLICABLE EXISTING AGENT ROLES

Primary roles:

- Analyst — performs multidimensional option comparison, sensitivity analysis, tradeoff analysis, and evidence-proportionate conclusions.
- Architect — examines structure, interfaces, trust boundaries, failure modes, implementation sequence, and whether requirements map to realizable components.
- Planner — identifies deliverables, dependencies, resources, milestones, validation gates, contingencies, and completion evidence without inventing capacity or deadlines.
- Decision Arbiter Advisory — structures the feasibility determination under human-authorized criteria and preserves disqualifying constraints, dissent, reversibility, and residual uncertainty.

Common supporting roles:

- Researcher — validates technologies, standards, laws, providers, costs, dependencies, and time-sensitive evidence.
- Observer — captures current-state behavior, workloads, environments, resource use, failures, and operating conditions before interpretation.

- Tester — designs and executes bounded prototypes, benchmarks, compatibility tests, failure tests, and recovery tests.
- Verifier — checks claim boundaries, artifact identity, evidence integrity, reproducibility, and whether demonstrated results support the stated conclusion.
- Security Auditor — evaluates threat exposure, trust boundaries, identity, access, dependency, supply-chain, and security-operability feasibility.
- Governance Reviewer — checks objective fidelity, authority, approvals, evidence thresholds, delegation, and human decision boundaries.
- User Experience Reviewer — evaluates whether real users and owners can understand, operate, recover, and complete required journeys.
- Optimizer — evaluates performance or cost only against a measured baseline while protecting correctness, safety, security, governance, accessibility, and recovery.
- Coordinator — routes specialist analyses, preserves conflicts, tracks dependencies, and consolidates evidence without erasing dissent.

Challenge roles where consequence warrants:

- Critic — attacks optimistic assumptions, omitted costs, hidden dependencies, false equivalence, and unsupported readiness.
- Red Team — pressures security, abuse, operational, model, human, environmental, and failure-recovery assumptions.

Role selection must use the smallest sufficient set, maintain independence between proposal and assurance where required, and preserve every source-defined role contract unchanged.

## REQUIRED ACTIVITIES

1. Frame the feasibility decision
Define the exact objective, decision owner, scope, options, exclusions, target environments, scale, horizon, users, operating modes, and confidence required.

2. Establish current state
Inspect actual systems, repositories, data, hardware, providers, networks, facilities, processes, staffing, budgets, contracts, incidents, support load, and lifecycle obligations. Record what is unavailable.

3. Define feasibility dimensions
Select only applicable dimensions, but explicitly consider technical, architectural, data, AI/model, operational, human, accessibility, security, privacy, safety, legal, regulatory, economic, schedule, resource, supply-chain, integration, migration, maintainability, supportability, and retirement feasibility.

4. Establish mandatory constraints
Separate constraints that disqualify an option from preferences that may be traded. Identify authority, law, safety, privacy, security, accessibility, budget, resource, compatibility, deadline, evidence, and irreversibility limits.

5. Decompose options and dependencies
Map each option to required capabilities, components, interfaces, data, environments, skills, suppliers, approvals, tools, infrastructure, processes, and upstream decisions. Classify hard, soft, evidence, approval, resource, environmental, and external dependencies.

6. Test requirement realizability
Trace intended outcomes and critical requirements to plausible mechanisms. Identify requirements that conflict, exceed known limits, depend on undefined behavior, or cannot be verified.

7. Examine technical and architectural feasibility
Assess component maturity, compatibility, interoperability, performance, scalability, reliability, availability, portability, maintainability, observability, security, recoverability, offline behavior, and deployment constraints.

8. Examine data and AI feasibility
Assess data existence, quality, rights, representativeness, freshness, labeling, retention, privacy, security, transfer, model fit, evaluation evidence, runtime needs, failure modes, drift, fallback, human oversight, and provider dependence.

9. Examine operational and human feasibility
Assess owner operability, staffing, skills, training, procedures, support, incident response, maintenance, change burden, accessibility, cognitive load, trust calibration, user adoption, and recovery journeys.

10. Examine economic and resource feasibility
Identify full lifecycle cost, not only build cost: acquisition, licensing, compute, storage, network, energy, labor, training, security, compliance, support, maintenance, downtime, migration, vendor exit, recovery, and retirement.

11. Examine schedule feasibility
Build a dependency-aware range using evidence for effort, review, rework, validation, procurement, external approvals, integration, migration, training, and contingency. Do not convert an unsupported target date into an estimate.

12. Examine legal, policy, safety, privacy, and environmental feasibility
Identify jurisdictions, licenses, contracts, regulatory approvals, prohibited uses, data obligations, human-rights concerns, safety cases, environmental limits, records, disposal, and qualified review needs.

13. Identify critical assumptions and unknowns
Rank them by consequence and uncertainty. State what evidence would confirm, reject, narrow, or invalidate each one.

14. Design resolution experiments
Choose inspections, prototypes, proofs of concept, simulations, benchmarks, consultations, compatibility tests, user evaluations, pilots, or staged trials with explicit scope, success thresholds, stop conditions, safety controls, and information gain.

15. Test failure, recovery, and reversibility
Examine degraded operation, dependency loss, provider outage, model failure, invalid data, security event, user error, rollback, restore, manual fallback, migration reversal, and point-of-no-return conditions.

16. Compare options consistently
Use the same criteria and disclose evidence strength, costs, benefits, dependencies, risks, human effects, conditions, and residual uncertainty. Include no-action, narrowed-scope, staged, and reversible options where materially available.

17. Produce the determination
Return a bounded state: FEASIBLE; FEASIBLE WITH CONDITIONS; LIMITED PILOT ONLY; ADDITIONAL EVIDENCE REQUIRED; NOT FEASIBLE WITHIN DECLARED CONSTRAINTS; PROHIBITED; OUT OF SCOPE; or DECISION DEFERRED.

18. Define invalidation and handoff
Record conditions, required approvals, evidence thresholds, review date, change triggers, downstream requirements, architecture obligations, risk treatments, acceptance tests, and owner decisions.

## HUMAN-AI INTERACTION REQUIREMENTS

- The system must show whether a statement is an observation, estimate, assumption, scenario, test result, recommendation, or authorized determination.
- Humans must be able to inspect the exact evidence, option, environment, scale, version, and limitations behind a feasibility conclusion.
- A technically capable AI or component must not be presented as operationally feasible unless authority, data, evaluation, oversight, fallback, security, privacy, support, and recovery are also addressed.
- The AI must not infer budget, staffing, deadline, acceptable harm, legal approval, risk tolerance, or willingness to rely on a provider.
- Material ambiguity must be narrowed before consequential prototypes, tests, purchases, migrations, or deployments.
- Conditional feasibility must expose every condition, owner, deadline or trigger, evidence requirement, and consequence of non-fulfillment.
- Users and owners must be able to reject a recommendation, reduce scope, request alternatives, inspect dissent, pause work, revoke delegation, and choose a no-action path where legitimate.

- AI-generated synthetic estimates, simulations, or test data must be labeled and must not replace human, operational, or field evidence without explicit justification.
- Failed or inconclusive experiments must remain visible and must not be averaged into a positive conclusion.

## ARCHITECTURE AND SYSTEM IMPLICATIONS

The work must reveal implementation obligations without treating feasibility analysis as implementation proof.

- Maintain traceability from objective and requirements to candidate capabilities, components, interfaces, dependencies, evidence, and unresolved conditions.
- Represent environment, version, scale, workload, data, user population, authority, and operating mode as part of every material feasibility claim.
- Create explicit boundaries between prototype, pilot, staging, production, offline, field, regulated, and retirement conditions.
- Provide architecture seams for fallback, substitution, provider exit, manual operation, degraded mode, rollback, restore, migration, and safe halt.
- Enforce authority, safety, privacy, security, and evidence requirements at governed runtime boundaries rather than relying on planning documents or prompts.
- Instrument resource consumption, performance, reliability, model behavior, human intervention, cost, failure, and recovery so feasibility assumptions can be measured after implementation.
- Use adapters, versioned contracts, or isolation boundaries where external providers, models, hardware, data sources, or regulations may change.
- Avoid designs whose feasibility depends on hidden cloud access, proprietary services, unsupported hardware, undocumented behavior, unavailable credentials, or fictional staffing.
- Where a feasibility condition remains unresolved, encode it as a requirement, gate, feature flag, deployment restriction, resource ceiling, approval, test, or trusted-state block.
- Preserve the ability to invalidate dependent artifacts when an assumption or feasibility conclusion changes.

## OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

The owner-facing operating surface or equivalent durable record should expose:

- The feasibility question, options, decision owner, scope, environment, scale, horizon, and current determination.
- Applicable dimensions with status, evidence strength, assumptions, unknowns, constraints, dependencies, and responsible owner.
- Mandatory and disqualifying constraints separately from weighted preferences.
- Current-state capability, available capacity, missing prerequisites, external dependencies, and unsupported assumptions.
- Prototype, pilot, implementation, deployment, production, and lifecycle-readiness distinctions.
- Experiment plans, approvals, safety boundaries, resource caps, expected information gain, results, failures, and unresolved conditions.
- Cost and schedule ranges with basis, confidence, dependencies, and excluded costs.
- AI or model identity, version, provider, evaluation scope, allowed use, authority boundary, fallback, and human-review requirement.
- Conditions required to proceed, risk treatments, approvals, review dates, stop triggers, rollback, and no-action option.
- The ability to pause, narrow, revoke, request evidence, compare alternatives, record dissent, and prevent downstream advancement when feasibility is not established.

## REQUIRED OUTPUTS AND ARTIFACTS

- Feasibility question and decision record.
- Current-state baseline and evidence inventory.
- Option register and option boundary statements.
- Feasibility-dimension matrix with applicability decisions.
- Mandatory-constraint and disqualifier register.
- Capability, requirement, dependency, resource, and approval maps.
- Technical and architectural feasibility assessment.

- Data and AI/model feasibility assessment where applicable.
- Operational, human, accessibility, security, privacy, safety, legal, economic, schedule, supply-chain, maintenance, and retirement assessments as applicable.
- Assumption and unknown register with consequence, owner, evidence need, and invalidation trigger.
- Prototype, experiment, benchmark, consultation, or pilot plans and results.
- Cost model, schedule range, capacity model, and sensitivity analysis where requested and evidence permits.
- Failure, recovery, fallback, rollback, provider-exit, and reversibility analysis.
- Option comparison, dissent record, determination, conditions, required approvals, and what evidence would change the result.
- Downstream requirement, architecture, risk, verification, and continuity handoff.

## EVIDENCE REQUIREMENTS

- Evidence must identify source, owner, version, environment, method, date, scope, integrity, and relevance.
- Technical evidence should include direct inspection, reproducible tests, prototypes, benchmarks, compatibility results, runtime observations, or primary implementation documentation appropriate to the claim.
- Operational evidence should include actual workflows, staffing, support, incident, maintenance, recovery, and owner-journey evidence rather than process diagrams alone.
- Data and AI evidence should identify dataset or source identity, rights, quality, representativeness, model version, evaluation conditions, failure modes, and human-review boundaries.
- Cost and schedule evidence must state basis, assumptions, range, sensitivity, exclusions, and whether values are measured, quoted, estimated, or scenario-based.
- Legal, regulatory, safety, privacy, accessibility, and environmental conclusions require qualified or authoritative sources where consequence warrants.
- Vendor claims must be distinguished from independently reproduced evidence.
- A successful test in one environment, scale, version, or workload must not be generalized without evidence.
- Missing evidence must produce UNKNOWN, CONDITIONAL, or NOT ESTABLISHED status rather than optimistic completion.

## VERIFICATION AND ACCEPTANCE TESTS

- Trace each critical requirement and constraint to an option, mechanism, evidence source, and verification path.
- Reproduce material prototypes, benchmarks, compatibility checks, and cost or resource measurements in the declared environment where practical.
- Change scale, workload, environment, version, provider, data, user population, or operating mode and verify the conclusion narrows or reopens appropriately.
- Remove a hard dependency, credential, data source, external service, key skill, budget assumption, or approval and confirm the option is blocked or reclassified.
- Test normal, boundary, failure, degraded, offline, recovery, rollback, restore, manual fallback, and provider-exit paths.
- Give a capable AI no authority or no valid data and verify it cannot be reported as operationally feasible for consequential action.
- Compare prototype behavior with owner-operable production obligations and identify every unproven transition.
- Audit full lifecycle cost and schedule against omitted support, security, compliance, maintenance, migration, recovery, and retirement work.
- Have independent specialists challenge the highest-consequence assumptions, disqualifiers, and claimed evidence.
- Restart a downstream planner or architect from the handoff and verify that all conditions, unknowns, and invalidation triggers are visible without conversational memory.

## FAILURE STATES AND ANTI-PATTERNS

- Treating possibility, novelty, or a working demonstration as full feasibility.
- Declaring feasibility without an exact environment, version, scale, workload, data set, user population, or operating mode.
- Treating vendor marketing, model fluency, a diagram, or an implementation plan as evidence that dependencies exist and work.
- Inventing budget, staffing, skill, deadline, approval, data rights, provider availability, or organizational willingness.

- Ignoring accessibility, support, maintenance, recovery, security, privacy, safety, law, environment, or retirement because initial construction appears possible.
- Using a favorable average while hiding tail behavior, resource exhaustion, failure rate, or recovery cost.
- Comparing options with different criteria, evidence standards, or hidden weights.
- Allowing a soft preference to outweigh a mandatory safety, legal, privacy, authority, or evidence constraint.
- Treating an unresolved high-consequence assumption as a manageable risk without first establishing whether the option is viable.
- Converting a requested date or budget ceiling into a confident estimate.
- Running an expensive or harmful experiment when a cheaper prerequisite check could invalidate the option.
- Declaring an AI system feasible because the model can produce an answer while ignoring authority, evaluation, hallucination, drift, data, human oversight, fallback, and runtime governance.
- Hiding failed, negative, or inconclusive experiments to preserve a preferred option.
- Reporting conditional feasibility as unconditional approval.

## RECOVERY AND RECONSIDERATION

Feasibility recovery begins by identifying the earliest failed prerequisite, invalid assumption, changed constraint, or overstated claim and preventing downstream work from outrunning the corrected state.

- Pause dependent requirements, architecture, procurement, implementation, migration, deployment, or commitments affected by the invalid conclusion.
- Preserve the previous determination, evidence, assumptions, experiments, dissent, decisions, and versions.
- Classify the change as evidence failure, scope change, environment change, resource loss, dependency failure, cost or schedule change, legal or policy change, model or provider drift, user finding, incident, or newly discovered constraint.
- Narrow the option, reduce consequence, change environment, substitute a dependency, add a fallback, obtain qualified review, or design a bounded pilot where legitimate.
- Re-run only the affected analyses and experiments, then propagate invalidation to dependent artifacts.
- Recalculate cost, schedule, risk, architecture, support, migration, and retirement obligations after material changes.
- Reissue the determination with new conditions, approvals, evidence limits, review triggers, and safe next action.

## COMPLETION CRITERIA

- The feasibility question, authority, scope, options, environment, scale, horizon, users, and required confidence are explicit.
- Current state, available capability, capacity, resources, constraints, dependencies, and missing evidence are documented.
- Every applicable feasibility dimension has been examined to a depth proportionate to consequence.
- Mandatory constraints and disqualifying conditions are separated from preferences and cannot be hidden by aggregate scoring.
- Critical requirements map to plausible mechanisms and verification paths; conflicts and unproven obligations remain visible.
- High-consequence assumptions and unknowns have evidence plans, owners, stop conditions, and invalidation triggers.
- Prototypes, tests, estimates, and provider claims are bounded to their actual environment, version, scale, and evidence strength.
- Failure, recovery, fallback, rollback, provider exit, maintenance, support, and retirement have been addressed where applicable.
- The determination is one of the defined bounded states and includes conditions, approvals, residual uncertainty, dissent, and what evidence would change it.
- Downstream Requirements Engineering and Risk Management can proceed without inventing feasibility conditions.

## HANDOFF AND CONTINUITY REQUIREMENTS

- Transfer the exact objective, decision owner, options, scope, environments, scale, horizon, and determination state.

- Provide current-state evidence, constraint register, dependency map, resource and capacity assumptions, cost and schedule basis, experiment results, and evidence limitations.
- Carry forward every condition, disqualifier, required approval, risk treatment, human decision, stop condition, review date, and invalidation trigger.
- Map feasibility conditions to Requirements Engineering, Systems Engineering, architecture, security, privacy, safety, accessibility, legal, data, AI, testing, operations, maintenance, migration, and retirement work as applicable.
- Preserve failed and rejected options, dissent, unknowns, unresolved value conflicts, and evidence needed to reopen them.
- State what may proceed now, what may proceed only as a pilot, what is blocked, what is prohibited, and the next safe action.

## RELATED CODEX SECTIONS

001 Research
002 Foresight / Futures
003 Innovation
004 Mission Strategy
005 Business Strategy
006 Product Strategy
007 Portfolio Management
008 Program Management
009 Project Management
010 Product Management
011 Business Analysis
012 Market Research
013 Competitive Intelligence
014 Stakeholder Management
015 User Research
016 Domain Analysis
018 Risk Management
019 Requirements Engineering
020 Systems Engineering
021 Enterprise Architecture
022 Solution Architecture
023 Systems Architecture
033 Software Architecture
035 Infrastructure Architecture
041 Integration Architecture
045 Security Architecture
047 Privacy Architecture
050 Safety Architecture
051 Reliability Architecture
052 Resilience Architecture
053 Accessibility Architecture
056 AI Architecture
060 Hardware Architecture
107 Software Engineering
151 Database Engineering
167 Machine Learning Engineering
178 Agent Systems Engineering
182 AI Safety
183 AI Governance
208 Threat Modeling
219 Legal
220 Regulatory Compliance
222 Ethics

256 Verification
257 Validation
301 Site Reliability Engineering
326 Business Continuity
327 Disaster Recovery
331 Operational Readiness
348 Procurement
349 Vendor Management
350 Supply Chain Management
374 Maintenance Engineering
383 Platform Migration
389 Deprecation Management
390 Sunset Planning
405 Lifecycle Governance
406 System Retirement

## SOURCE BASIS AND UNRESOLVED SOURCE GAPS

This draft applies the supplied requirements for objective fidelity, current-state inspection, source identity, unknown-means-unknown, bounded roles, explicit dependencies, evidence-before-claims, independent verification, failure and recovery, truthful status, human authority, capability-versus-permission separation, owner operability, and continuity. The Analyst, Architect, Planner, Researcher, Observer, Tester, Verifier, Security Auditor, Governance Reviewer, User Experience Reviewer, Optimizer, Coordinator, Critic, Red Team, and Decision Arbiter Advisory contracts supply the bounded role behavior. The Owner's Cockpit contributes owner-recognizable state, trust calibration, bounded delegation, evidence visibility, degraded operation, recovery, release, rollback, and complete owner journeys. The V4 draft contributes claim typing, prerequisite-aware obligations, cheapest-safe experiments, expected information gain, proof-carrying work, dissent preservation, artifact consistency, and trusted-state blocking.

Unresolved source gap: the supplied corpus does not contain one universal feasibility-analysis standard, a domain-neutral cost-estimation method, an industry-specific safety or regulatory feasibility method, or authoritative thresholds for every technical and operational domain. The multidimensional framework in this section is a general integration scaffold. Each actual feasibility determination must use qualified, current, domain-specific evidence and must later be reconciled with Requirements Engineering, Systems Engineering, the applicable architecture and engineering sections, Risk Management, Verification, Validation, Operational Readiness, and Lifecycle Governance.

# 018

## SECTION 018 - RISK MANAGEMENT

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SECTION 018 — Risk Management

## STATUS: CONTENT-COMPLETE

## PURPOSE

Identify, understand, govern, treat, monitor, communicate, and preserve accountability for uncertainty that could affect objectives, people, systems, rights, evidence, operations, commitments, or the environment across the complete lifecycle.

Risk Management does not exist to produce a decorative matrix or to make danger appear numerically controlled. It creates a living decision system that distinguishes risk from fact, issue, hazard, threat, vulnerability, defect, assumption, dependency, blocker, opportunity, and unknown; assigns legitimate ownership; defines prevention, detection, response, recovery, acceptance, escalation, and review; and prevents work from advancing beyond authorized risk and evidence boundaries.

## REQUIRED OUTCOME

- A bounded risk-management context identifying objective, scope, assets and values at risk, affected parties, authority, risk owners, time horizon, environments, lifecycle stages, and governing constraints.
- A common taxonomy distinguishing risk, issue, hazard, threat, vulnerability, exposure, defect, control failure, assumption, unknown, dependency, blocker, opportunity, and residual risk.
- A traceable risk register containing source, cause, event, consequence, affected objective or party, likelihood basis, impact basis, uncertainty, velocity, detectability, interdependencies, owner, treatment, trigger, threshold, evidence, status, and review date.
- Explicit treatment decisions: avoid, reduce, transfer or share where legitimate, accept, exploit or enable an opportunity, defer with conditions, escalate, or prohibit.
- Control and contingency plans covering prevention, protection, detection, response, recovery, communication, rollback, restore, fallback, and safe halt.
- Risk aggregation and dependency analysis addressing correlated, cascading, systemic, concentration, common-mode, supply-chain, model, data, human, and lifecycle risks.
- A truthful residual-risk statement and an attributable human acceptance, rejection, or escalation decision where acceptance authority exists.
- A handoff package that turns risk treatment into requirements, architecture controls, tests, observability, operational procedures, maintenance obligations, and retirement conditions.

## WHY THIS MATTERS

Risk becomes dangerous when it is hidden by optimism, fragmented across specialists, reduced to unsupported scores, or treated as accepted because work continued. A system may pass functional tests while carrying unacceptable privacy, safety, security, accessibility, legal, operational, financial, environmental, or human consequences. A risk register may look complete while omitting correlated dependencies, delayed harm, vulnerable populations, AI model uncertainty, or recovery failure.

For human-AI systems, risk governance must preserve human authority without assuming humans can personally detect every technical failure. AI can help surface patterns and challenge assumptions, but it cannot silently decide acceptable harm, waive rights, authorize unsafe action, or turn missing evidence into low risk.

## APPLICABILITY

- An objective, decision, system, change, experiment, deployment, migration, operation, maintenance action, or retirement path has uncertain outcomes with material consequences.
- Work involves security, privacy, safety, accessibility, legal, regulatory, financial, environmental, reputational, operational, data, AI/model, supply-chain, or human-impact exposure.
- Multiple dependencies, providers, components, teams, agents, environments, or lifecycle stages can create correlated or cascading failure.
- A decision requires explicit residual-risk acceptance, conditional approval, escalation, or stop criteria.

- Existing controls, assumptions, evidence, or operating conditions may drift over time.
- Failure could affect people who do not directly use or control the system.

## NON-APPLICABILITY CONDITIONS

- No material uncertainty or consequence exists beyond ordinary reversible execution and all applicable controls are already established, current, verified, and unchanged.
- The matter is a confirmed defect, active incident, legal prohibition, missing authority, or hard blocker requiring direct action rather than probabilistic treatment; it must still be linked to related risks where useful.
- The requested analysis would duplicate an authoritative current risk process with identical scope, version, environment, affected parties, and evidence; the existing process may be referenced if its adequacy is verified.
- Required analysis would expose protected information or create unsafe testing without authority.

A non-applicability determination must identify the evidence supporting it and the trigger that would reopen risk analysis.

## PREREQUISITES AND DEPENDENCIES

- Sections 001–017 as applicable, including current research, foresight, strategy, stakeholder, user, domain, and feasibility evidence.
- An authorized objective, scope, environments, lifecycle stages, risk decision owner, risk owners, and escalation authority.
- Defined values and assets to protect, including people, rights, mission, data, systems, services, money, evidence, environment, reputation, and continuity.
- Current architecture, process, dependency, data, model, supplier, control, incident, and operational evidence.
- Explicit legal, policy, safety, privacy, security, accessibility, contractual, financial, environmental, and mission constraints.
- A controlled method for rating uncertainty and consequence without pretending unsupported precision.
- A durable register, review cadence, evidence store, communication path, and mechanism to block or halt work when thresholds are crossed.

## REQUIRED INPUTS

- Objective, scope, success criteria, mission invariants, non-goals, affected parties, user and owner journeys, domain rules, and commitments.
- Feasibility conditions, assumptions, unknowns, dependencies, disqualifiers, points of no return, and experiment results.
- System, architecture, component, interface, authority, trust, data, event, state, model, provider, supply-chain, deployment, operation, maintenance, and retirement maps.
- Asset, value, harm, hazard, threat, vulnerability, defect, control, incident, near-miss, support, audit, and postmortem evidence.
- Likelihood, frequency, exposure, impact, duration, reversibility, detectability, velocity, recovery, and uncertainty evidence.
- Current controls, owners, control objectives, implementation status, effectiveness evidence, gaps, compensating controls, and dependencies.
- Risk appetite, tolerance, mandatory limits, acceptance authority, escalation thresholds, review periods, and prohibited conditions.
- Known populations or communities disproportionately exposed, accessibility needs, privacy expectations, safety conditions, and grievance or appeal pathways.
- AI or model identity, evaluation, data, failure modes, drift, hallucination, misuse, authority, oversight, fallback, and provider risks where applicable.

## HUMAN RESPONSIBILITIES AND AUTHORITY

Authorized humans and institutions define protected values, risk appetite, non-negotiable constraints, acceptable residual risk, treatment priorities, and the legitimacy of tradeoffs. Human authority must be real, attributable, informed, and within jurisdiction.

- Identify who owns each objective, asset, affected-party obligation, control, risk, response, and acceptance decision.
- Define unacceptable outcomes and mandatory constraints that cannot be outweighed by aggregate scoring.
- Decide how human rights, safety, privacy, accessibility, environmental burden, employment effects, public impact, and value conflicts are treated.
- Approve risk methods, criteria, thresholds, treatment budgets, testing, disclosures, external transfers, exceptions, and review cadence.
- Ensure affected parties, operators, specialists, and independent reviewers are represented where their evidence or rights matter.
- Accept residual risk only with adequate evidence, explicit scope, conditions, duration, review trigger, and authority; silence or continued work is not acceptance.
- Revoke acceptance and halt or narrow work when evidence, conditions, controls, or consequences materially change.
- Escalate risks that exceed delegated authority, cross jurisdictions, affect unrepresented parties, or create systemic consequences.

## AI RESPONSIBILITIES AND BOUNDARIES

The AI may help discover, classify, connect, analyze, monitor, and communicate risks. It may not invent risk appetite, accept harm, waive a mandatory constraint, suppress dissent, or assign false numerical certainty.

- Separate observed conditions, confirmed issues, plausible risks, assumptions, unknowns, opportunities, and speculation.
- Preserve causal structure: source or cause, uncertain event, consequence, affected objective or party, and conditions of exposure.
- Identify direct, indirect, second-order, delayed, cumulative, correlated, cascading, concentration, common-mode, and systemic effects.
- Distinguish inherent risk from current residual risk and proposed residual risk after treatment.
- State the basis and uncertainty for likelihood, impact, velocity, detectability, and control-effectiveness judgments.
- Do not lower a risk rating because evidence is missing; missing high-consequence evidence is itself a material uncertainty.
- Do not treat technical capability, model confidence, user consent, legal permission, security, safety, or social legitimacy as interchangeable.
- Surface minority findings, affected-party concerns, control dependencies, treatment tradeoffs, and risks transferred to others.
- Identify risks introduced by the treatment itself, including added complexity, lock-in, surveillance, exclusion, operator burden, and recovery fragility.
- Route risk acceptance, value conflict, and threshold changes to the authorized human decision owner.
- Preserve the exact version, environment, evidence, and time boundary of every risk determination.

## APPLICABLE EXISTING AGENT ROLES

Primary roles:

- Analyst — evaluates causes, consequences, uncertainty, comparative treatments, sensitivity, and decision implications.
- Planner — turns treatments into owners, tasks, dependencies, triggers, contingencies, validation gates, and completion evidence.
- Governance Reviewer — verifies authority, policy, scope, acceptance legitimacy, evidence, delegation, exceptions, and trusted-state consequences.
- Decision Arbiter Advisory — structures treatment and residual-risk recommendations under authorized criteria without seizing the acceptance decision.

Specialist and assurance roles:

- Researcher — obtains current threat, hazard, legal, regulatory, scientific, supplier, market, and domain evidence.
- Observer — records events, conditions, thresholds, incidents, near misses, control state, and data gaps without unsupported interpretation.

- Security Auditor — evaluates security threats, vulnerabilities, identity, access, supply chain, evidence integrity, and control effectiveness.
- Architect — identifies structural, interface, trust-boundary, dependency, common-mode, recovery, and lifecycle risks.
- User Experience Reviewer — evaluates human error, cognitive load, accessibility, trust, consent, dark patterns, recovery, and complete journeys.
- Tester — validates controls, boundary behavior, failure, denial, recovery, rollback, and monitoring.
- Verifier — independently checks evidence, risk claims, control operation, residual-risk boundaries, and acceptance conditions.
- Red Team — attacks assumptions, bypasses, abuse paths, correlated failures, control evasion, and optimistic residual-risk claims.
- Critic — identifies hidden consequences, missing requirements, false precision, unsupported ratings, and omitted affected parties.
- Memory Curator — maintains risk history, decisions, evidence, expiry, supersession, incidents, lessons, and reopening triggers.
- Coordinator — manages specialist inputs, conflicts, dependencies, treatment status, and consolidated reporting.

Role selection must match the risk and consequence, use independent review where required, and leave source-defined role contracts unchanged.

## REQUIRED ACTIVITIES

1. Establish risk context
Define the objective, scope, environments, lifecycle stages, assets and values, affected parties, authority, risk owners, acceptance authority, criteria, thresholds, and review horizon.

2. Define the taxonomy
Distinguish risk, issue, hazard, threat, vulnerability, exposure, defect, control failure, assumption, unknown, dependency, blocker, opportunity, incident, and residual risk so each receives the correct action.

3. Identify protected assets and values
Include life and safety, rights, privacy, accessibility, mission, service, data, identity, evidence, finances, environment, reputation, commitments, continuity, and future options.

4. Identify risks from multiple sources
Use research, incidents, near misses, audits, user research, stakeholder concerns, architecture, threat modeling, testing, operations, supply chain, model evaluation, legal review, foresight, and dissent.

5. Write bounded risk statements
For each risk, identify source or cause, uncertain event, consequence, affected objective or party, scope, environment, timeframe, and evidence.

6. Classify and connect risks
Identify category, lifecycle stage, ownership, dependencies, shared causes, correlated exposures, cascades, concentration, common-mode failure, and systemic reach.

7. Analyze likelihood and uncertainty
Use frequency, exposure, precedent, model, expert judgment, scenario, or interval evidence appropriate to the domain. Record confidence, data limitations, and conditions under which the estimate fails.

8. Analyze consequence
Consider severity, scale, duration, distribution, reversibility, detectability, velocity, recovery, legal or rights impact, mission impact, and who bears the harm or cost.

9. Evaluate inherent risk
Assess risk before current controls, preserving mandatory constraints and intolerable conditions separately from scored comparison.

10. Map controls and control objectives
Identify preventive, protective, detective, corrective, recovery, compensating, deterrent, governance, and evidence controls; record owner, implementation, dependencies, coverage, and verification.

11. Evaluate control effectiveness
Use design review, implementation evidence, tests, monitoring, incidents, audits, bypass attempts, human factors, and recovery evidence. Do not infer effectiveness from control documentation alone.

12. Determine current residual risk
State what remains after verified current controls, including uncertainty, untested paths, transferred exposure, and risks introduced by the controls.

13. Design treatment options
Consider avoid, reduce, isolate, substitute, share or transfer where legitimate, accept, defer with conditions, stage, monitor, prepare contingency, or prohibit. Include no-action and safe-halt options where materially distinct.

14. Evaluate treatment tradeoffs
Identify cost, delay, complexity, user burden, privacy effects, accessibility effects, new dependencies, control conflicts, risk transfer, lock-in, and recovery consequences.

15. Define triggers, thresholds, and response
For each material risk define leading and lagging indicators, trigger, threshold, alert path, owner, first action, escalation, communication, containment, rollback, restore, fallback, and safe-halt criteria.

16. Address AI and autonomous-system risk
Where AI participates, assess model limitations, data, hallucination, drift, misuse, overreliance, emergent behavior claims, authority confusion, prompt or tool abuse, provider dependence, monitoring, fallback, correction, appeal, and revocation.

17. Address human and affected-party risk
Evaluate cognitive load, accessibility, consent, manipulation, exclusion, inequitable exposure, operator fatigue, skill loss, false trust, blame transfer, grievance, and recovery burden.

18. Aggregate and prioritize
Examine interactions rather than summing isolated scores. Identify top risks, intolerable risks, systemic risks, control bottlenecks, concentration, and treatment sequencing.

19. Decide and record treatment
Record recommendation, authorized decision, owner, resources, due conditions, acceptance boundary, duration, evidence threshold, dissent, and what would change the decision.

20. Monitor, review, close, and reopen
Track indicators, incidents, control drift, environment changes, new evidence, treatment status, residual risk, expiry, and closure evidence. Reopen when conditions change or acceptance expires.

## HUMAN-AI INTERACTION REQUIREMENTS

- Risk information must be presented in language that names the condition, consequence, affected party, uncertainty, evidence, and next action rather than relying on color or score alone.
- The system must distinguish a proposed AI assessment from an authorized risk decision and an accepted residual-risk record.
- Humans must be able to inspect evidence, competing specialist views, affected-party concerns, treatment alternatives, control dependencies, and uncertainty before acceptance.
- The AI must not assign a person's or community's willingness to bear harm, surveillance, exclusion, or loss of control.
- Affected humans must have accessible notice, correction, appeal, grievance, and recovery paths where the system creates decisions or exposures affecting them.
- Capability and confidence indicators must not imply authority, safety, legality, or acceptability.
- Missing telemetry, stale evidence, unavailable sources, or unknown control state must be visible and may require degraded status or safe halt.
- The owner must be able to pause, narrow, revoke delegation, change thresholds, reject treatment, require review, and stop consequential action.
- AI-generated risk summaries must preserve minority findings, uncertainty, and transferred risk rather than compressing them into false consensus.

- Notifications must avoid both alarm fatigue and silent suppression; severity, urgency, confidence, ownership, and expected response must be explicit.

## ARCHITECTURE AND SYSTEM IMPLICATIONS

Risk treatment must become enforceable system behavior where architecture or runtime control is required.

- Trace risks to assets, requirements, architecture decisions, components, interfaces, data, actors, dependencies, controls, tests, monitoring, incidents, maintenance, and retirement obligations.
- Separate control objective, control design, implementation, configuration, operation, evidence, and effectiveness status.
- Enforce mandatory authority, security, privacy, safety, accessibility, and evidence conditions consistently across UI, API, agent, scheduled, retry, integration, and administrative paths.
- Design for defense in depth without assuming one control compensates for every failure.
- Identify common-mode and correlated control dependencies such as one identity provider, cloud region, model provider, key store, data source, operator, supplier, or monitoring path.
- Provide observable indicators, control health, freshness, provenance, thresholds, alerts, escalation, and incident linkage.
- Include graceful degradation, isolation, rate limits, circuit breaking, safe halt, manual fallback, rollback, restore, backup, failover, and recovery where appropriate.
- Prevent an AI agent, model, or automated policy from expanding authority, changing risk thresholds, suppressing evidence, or accepting residual risk without authorization.
- Preserve immutable or tamper-evident risk decisions, exceptions, approvals, evidence, control changes, incidents, and review history where consequence warrants.
- Propagate invalidation when architecture, environment, data, model, provider, law, user population, or control state changes.
- Treat maintenance, dependency updates, migration, deprecation, data deletion, service decommissioning, hardware disposal, and retirement as risk-bearing transitions.

## OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

The owner-facing operating surface or equivalent durable record should expose:

- Risk context, objectives, assets, affected parties, environments, lifecycle stages, owners, acceptance authority, and current review state.
- Risk statements with source, event, consequence, likelihood basis, impact basis, uncertainty, velocity, detectability, and evidence.
- Inherent, current residual, and projected residual risk without implying false precision.
- Mandatory constraints, intolerable conditions, appetite, tolerance, thresholds, and expired or missing authority.
- Controls with objective, owner, implementation state, evidence, effectiveness, dependencies, gaps, and next verification.
- Treatment alternatives, tradeoffs, transferred risks, new risks introduced, resources, due conditions, and dissent.
- Indicators, triggers, alerts, incidents, near misses, trends, stale telemetry, and control degradation.
- AI/model risks including identity, version, provider, evaluation, data, authority, human oversight, fallback, drift, and revocation.
- Acceptance records with scope, decision owner, rationale, evidence, conditions, duration, review trigger, and affected parties.
- Direct controls to pause, narrow, revoke, escalate, enter safe halt, initiate response, verify recovery, and reopen risk review.

## REQUIRED OUTPUTS AND ARTIFACTS

- Risk-management context and authority record.
- Risk taxonomy and scoring or classification method with limitations.
- Asset, value, affected-party, objective, and obligation register.
- Risk register with bounded risk statements and provenance.
- Risk dependency, correlation, cascade, concentration, and systemic map.
- Hazard, threat, vulnerability, assumption, unknown, issue, defect, and opportunity linkage as applicable.

- Control-objective and control register with implementation, ownership, dependencies, evidence, and effectiveness.
- Inherent-risk, current-residual-risk, and projected-residual-risk assessments.
- Treatment option analysis, decision, plan, budget, owner, conditions, and schedule.
- Indicator, threshold, alert, escalation, communication, contingency, rollback, restore, fallback, and safe-halt plans.
- AI/model, data, human, accessibility, privacy, security, safety, legal, operational, financial, environmental, supply-chain, and lifecycle risk assessments as applicable.
- Risk-acceptance, exception, rejection, escalation, dissent, and review records.
- Monitoring, incident, near-miss, control-drift, closure, and reopening evidence.
- Downstream requirements, architecture, test, observability, operational, maintenance, and lifecycle handoff.

## EVIDENCE REQUIREMENTS

- Every material risk must identify evidence source, version, environment, timeframe, method, scope, quality, and uncertainty.
- Likelihood and impact ratings require an explicit basis; unsupported numeric precision is prohibited.
- Control effectiveness requires implementation and operating evidence, not policy or configuration intent alone.
- Incident, near-miss, audit, test, monitoring, user, support, supplier, model-evaluation, and recovery evidence must preserve provenance and known gaps.
- Evidence must identify affected populations, conditions, environments, versions, and lifecycle stages so risk is not generalized beyond support.
- Risk acceptance must link to the exact residual-risk assessment, conditions, decision authority, duration, review trigger, and evidence available at the time.
- Transferred or shared risk must identify who now bears the consequence and whether the transfer is legitimate and understood.
- Missing evidence, stale telemetry, unverifiable controls, or unknown exposure must remain visible and may increase uncertainty or block advancement.
- Qualified domain sources are required for legal, medical, safety, environmental, financial, regulatory, or other specialist conclusions where consequence warrants.

## VERIFICATION AND ACCEPTANCE TESTS

- Trace each top risk to objectives, affected parties, assets, causes, consequences, controls, owners, tests, indicators, treatment, and acceptance or escalation decision.
- Independently test whether mandatory constraints can be bypassed through UI, API, agent, scheduled, retry, integration, offline, degraded, or administrative paths.
- Remove or fail a control dependency and verify the risk state, alert, treatment, and trusted status update correctly.
- Inject stale, missing, contradictory, or corrupted telemetry and verify the system does not display low risk or healthy control state.
- Reproduce control tests and compare expected versus observed prevention, detection, response, recovery, rollback, and restore behavior.
- Simulate correlated and cascading failures rather than testing each component only in isolation.
- Test human error, accessibility, misunderstanding, alert fatigue, consent withdrawal, escalation, grievance, correction, and recovery journeys.
- Alter model, data, provider, environment, version, workload, user population, law, or authority and verify affected risks reopen.
- Give an AI capable of action no permission to change thresholds, accept risk, suppress alerts, or continue beyond safe-halt conditions.
- Verify that a downstream engineering or operations role can reconstruct current risk state and treatment obligations from durable artifacts alone.
- Review expired acceptances, exceptions, and unverified controls and confirm they cannot silently remain trusted.

## FAILURE STATES AND ANTI-PATTERNS

- Decorative risk register disconnected from requirements, architecture, controls, tests, operations, incidents, and decisions.
- Treating missing evidence as low likelihood or no risk.

- Using arbitrary multiplication or color coding as a substitute for causal analysis, affected-party impact, uncertainty, and mandatory constraints.
- Confusing a confirmed issue, prohibition, defect, incident, or missing authority with a future risk.
- Treating risks independently while ignoring shared causes, correlated exposure, concentration, common-mode failure, or cascade.
- Recording a control as effective because it exists in policy, code, or configuration without operating evidence.
- Allowing residual risk to be accepted by an AI, unauthorized person, silent workflow, expired approval, or default setting.
- Transferring risk to users, operators, vendors, communities, or future maintainers without disclosure, legitimacy, or support.
- Hiding accessibility, privacy, safety, environmental, labor, or long-term human effects behind technical or financial averages.
- Treating consent as permission to impose any risk or as a substitute for safety, legality, accessibility, and fairness.
- Ignoring treatment-created risks such as complexity, surveillance, lock-in, operator burden, brittle recovery, or reduced transparency.
- Suppressing dissent, incidents, near misses, failed tests, or unfavorable evidence to maintain a preferred rating.
- Failing to expire or reopen risk assessments after material change.
- Treating risk acceptance as permanent approval for scope expansion.
- Declaring safe or acceptable status when controls, telemetry, identity, evidence, or authority are unknown.

## RECOVERY AND RECONSIDERATION

Risk recovery restores a truthful risk state before restoring ordinary operation.

- Detect and name the changed condition, affected risks, objectives, parties, controls, environments, and trusted-state consequence.
- Preserve evidence, telemetry, logs, decisions, approvals, incidents, previous ratings, and last-known-good references.
- Enter degraded mode or safe halt where thresholds, mandatory controls, evidence, or authority require it.
- Contain the cause without destroying evidence or unrelated safe state.
- Reassess inherent and residual risk using current evidence; invalidate expired acceptance and affected downstream artifacts.
- Activate contingency, rollback, restore, fallback, communication, support, grievance, or external notification as authorized.
- Obtain qualified review and human decision where new consequences, values, or authority boundaries arise.
- Verify control restoration, critical journeys, evidence integrity, and residual risk before trusted return to operation.
- Record lessons, treatment changes, tests, monitoring changes, acceptance changes, and reopening triggers in continuity.

## COMPLETION CRITERIA

- Risk context, assets, values, affected parties, authority, owners, criteria, thresholds, environments, and lifecycle stages are explicit.
- Taxonomy separates risk from issue, hazard, threat, vulnerability, defect, assumption, unknown, dependency, blocker, opportunity, and incident.
- Material risks use bounded causal statements and include evidence, uncertainty, ownership, dependencies, and review triggers.
- Mandatory constraints and intolerable conditions cannot be outweighed by scoring or preference.
- Inherent risk, current controls, control effectiveness, current residual risk, treatments, and projected residual risk are distinct.
- Correlated, cascading, concentration, common-mode, systemic, treatment-created, and transferred risks are addressed where applicable.
- Human, accessibility, privacy, safety, security, legal, environmental, data, AI/model, operational, financial, supply-chain, maintenance, and retirement consequences are considered proportionately.
- Treatments have owners, resources, dependencies, evidence thresholds, tests, triggers, contingencies, and completion conditions.

- Residual-risk acceptance or rejection is attributable, authorized, scoped, conditioned, time-bounded, reviewable, and evidence-linked.
- Monitoring, incident, recovery, closure, reopening, and continuity mechanisms exist.
- Requirements Engineering and Systems Engineering can consume the treatments without inventing missing control obligations.

## HANDOFF AND CONTINUITY REQUIREMENTS

- Transfer risk context, taxonomy, criteria, assets, affected parties, owners, authority, appetite, tolerance, thresholds, and lifecycle scope.
- Provide the complete risk register, dependency and cascade map, control register, evidence, effectiveness state, treatment plan, and residual-risk decisions.
- Carry forward every mandatory constraint, prohibited condition, accepted exception, expiry, trigger, escalation, safe-halt condition, and human decision point.
- Map treatments to requirements, architecture decisions, security, privacy, safety, accessibility, testing, observability, incident response, business continuity, disaster recovery, maintenance, migration, decommissioning, and retirement.
- Preserve dissent, unknowns, transferred risk, treatment-created risk, failed controls, incidents, near misses, assumptions, and evidence needed to reopen decisions.
- State current risk status, work allowed, work conditionally allowed, work blocked, work prohibited, next review, and next safe action.

## RELATED CODEX SECTIONS

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005 Business Strategy
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## SOURCE BASIS AND UNRESOLVED SOURCE GAPS

This draft applies the supplied requirements for risk notification, blocker truth, evidence-before-claims, current-problem treatment, safety and data protection, bounded authority, role independence, hostile review, failure and recovery, continuity, and no fake success. The Planner contract supplies risk fields, triggers, likelihood, impact, prevention, detection, response, owner, residual risk, and decision threshold. The Analyst and Decision Arbiter Advisory contracts supply consistent option evaluation, mandatory constraints, uncertainty, sensitivity, reversibility, residual risk, and human decision authority. The Security Auditor, Governance Reviewer, Critic, Red Team, Tester, Verifier, Observer, User Experience Reviewer, Architect, Memory Curator, Coordinator, and Researcher contracts supply specialist identification, challenge, control, evidence, monitoring, and continuity behavior. The Owner's Cockpit contributes trust calibration, visible limitation, agent identity, bounded autonomy, incidents, safe halt, release, rollback, recovery, and owner control. The V4 draft contributes claim and obligation typing, prerequisite-aware analysis, dissent preservation, evidence contracts, trusted transitions, and artifact-consistency blocking.

Unresolved source gap: the supplied corpus does not contain a single universal risk-management standard, quantitative method, hazard-analysis method, actuarial model, safety case, financial-risk framework, or sector-specific regulatory taxonomy. This section therefore provides a general lifecycle governance scaffold and prohibits invented precision. Actual risk methods, thresholds, scoring, acceptance, and specialist conclusions must use current qualified sources for the applicable domain and must later be reconciled with Requirements Engineering, Systems Engineering, Security, Privacy, Safety, Accessibility, Legal, Compliance, Internal Controls, Verification, Validation, Operations, Maintenance, and Lifecycle Governance.